Index Calculus for the Elliptic-Curve Discrete Logarithm:

Theory, Algorithms, and a Measured Frontier

A treatise from a four-month research programme,

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Abstract

Pollard rho remains the fastest known algorithm for the discrete logarithm on elliptic curves of cryptographic shape. Index calculus could in principle exploit more structure, but it has never beaten rho there. This treatise collects everything a four-month research programme (July–October 2026) learned about why. Every number is drawn from committed evidence and carries its boundary: oracle stage, complete solve, cold one-target cost, or online cost.

Theory. We give the Galois group of Semaev's summation cover. It is $(\mathbb{Z}/2)^{m-2}$ for every curve, field and characteristic, with arithmetic monodromy equal to geometric. We also give an exact factorisation law for its fibres over $\mathbb{F}_q$. Two consequences follow:

- fibre statistics are constant on the factor-base locus;
- at fixed factor-base size the mean decomposition yield is fixed.

So levers that rely on curve-dependent monodromy or on raising mean yield at matched base size are closed without experiment. Coverage, oracle cost and solving difficulty remain free.

On fall degrees, we prove for the binary Weil descent of S_3 that:

- the first fall degree is 2 for every n, including Semaev's chained systems, where 4 was stated;
- the last fall degree is at most $k+2$, or $(t-1)k+t$ for chains.

We also certify that the last fall degree is at least 5 on named instances at $n = 33$–45 and in the $m = t = 3$ chain, with certificates that an independent program checks. Every F4 run on those systems therefore exceeds degree 4, and Semaev's Assumption 1 fails as stated. For an explicit-membership chain we prove $(d_{\mathrm{ff}}, d_F) = (4, 5)$ exactly at $B = 3$ for every $p > 3$, and $\max(5, B+1) \le d_F \le 3B-4$ in general. For PKM Kummer towers we prove a constant first fall degree of 4 and show that the last fall degree depends on the presentation, not the ideal. Throughout, the first fall degree is a constant of the presentation and the last is not.

Algorithms. The programme engineered:

- root-finding and meet-in-the-middle decomposition oracles, two to five orders of magnitude cheaper than Gröbner solves where they apply;
- inherited and gated F4 engines, and an $\mathbb{F}_p$ F4 at msolve parity;
- compact Frobenius-orbit and known-logarithm-closed factor bases;
- FGHR's D_4 symmetry at $k = 4$, cutting one decomposition 370–481×;
- double-large-prime genus-3 index calculus after the cubic GHS transfer, from $q^{1.91}$ to $q^{1.36}$;
- Diem's plane-quartic index calculus on the same family, at exponent ≈ 1.

Frontier. A priced campaign of 4,848 executions found no index calculus below rho on any prime or prime-degree binary curve at any weight of a pricing band applied to every arm. The distance at a matched reference is 4.7–18× on Koblitz curves of 37–47 bits and 10.5–102× on prime curves of 20–40 bits. Only in weak extension-field families does index calculus reach or cross rho:

- genus 3 via GHS: below the strongest rho at one size, unresolved at the others;
- plane quartics: 0.005× a negation rho at $q \approx 2^{21}$;
- Joux–Vitse covers: 1.6–2.0×;
- $k = 4$ in characteristic two: unsupported beyond $m = 15$.

Every online win was retired by an equal-advice audit against Bernstein–Lange rho. We record the corrections that matching references, pricing native work and replaying records forced. Nothing here concerns a deployed curve or key recovery.

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Introduction

For a subgroup $\langle G \rangle$ of prime order r on an elliptic curve $E/\mathbb{F}_q$, Pollard rho [32, 38] computes discrete logarithms in about $\sqrt{\pi r/(2A)}$ group operations, where A is the number of efficiently computable automorphisms [18, 41]. No algorithm is known to beat it on curves of cryptographic shape.

Index calculus breaks finite-field discrete logarithms in subexponential time. It reached elliptic curves through Semaev's summation polynomials [33] and the extension-field analyses of Gaudry [19] and Diem [9]. It is the method that should exploit the most structure. The question is *where it crosses rho*: on which curves, at which sizes, under which accounting.

This treatise records what one research programme learned about that question between July and October 2026. The programme spanned three repositories (`crypto`, `crypto-autoresearcher`, `cryptanalysis`), several thousand commits and several hundred preregistered studies. Two companion documents organise the same work differently:

- a programme record [3], chronological and exhaustive, with study catalogues;
- an improvement ledger [2], organised as a ratchet of incumbents.

This document organises it as a treatise:

- the theory that constrains index calculus on elliptic curves;
- the algorithms engineered against those constraints;
- the measured frontier against a fairly priced rho;
- what remains open.

It runs no new experiment. It re-verified one certificate (§6.3), adds no improvement of its own, and cites the committed artefact behind every number.

The argument in brief

1. Where the cost lives (Part I). An index calculus pays for factor-base set-up, relation collection, linear algebra and the target's descent. Relation collection costs (trials per relation) × (cost of one decomposition). Chapter 2 writes this out and maps every lever the programme pulled to the term it moves.

2. What is fixed for every curve (Chapter 5). The Galois group of the summation cover is $(\mathbb{Z}/2)^{m-2}$ for every curve, field and characteristic. Its fibres over $\mathbb{F}_q$ split by an exact law, and they split completely on the factor-base locus. The mean decomposition yield at fixed base size is a double count. Three consequences follow:

- no curve has anomalous monodromy to exploit;
- no fibre census can find an anomalous relation rate;
- no factor-base geometry can raise mean yield.

The design space that remains is the cost of the oracle, the algebraic difficulty of the decomposition system, the linear algebra, and where the problem lives.

3. How hard is the decomposition system (Chapter 6)? Every subexponential claim for prime-degree binary curves assumes that the solving degree stays near the first fall degree. In three presentations we prove that the first fall degree is a constant of the presentation, and show that the last fall degree, which equals the solving degree for unsatisfiable systems, is not bounded by it:

- it is proved to differ for an explicit-membership chain ($(4, 5)$ exactly at $B = 3$);
- it is certified to differ, by independently checked closure certificates, for binary Weil descent (≥ 5 against 2 at $n = 33$–45);
- it is measured to differ for prime-field Kummer towers (6 against 4).

A proof that the last fall degree grows without bound remains the open problem, as Huang, Kosters and Yeo [22] and Kousidis and Wiemers [27] left it.

4. What engineering bought (Part III). Root-finding oracles made the decomposition step two to five orders of magnitude cheaper where they apply. Gröbner engines improved one to two orders in counted work. Factor bases closed under maps of known logarithm cut linear algebra by constant factors.

On the cold one-target frontier none of this changed the conclusion: the stages these changes leave alone (relation yield per probe, table set-up) now dominate.

5. Where index calculus wins (Chapter 13 and Chapter 14). Only in weak extension-field families does the measured cost cross or approach a fairly priced rho:

- $E_0/\mathbb{F}_{q^3}$ through a genus-3 GHS cover, by double-large-prime hyperelliptic index calculus or by Diem's plane-quartic index calculus;
- the Joux–Vitse cover of $E(\mathbb{F}_{p^6})$;
- possibly $k = 4$ over $\mathbb{F}_{(2^m)^4}$.

These are known theoretical results. The contribution here is matched, priced, verified measurement. None of these families is deployed.

6. What the measurements say (Part IV). At a matched reference, rho is cheaper on every prime and prime-degree binary curve measured: 4.7–18× on Koblitz curves of 37–47 bits, and 10.5–102× on prime curves of 20–40 bits, with the gap widening as $r^{1/6}$. Every apparent online win disappeared against Bernstein–Lange rho with equal advice [4]. Matching the reference, pricing native work and replaying records each moved recorded gaps upward. Chapter 18 lists every such correction.

Conventions

- *Unit.* Costs are group-addition equivalents per $\sqrt{r}$ (S) unless a family uses its own counted unit, such as $\mathbb{F}_q$ multiplications.
- *Ratios.* "$X\times$" is baseline over candidate when it describes an improvement, and IC over rho when it describes a distance. Each sentence says which.
- *Intervals.* Intervals are 95%, as the source recorded them.
- *Change classes.* Every change is an ADVANCE (an exponent or crossover moves under complete accounting), ENGINEERING (a constant moves), RELABELLING or ACCOUNTING. Against rho, the programme produced no ADVANCE on any prime-order curve of cryptographic shape.
- *Paths.* Artefact paths are relative to the `crypto` repository, unless prefixed `crypto-autoresearcher:` or `cryptanalysis:`.

What this treatise does not claim. No speed-up on a deployed curve, no key recovery, no asymptotic sub-rho algorithm for prime fields, and no statement about the security of ECC2K-130, P-256 or secp256k1. The programme's GPU Pollard-rho work on ECC2K-130 is published separately (IACR ePrint 2026/2469) and is not repeated here.

Part I

Foundations

1 Background

Rho and its floor. With A efficiently computable automorphisms acting on $\langle G\rangle$, a rho walk on orbit representatives has expected cost $\sqrt{\pi r/(2A)}$ group additions. $A = 2$ (negation) on every curve; $A = 2n$ on a Koblitz curve $y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$, where the Frobenius $\pi(x, y) = (x^2, y^2)$ acts on $\langle G\rangle$ as a scalar λ [18, 41]. ECC2K-130 is the case $a = 0$, $n = 131$ of this family; its record GPU rho attempt is a separate paper in this repository (`paper/ecc2k130/`).

Index calculus on elliptic curves. Fix a factor base $\mathcal{F} \subset E(\mathbb{F}_q)$. A *relation* is a decomposition $aG + bQ = R = P_1 + \cdots + P_m$ with $P_i \in \mathcal{F}$; once about $|\mathcal{F}|$ relations are found, linear algebra over $\mathbb{Z}/r$ yields $\log_G Q$. The pipeline has five stages (factor base, decomposition oracle, relation yield, rank/linear algebra, verification) and a sixth, the comparison with rho on the same target. The *decomposition oracle* decides whether R splits over $\mathcal{F}$. Semaev [33] reduced it to the roots of the summation polynomial $S_{m+1}(x_1, \ldots, x_m, x_R)$, which vanishes exactly when $\pm P_1 \pm \cdots \pm P_m = R$ for points with those abscissae. Over $\mathbb{F}_{2^n}$ or $\mathbb{F}_{p^k}$ the factor base is chosen as the points whose abscissa lies in a subspace or subfield, and Weil restriction turns $S_{m+1} = 0$ into a polynomial system over the base field; the textbook oracle solves that system with a Gröbner-basis engine (F4/F5, [11, 12]) or a SAT solver [37]. Gaudry [19] and Diem [9] give the asymptotics over extension fields; for prime fields no factor base is known that makes the method competitive, and summation-polynomial systems over $\mathbb{F}_p$ are the subject of the Kummer-tower and rational-map bases of Petit–Kosters–Messeng [31].

Frobenius-invariant bases. Galbraith, Granger, Merz and Petit [17] observe that a factor base with $\pi(\mathcal{F}) = \mathcal{F}$ splits into Frobenius orbits of size dividing n, with $\log \pi^k(P) = \lambda^k \log P$. The unknowns collapse by $\approx n$, the matrix by $\approx n^2$, and ordered decompositions save $m!$—more than rho's $\sqrt{2n}$, but from a base too far behind to overtake it at deployed sizes. Invariant bases come from linearised polynomials attached to the factors of $x^n - 1$ over $\mathbb{F}_2$; the subspace dimension is $\ell = \mathrm{ord}_n(2)$, so the base size is fixed by n (Table 1 of `RESEARCH_KOBLITZ_INDEX_CALCULUS.md`).

Symmetries of the system. The decomposition system is invariant under the symmetric group S_m; rational 2-torsion adds sign changes, and Faugère, Gaudry, Huot and Renault [13] show that in the coordinate Y with $Y(P + T) = -Y(P)$ the invariance group becomes the Weyl group D_m, with a much smaller quotient. We measure this at $m = 4$ in §12.9.

2 Where the cost of index calculus lives

An improvement is only useful if it moves a term that matters. This chapter writes the cost of one index-calculus solve as a sum of terms, and maps each lever of the programme to the term it acts on.

2.1 The pipeline

Fix a factor base $\mathcal{F} \subset E(\mathbb{F}_q)$ of size N. A *relation* is a decomposition $aG + bQ = P_1 + \cdots + P_m$ with every $P_i \in \pm\mathcal{F}$. Once about N independent relations are known, linear algebra over $\mathbb{Z}/r$ yields $\log_G Q$. A heuristic count is enough to place the levers. A uniformly random point decomposes into m base points with probability about $N^m/(m!\,r)$. Collecting N relations therefore needs about $m!\,r/N^{m-1}$ trials. The cold cost of one target is then

$$T \;\approx\; \underbrace{T_{\mathrm{setup}}(N)}_{\text{base, tables}} \;+\; \underbrace{\frac{m!\,r}{N^{m-1}}\,\frac{1}{\rho_{\mathrm{yield}}}\,C_{\mathrm{dec}}(N,m)}_{\text{relation collection}} \;+\; \underbrace{T_{\mathrm{LA}}(N,m)}_{\text{linear algebra}} \;+\; \underbrace{T_{\mathrm{target}}}_{\text{descent, check}} \quad . \tag{2.1}$$

Here C_{dec} is the cost of one decomposition attempt (the *oracle*). The factor $\rho_{\mathrm{yield}} \geq 1$ collects everything that makes a trial more productive than the heuristic: free relations, large primes, symmetry, re-randomised targets. T_{LA} is roughly mN^2 for sparse elimination. With rho's cost $\sqrt{\pi r/(2A)}$ on the other side, the classical balance at $m = 2$ with a constant-cost oracle gives $N \approx r^{1/3}$ and $T \approx r^{2/3}$. That is why two-summand prime-field index calculus loses ground to rho as $r^{1/6}$ (Chapter 17).

2.2 The levers

Table 2.1 lists, for each term of (2.1), the levers the programme pulled and the sections that report them.

2.3 Which term dominates now

Early in the programme, C_{dec} dominated. A Gröbner solve per trial cost milliseconds to seconds. After root finding (Chapter 8), the oracle is a handful of field operations and table probes, and the cost has moved elsewhere.

- On the Koblitz descent of the leaderboard's Table A, reusable set-up is 98.5–99.7% of cold S, and descent is 0.3–1.4% (`docs/ic/LEADERBOARD.md`, phase table).
- On the $n = 83$ cold pipeline with base K1107, a mean 113.755 s process splits into 47.166 s of set-up and 66.486 s of rank stage. The target stages take 0.099 s (`research/n83_cold_base_confirm_20261021/RESULT.md`).

So the relation-collection and set-up terms dominate. Their cost is set by N and by the yield per probe, not by the oracle. This is the main structural fact behind the ledger. Oracle speed-ups of three to five orders of magnitude moved the end-to-end gap by one order at most.

Table 2.1: The terms of (2.1) and the levers that act on them.

Term	Levers pulled	Where
C_{dec}	root finding of S_3; pair and meet-in-the-middle tables; inherited F4, F5, F6-IC; gate-first order; harvest gates; $\mathbb{F}_p$ F4; D_4 invariants	Ch. 8, 9, 13
ρ_{yield}, N	compact Frobenius orbits; free relations; stride-renewed targets; greedy coverage; $n = 83$ base selection; double large primes	Ch. 10, 12, 14
T_{setup}	stacked producer; cache-blocked pair table; u64 root batching	Ch. 10, 12
T_{LA}	CADO-style filter and Wiedemann; structured Gaussian elimination; block Wiedemann; first-dependency stop	Ch. 10, 12, Ch. 13, 14
the exponent	genus-three transfer (cubic GHS); $k = 4$ extension fields	Ch. 13, 14

3 The theory, by field type

3.1 What a decomposition costs

Every index-calculus algorithm for elliptic curves in the literature spends its time in three places: finding about $|\mathcal{F}|$ relations (each a decomposition of a random combination $aG + bQ$ over $\mathcal{F}$), solving the sparse linear system over $\mathbb{Z}/r$, and, in some variants, a descent of the target. The second and third are well understood. The first is the crux: a uniformly random point decomposes over a base of size N into m summands with probability about $N^m/(m!\,r)$, so relations need about $m!\,r/N^{m-1}$ trials, and each trial costs whatever the decomposition oracle costs. Every result below is a statement about one of three things: how large N can be while membership stays algebraically testable, how cheap the oracle is, and how the two trade against the linear algebra.

3.2 Extension fields $\mathbb{F}_{q^k}$ with $k > 1$ fixed or small

Gaudry [19] takes $\mathcal{F}$ to be the points with abscissa in the base field $\mathbb{F}_q$ and decomposes into $m = k$ summands by Weil restriction of S_{k+1}, a system of k equations in k unknowns over $\mathbb{F}_q$. For fixed k this gives $\tilde{O}(q^{2-2/k})$, asymptotically below rho's $q^{k/2}$ for $k \geq 3$; the constant hidden in the Gröbner solve grows very quickly with k, which is what makes the crossover large in practice; the measured handovers at $k = 4$ are in Chapter 13. Diem [9] extends the analysis to families where k grows, proving subexponential complexity over sequences of fields with suitably balanced q and k. Joux and Vitse [25] combine Weil descent with decomposition-based index calculus into one algorithm, well suited to curves over $\mathbb{F}_{p^6}$, and computed logarithms on a 149-bit group over a degree-6 extension in about 110,000 CPU-hours; they also give an $(n-1)$-decomposition variant. FGHR [13] exploit the action of S_m and of the 2-torsion to replace S_{m+1} by its invariant form, cutting the Gröbner work by large factors; our measurement at $k = 4$ (§12.9) is a direct instance.

3.3 Binary fields $\mathbb{F}_{2^n}$ with n prime

Prime n leaves no subfield, so the factor base is taken to be the points whose abscissa lies in an $\mathbb{F}_2$-vector subspace V of dimension ℓ. Weil restriction then gives a quadratic-to-low-degree Boolean system whose difficulty is governed by its degree of regularity. The whole debate about binary curves is a debate about that degree.

The subexponential claims. Petit and Quisquater [30] conjectured, from experiments and heuristic evidence, that these systems have degrees of regularity only slightly above the degrees of their equations (the *first fall degree assumption*), and deduced a heuristic $2^{c\,n^{2/3}\log n}$, $c < 2$, for a variant of Diem's algorithm, beating generic methods only for $n \gtrsim 2000$. Semaev [34] split summation polynomials into cubic Boolean systems, argued for regularity degree at most 4 under the same kind of assumption, and obtained a heuristic $2^{c\sqrt{n \ln n}}$ with $c \approx 1.69$; its first version concluded that several FIPS binary curves fall faster than Pollard rho, a conclusion its revision says does not hold under the weaker assumption that the regularity degree is $o(\sqrt{n/\ln n})$.

The evidence against. Shantz and Teske [35] tested the Petit–Quisquater heuristics experimentally. Huang, Kosters and Yeo [22] introduced the *last fall degree*, independent of the monomial order, gave computational evidence against the first fall degree assumption in characteristic 2, and constructed summation-polynomial Weil descent systems where it is unlikely to hold. Galbraith and Gebregiyorgis [16] improved the variable choice (binary Edwards curves), found symmetry-breaking factor bases and SAT solvers inconclusive, and concluded that rho remains much faster for prime-degree binary fields of reasonable size. The Galbraith–Gaudry survey [15] reaches the same verdict for characteristic-two curves over prime-degree fields.

Where it stands. No proof or refutation of a subexponential algorithm for prime-degree binary curves exists. The open quantity is the solving degree of Weil-descended S_{m+1} as n and m grow; recent work supplies diagnostics (saturation-exponent bounds, multi-graded regularity, tests for where systems stop looking generic) but no bound for these systems: rigorous last-fall-degree bounds for summation-polynomial descents are named as open by Huang, Kosters and Yeo [22] and by Kousidis and Wiemers [27], and Kosters and Yeo [26] show that the first fall degree of the descent of S_3 is usually 2, through a trace identity. Chapter 6 proves and certifies what the programme could add. Our own measurements on toy fields (§11.1) find first and last fall degrees of 3–4 at every reachable size while the solving degree grows with ℓ and stays within one of the semi-regular prediction; closure algorithms refute a degree or two earlier but rise too. That supports no subexponential claim and refutes none.

3.4 Subfield (Koblitz) curves

When the curve is defined over $\mathbb{F}_2$ the Frobenius acts on the group, and Galbraith, Granger, Merz and Petit [17] show that a Frobenius-invariant factor base saves a factor $\approx n$ in relations and $\approx n^2$ in linear algebra—more than rho's $\sqrt{2n}$—while concluding that Koblitz-curve security is not threatened at deployed sizes. Our Koblitz work (§10) builds on this.

3.5 Prime fields

Over $\mathbb{F}_p$ there is no subfield and no subspace, and no factor base is known for which membership is both dense enough and algebraically testable. Semaev's original small-abscissa base leads to systems of very high degree. Petit, Kosters and Messeng [31] construct bases from rational maps (including the Kummer tower μ_{2^t}) that give the summation system extra structure; Amadori, Pintore and Sala [1] build the base from the instance and need only one relation search, improving on both; McGuire and Mueller [28] avoid Gröbner bases altogether by fast evaluation of S_m through iterated resultants, and in a second variant by summing $m-1$ factor-base points and testing membership of the result—the same mechanism as the two-summand root-finding oracle of §8. All of these remain exponential and slower than rho; a 2024 analysis again concludes that no subexponential prime-field method is known.

3.6 Lower bounds

Generic-group lower bounds bound everything that does not use curve structure. With preprocessing, Corrigan-Gibbs and Kogan [7] show $ST^2 = \tilde{\Omega}(\varepsilon N)$ for S bits of advice and online time T; this is the bound against which the retained-set-up (online) results of §10 must be read. A 2026 structured-generic-group bound shows that an algorithm exploiting structure on only a δ fraction of elements needs $\Omega(\min(\sqrt{q}, 1/\delta))$ time—for index calculus, the decomposable set must be dense *and* cheaply testable.

4 Measuring index calculus against rho

The programme's first lesson was that index-calculus numbers from different studies did not compose: one study timed a decomposition, another a relation, a third a whole multi-target batch with shared tables, each on a different host and in a different unit. Four mechanisms, built in this order, fixed that.

4.1 The boundary ledger (10 September)

`docs/ic/BOUNDARY_TARGETS.md` and its machine twin `docs/ic/boundary_targets.json` give each regime (Koblitz, binary, prime) and each pipeline stage (`factor_base`, `decomposition`, `relation_yield`, `rank`, `end_to_end_dlp`, `vs_rho`) a current record, a next target and acceptance gates. A row promotes only when its evidence JSON carries every field the measurement schema requires for that stage—for example, a `decomposition` claim that beats an algebraic frontier must report the first-fall degree, and a `vs_rho` claim needs one target, the identical public point for both methods, verified scalars, five exclusive IC online phases, the automorphism discount, matched resources and a host-isolation receipt. Missing fields fail closed. The primary `vs_rho` question is the online cost of *one* previously unseen target after reusable IC preparation; multi-target and amortised results are secondary and must be declared as such.

4.2 ecbench (from 2 October)

ecbench (`src/cryptanalysis/ecbench/`, `src/bin/ecbench.rs`, documented in `docs/ecbench/README.md`) runs every ECDLP method behind one interface, on the same single-target workloads, in one unit, into sealed records.

Unit.

$$S \;=\; \frac{\text{total group-addition equivalents (GAE)}}{\sqrt{r}}, \qquad S_{\text{floor}} = \sqrt{\pi/(2A)}.$$

An addition or doubling is one GAE; a scalar multiplication is charged as the additions it performs; native work (lookups, row operations, square roots, Artin–Schreier solves, word XORs) is priced at a pinned conversion where the repository has one and is otherwise recorded as *uncharged*, which marks the record's S as a *lower bound*. Plain rho has $S \approx 1.25$.

What a measurement is. One workload is one curve, subgroup, generator and target; nothing survives between executions, so every table a method builds is inside its cost (*cold*). Beside the cold S every record carries the one-target *online window*, split for index calculus into the five exclusive phases `target_query`, `target_PDP`, `target_relation_check`, `target_descent`, `target_recovery_check`. Targets are either planted or *public*—hashed to the subgroup so that nobody knows the logarithm—and the runner verifies $[k]G = Q$ in its own process; `ecbench verify` checks it again from the files. Failed, exhausted and timed-out runs are rows, never dropped.

Identity, sealing and replay. Every object (curve, factor base, method parameters, spec, session, host environment class) is named by a hash of what it is (`ECBS1h...` for sessions, `ECBND1h...` for bounds). Each execution appends one sealed record; `ecbench verify` replays measured records and emits an audit receipt whose SHA-256 is the replay certificate. Operation counts are deterministic, so a replay must match to the last digit; a mismatch is preserved as a failed audit (§4.5).

Isolation. A run earns L0 (recorded under a host capsule), L1 (pinned, affinity read back, no foreign SMT sibling, CPU/wall ≤ 1.05), L2 (quiet: zero steal, foreign busy time $\leq 1\%$, run-queue delay $\leq 0.5\%$, ≤ 50 preemptions/s) or L3 (isolated cores, fixed governor, bare metal). Wall time is admitted only at the spec's required level (default L2) with at least five admitted pairs and an A/A interval; otherwise it is printed as *descriptive*. Operation counts need no isolation. macOS hosts earn L0; a GitHub-hosted runner earned L1 on 72/72 runs and matched a macOS session's operation counts to the last digit. *isolab* prepares lab hosts for L3 and cross-host replay.

Statistics. The comparable statistic is the mean S (the floor is an expectation). Comparisons are paired by (workload, round) with shared seeds, reported as a ratio of totals $\sum S_B / \sum S_A$, never a mean of ratios, with a two-stage cluster bootstrap that resamples workloads, then pairs. An operations A/A is exactly 1 by construction and checks determinism. A claim about scaling fits exponents over at least four sizes.

Bounds and the frontier. `ecbench` also records sealed *bounds*—constant and exponent with intervals, per phase—and a challenge protocol whose verdict says whether a candidate advances, trades, matches or regresses an incumbent and at which level (exponent, constant, primitive). The incumbent for cold one-target Koblitz operations is `rho.signed_frobenius_strong` (bound `ECBND1h05a2ea918f3c`): $1.089\times$ the floor $[0.927, 1.261]$, fitted exponent $\alpha = 0.451$ $[0.399, 0.512]$, 144 runs at six sizes (`research/ecbench_frontier_20261013/`).

Definition 4.1 (comparability)**.** A performance point is keyed by the tuple

$$D = \big(\text{curve and subgroup},\ \text{target law},\ \text{cold} \mid \text{online} \mid \text{stage},\ \text{unit and unpriced work},\ \text{set-up amortisation } q,\ \text{memory envelope},\ \text{correctness/replay/isolation class}\big).$$

Two points are ratio-comparable only when every component of D except the method matches; a method is on the frontier of a domain when no admitted point is no worse on every axis and strictly better on one.

This rules out the four collapses we saw most often: online wall as cold cost; word XORs, probes, field operations and GAE as interchangeable; a lower bound as a complete cost; an L0 wall ratio as an admitted speed-up.

4.3 The pricing band

Almost no cost in a record is complete, and the rho baselines are no exception. Rho, BSGS and the kangaroo leave canonicalisations, partition hashes and table inserts and lookups unpriced. An index-calculus arm is complete only on curves that have a pinned timing calibration (`docs/ic/calibration.json`), which converts lookups, row operations, Artin–Schreier solves, Frobenius maps and solver word XORs to GAE at measured ratios of roughly 10^{-3} to 10^{-1}. A "complete" index calculus set against a lower-bound rho is therefore lopsided, and several earlier comparisons in this repository were of that kind.

Since 15 October every cross-method comparison uses a *pricing band* (repository rule, `AGENTS.md` §5a; native implementation `ecbench band`). For a record with charged cost G, calibration-priced native cost C over n_C operations, and U operations behind its unpriced counters,

$$S(w) = \frac{G - C + w\,(U + n_C)}{\sqrt{r}}, \qquad w \in \{0,\ 0.1,\ 1\} \text{ GAE per native operation},$$

applied alike to every arm, references included. Two arms are ranked only if their order is the same at every weight; otherwise the pair is reported as unresolved. An arm that names an unpriced counter without counting it is a lower bound and cannot be ranked. The band brackets the pinned calibrations from below and above, so a ranking that survives it does not depend on the host the calibration came from.

4.4 Claim hygiene and change classes

Every study directory states a protocol before it runs (frozen inputs are committed first), keeps failures, and closes with a decision in one of the four classes of [3]. A controlled-experiment rule requires a negative control from the right null population, a positive control that proves the detector can see the effect, and the same instrument on every arm before any comparative statement. Blockers are recorded as named problems (`problems/`) with exit criteria rather than silently dropped. Measurement code is native (Rust); Python only lays out recorded values.

Generic baselines. The rho, BSGS and Pohlig–Hellman references were engineered alongside index calculus, which is why each round's distance is quoted against the strongest reference at the time. On Koblitz curves the strong signed-Frobenius rho fell from 1,196 to 679.7 instructions per step over four rounds with no established native-time gain (§16.4). On hyperelliptic Jacobians a negation fold and hash mixing each fell short of their registered gains, and full-width branch coefficients showed that the earlier 8-bit reference walk had been slow at large N (`benchmarks/20261018-hec-rho-negation/`, `benchmarks/20261019-hec-rho-branch-width/`). Generic DLP fixes cut Pohlig–Hellman steps 26.6× and BSGS time 1.3–2.0×, and the candidates run at 0.020–0.167 of PARI's `elllog` time, in shared-host wall time the validation sweep left unvalidated (`research/generic_dlp_memory_parallelism_20261017/`).

4.5 Corrections the accounting forced

Building the harness and the ledger exposed errors in earlier claims, including our own. We list them because they change how earlier numbers must be read.

1. *Results below rho that reversed against a stronger rho* (September; §10.3). A tournament candidate at 0.5347 of rho's instructions lost all five cells once rho used the same canonical form (`research/ic_triple_counted_20260923/`); an orbit-DLP ratio of 0.2403 became 1.923 against a strengthened rho and was withdrawn; a compact-orbit batch lead of 0.309 reversed against a batched-inversion rho.
2. *Online Koblitz ratios relabelled* (7–8 October, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`). Same-point online ratios of 124.85× ($n = 61$), 2,864.1× ($n = 71$), 1,189.7× ($n = 73$) and 8.2× ($n = 83$) had verified answers but no host-isolation receipt; they are *exploratory*. Only the later 64-target pinned panel at $n = 61$ is admitted (§10).
3. *ECC2K-130 extrapolation constants corrected* (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`): the structural model predicts $C = 0.25$ in probes/relation $\approx C\,r/(n^2K^2)$; the fitted C is 0.27 at $n = 61$ but 1.2–1.5 at $n = 71, 73$, an unexplained gap now stated rather than absorbed.
4. *"F4" that was Buchberger* (`research/ecbench_rootfinding_assessment_20261012/REPORT.md`): the NIST benchmark of the root-finding manuscript printed its baseline as F4 but called `groebner_f4::buchberger`; its 3,021–5,802× are ratios to Buchberger. The manuscript's claim that no implementation can use fewer than linear field operations per ideal is too broad once a reusable pair table is allowed.
5. *Unpriced word XORs* (`research/ecbench_rootfind_faster_20261014/`). Inherited F4's Boolean word XORs were counted but not charged, so its $S = 16.7$ at Koblitz $n = 13$ was a lower bound that looked competitive with rho (14.4). Pricing them gives $S = 421.4$.
6. *Cross-host counter drift* (`research/ecbench_frontier_20261013/audits/`): a replay-all audit of the root-finder session on x86_64 matched 120/168 records; all 48 inherited-F4 rows differed in solver word-XOR counts (e.g. 301,961 versus 300,531), with every scalar, trial count and charged GAE unchanged. The failed receipt is kept; the F4 stage comparison is historical until the drift is explained. A replay of the same session on current `main` (20 October) matches only 72/168, with mismatches in all four IC arms, root finding included. The cause was then isolated: an x86-only BMI2 rewrite in inherited F4 counts word XORs differently from the portable path; with it disabled the replay is 168/168 on x86 and on emulated aarch64 (`research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`). F4 counts are comparable across instruction sets only when the path is stated, which the method identity does not yet record.
7. *A soundness bug* (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md` §2.1): a degree-bounded `f4_fp::solve` could certify solutions of an inconsistent system (over $\mathbb{F}_5$, $\{x^3 + 1, xy - 1, x + y\}$ returned two "solutions"). Fixed by evaluating every candidate on the input.
8. *F6-IC charged cost* (§9): a stage win in word XORs became a 64.6–1,353× loss in charged S once its batched geometry was counted—recorded as an accounting result, not an advance. With inherited F4's own word XORs priced the $m = 13$ loss is about 2.5× (§9).
9. *Lopsided pricing* (§4.3): comparisons of a calibration-priced index calculus with a lower-bound rho. On the frontier campaign's train panel at Koblitz $n = 13$ the compact-orbit arm, earlier quoted at 1.08× rho as a lower bound, is 0.93×, 1.53× and 5.7× the strong rho at $w = 0$, 0.1 and 1 (columns 2, four train targets, `research/ic_frontier_campaign_20261015/select/SELECTION.md`): below rho with native work free, above it otherwise, so the pair is unresolved inside the band rather than a win for either side.
10. *Wall-time figures recomputed* (validation sweep). Recomputing from the committed per-target records gives 2,321× for the $\ell = 12$ binary MITM oracle (quoted 2,187×), 98.1× for its $\ell = 8$ run 3 (quoted

100.7×), and 6.08× and 6.14 s for the quartic solve at $d = 6$ (quoted 6.4× and 5.9 s); all are shared-host wall times left unvalidated, and the counted evidence behind them stands (`research/validation_sweep_20261019/B-benchmarks/VERDICT_TABLE.md`).

11. *An unmatched rho reference* (21–24 October). The Koblitz leaderboard's rule v3 priced its rho as built rather than at the canonical batched step. Re-pricing rho in the same process raised the cold distance 1.76–1.94×, to 9.9–43× (`research/ic_tool_program/rule/v4/README.md`); the ledger's rule v8, matching the reference its own way, raised it 1.30–1.58× and found that an apparent flattening of the gap's slope was mostly this artefact (`papers/ecdlp-ic-improvement-ledger/`). The same week withdrew a volcano-crater claim (IC 22–25% cheaper) whose reference should have been the cubic-automorphism rho, against which IC is 84.9× rho, and marked the plane-quartic S values unverified because their degree-3 reduction merges distinct classes. Canonical Pic^0 keys later separated those classes, and a re-run found that the earlier toy logarithms came from enumerating the subgroup (§14).

Part II

Theory

5 The summation cover: what is fixed for every curve

Every index calculus on an elliptic curve asks one question many times: does a target R decompose as a sum of $m-1$ factor-base points? Semaev's summation polynomial answers it algebraically, and every design lever of the programme acted on this question. This chapter collects three facts about it that hold for *every* curve. They closed, without an experiment, the levers that rely on curve-dependent monodromy, on fibre censuses, or on raising mean yield at matched base size. They also mark what stays free (§5.6).

The statements and proofs are those of `crypto-autoresearcher:papers/semaev-conservation-specialization/` (15 pages, anonymized draft, verification script `verify.py`, seed 20260807, which replays byte for byte), and of the internal finding `KN-FIND-bdcd60`. An integrity audit (`crypto-autoresearcher:research/sheriff/SHF-20261023-7b9227/`) upheld the mathematics within its scope. It blindly re-derived the group orders $2, 24, 192, 1920, 23040$ and checked 2.1 million admissible fibres with zero violations. It also rated as overclaimed a statement that three *independent* derivations agree: they came from one model family. **Novelty is not adjudicated** (§5.7).

5.1 Setting

Let K be a field, E/K an elliptic curve with inversion $\iota = [-1]$, and $\pi : E \to E/\langle\iota\rangle \cong \mathbb{P}^1$ the quotient (the x-coordinate in odd characteristic). The m-th summation polynomial $S_m \in K[X_1, \dots, X_m]$ is the reduced defining polynomial of the image of $\{(Q_1, \dots, Q_m) : \sum Q_i = O\}$ under π^m. Equivalently, $S_m(\xi_1, \dots, \xi_m) = 0$ if and only if there are points Q_i with $\pi(Q_i) = \xi_i$ summing to O [33]. For $m \geq 3$ it is symmetric, absolutely irreducible, and of degree 2^{m-2} in each variable. It satisfies the resultant recursion $S_m = \mathrm{Res}_X(S_{m-1}(\dots, X), S_3(X_{m-1}, X_m, X))$.

Fix $m \geq 3$, put $n = m-1$, and let $t_1, \dots, t_n$ be independent transcendentals. Write $L = K(t_1, \dots, t_n)$ for the function field of $(\mathbb{P}^1)^n$ and $M = K(E^n)$ for that of E^n. The group $\Gamma = \langle\iota\rangle^n \cong (\mathbb{Z}/2)^n$ acts on E^n coordinatewise; $\Delta \subset \Gamma$ is its diagonal. M/L is Galois with group Γ. With P_i the generic points ($\pi(P_i) = t_i$), the roots of $S_m(t_1, \dots, t_n, T)$ in T are the abscissae of the signed sums $\theta_\varepsilon = \pi(\varepsilon_1 P_1 + \dots + \varepsilon_n P_n)$, $\varepsilon \in \{\pm 1\}^n$, with $\theta_\varepsilon = \theta_{-\varepsilon}$.

5.2 The Galois group

Theorem 5.1 (Monodromy of the summation cover)**.** *For every field K, every elliptic curve E/K and every $m \geq 3$:*

1. *$S_m(t_1, \dots, t_n, T)$ is separable of degree 2^{m-2} over L, with root set $\{\theta_\varepsilon\}$;*
2. *its splitting field is $N = M^\Delta = K(E^n/\Delta)$;*
3. *N/L is Galois with group $\Gamma/\Delta \cong (\mathbb{Z}/2)^{m-2}$, acting simply transitively on the roots;*
4. *K is algebraically closed in N, so arithmetic and geometric monodromy coincide, $\mathrm{Mon}^{\mathrm{arith}} = \mathrm{Mon}^{\mathrm{geom}} \cong (\mathbb{Z}/2)^{m-2}$.*

None of the four statements depends on E, on K or on its characteristic.

Proof outline. A root-identification lemma shows that the roots are among the θ_ε. A generic-separability lemma shows that the 2^{m-2} classes $\{\pm\varepsilon\}$ give distinct values. Γ permutes the roots by $\sigma \cdot \theta_\varepsilon = \theta_{\sigma\varepsilon}$, and the kernel of this action is exactly Δ. So Γ/Δ acts faithfully and regularly on 2^{m-2} roots. The stabiliser of one root is Δ, which gives $N = M^\Delta$. Finally, E^n/Δ is geometrically integral, so K is algebraically closed in its function field. □

For a symmetric (unordered) factor base the relevant group is the wreath extension $(\mathbb{Z}/2)^{m-2} \rtimes S_{m-1}$, of order $2^{m-2}(m-1)!$ for $m \geq 4$. It is primitive if and only if $m-1$ is odd (`KN-FIND-bdcd60`). Statement (4) is the operative one: a cover whose arithmetic monodromy exceeds its geometric monodromy can split differently over different base fields, and hence on different curves. Here no such room exists.

5.3 An exact factorisation law over a finite field

Over $K = \mathbb{F}_q$, define for $a \in \mathbb{P}^1(\mathbb{F}_q)$ the *splitting invariant*:
- $c(a) = 0$ if both points above a are rational;
- $c(a) = 1$ if neither is;
- $c(a) = *$ at a branch point.

For odd q and $E : y^2 = f(x)$ this is the quadratic character of $f(a)$.

Theorem 5.2 (Factorisation law)**.** *Let $a = (a_1, \dots, a_n) \in \mathbb{F}_q^n$ lie outside the degenerate locus, which has $O_m(q^{n-1})$ points. Then $S_m(a_1, \dots, a_n, T)$ is squarefree of degree 2^{m-2}, and:*
1. if $c(a_1) = \cdots = c(a_n)$, it splits into 2^{m-2} distinct linear factors, with roots $x(\pm P_1 \pm \cdots \pm P_n)$;
2. otherwise it is a product of 2^{m-3} distinct irreducible quadratics and has no root in $\mathbb{F}_q$.
No irreducible factor of degree ≥ 3 ever occurs.

Proof. On the étale locus, the factorisation type is the cycle type of Frobenius in Γ/Δ (Dedekind's theorem for specialisations of a Galois cover [14]). The Frobenius element is $((-1)^{c(a_i)})_i$. It is trivial in Γ/Δ exactly when every $c(a_i)$ agrees. Otherwise it is a non-trivial element of order 2 of a regular group, and so has cycle type $2^{2^{m-3}}$. □

The density of totally split fibres is $2^{-(m-2)} + O(m^2/(2^{m-1}q))$. The first-order Hasse terms $\pm t/q$ cancel, so the statistic carries no first-order information about the curve.

5.4 Specialisation to the factor-base locus

Theorem 5.3 (Complete splitting on the factor-base locus)**.** *If each $a_i = x(P_i)$ with $P_i \in E(\mathbb{F}_q)$ and a is non-degenerate, then $S_m(a_1, \dots, a_n, T)$ splits into 2^{m-2} distinct linear factors over $\mathbb{F}_q$, with roots $\{x(\varepsilon_1 P_1 + \cdots + \varepsilon_n P_n)\}$.*

This is Theorem 5.2(1) with every $c(a_i) = 0$. It is the statement behind the root-finding oracle of Chapter 8: given $m - 1$ base points, the admissible last coordinates are the signed sums, computable in $O(2^{m-2})$ group operations. It is *not* an algorithm for relation search. Relation search fixes the target and asks for the base points. The theorem says nothing about how many tuples must be tried, nor about the algebraic difficulty of the system posed in that direction. Its value is negative:

Corollary 5.4 (Census obstruction)**.** *Every statistic of the form "factorisation type, cycle type or Frobenius class of the fibre of S_m" is constant on the factor-base locus, for every curve and every m. A census of generic fibres measures a quantity whose value where relation search operates is known in advance and is the same for all curves.*

The programme ran such censuses before this corollary was proved. The census-as-relation-rate probe was closed by ledger decision `DEC-20261017-435b01` (`reject_scoped`, replicated).

5.5 Conservation of decomposition yield

Proposition 5.5 (Conservation)**.** *Let G be a finite abelian group of order N and $\mathcal{D} \subseteq G \setminus \{0\}$ a set of size B. The number of size-m multisets from $\mathcal{D}$, summed over all targets, is $\binom{B+m-1}{m}$. The mean number of decompositions per target is therefore $\binom{B+m-1}{m}/N$, for every $\mathcal{D}$ of size B.*

The proof is a double count, and no novelty is claimed for it. Its consequences shaped the programme:
- at matched base size, no factor-base geometry can raise the mean yield;
- only the distribution of decompositions is free, and coverage (the fraction of targets with at least one decomposition) is at most $\min(1, \mu)$;
- additive structure that repeats sums *lowers* coverage;
- typing the summands, each from its own sub-base, is a fixed penalty: $\prod B_i \leq (B/m)^m < \binom{B+m-1}{m}$.

Gaudry and Diem's $1/m!$ decomposition probability over extension fields *is* this mean. The extension-field advantage lives in the solve, not in the yield.

5.6 Which levers are fixed and which are free

Together the results above divide the design space as follows:

- *Fixed for every curve:*
  - the monodromy group of the summation cover (Theorem 5.1);
  - the fibre statistics on the factor-base locus (Corollary 5.4);
  - the mean yield at fixed base size (Proposition 5.5).
- *Free:*
  - the coverage distribution (bounded by the mean);
  - the cost of deciding membership (the oracle, Part III);
  - the algebraic difficulty of the decomposition system when it is posed as a polynomial system (Chapter 6);
  - the linear algebra;
  - where the problem lives (transfers, Chapter 14).

Every measured improvement in Part III acts on a free item. Chapter 14 shows that transfers by isogeny conserve the counts as well.

5.7 Prior work and the limits of the novelty assessment

Absolute irreducibility of S_m is classical [19, 33]. Corollary 5.4's ingredient, that the roots are the signed sums, is the defining property of S_m. The action of 2-torsion and of sign changes on summation systems is used by Faugère, Gaudry, Huot and Renault [13] and by Galbraith and Gebregiyorgis [16]. Theorem 5.1's group may be folklore: it is the deck group of $E^n \to (E/\pm)^n$ modulo the diagonal. Theorem 5.2 is in essence a 2-descent statement. The draft could not reach ePrint during its literature search, and says so. Corollary 5.4 and the exact dichotomy of Theorem 5.2 are the parts most likely to be new. Two items must be settled before any of this is submitted:

- an expert check of the separability lemma in characteristic 2, where $E[2]$ is non-reduced;
- a full literature search.

6 Fall degrees of point-decomposition systems

Every claim that elliptic-curve index calculus can be subexponential rests on the cost of solving the point-decomposition system. In practice that means its *solving degree*: the largest degree a Gröbner-basis computation reaches. Petit and Quisquater [30] conjectured that the solving degree stays close to the *first fall degree*, which is small and easy to compute. Semaev [34] assumed it is at most 4 for his chained systems. Kosters and Yeo [26], Huang, Kosters and Yeo [22] and Kousidis and Wiemers [27] gave reasons and experiments to doubt this. All three name rigorous bounds on the *last* fall degree, which equals the solving degree for the systems that matter, as the open problem.

This chapter collects what the programme proved and certified about these degrees, in three presentations of the decomposition system:

- binary Weil descent;
- explicit membership over any field;
- prime-field Kummer towers.

The message is the same in all three. The first fall degree is a constant of the presentation, whereas the last fall degree is not. Where the last fall degree rises above 4, the chapter gives a certificate that a separate program checks without trusting the producer.

6.1 Definitions

Let $F = \{f_1, \dots, f_M\} \subset R = k[x_1, \dots, x_N]$, graded by total degree.

- *Closure.* $V_{F,d}$ is the smallest k-subspace of $R_{\le d}$ that contains $F \cap R_{\le d}$ and is closed under $g \mapsto hg$ whenever $\deg(hg) \le d$ [5, 22].
- *Last fall degree.* $d_F = \min\{d : f \in V_{F,\max(d,\deg f)}\ \forall f \in (F)\}$.
- *First fall degree.* We use the Caminata–Gorla convention [5]: $d_{\mathrm{ff}}(F)$ is the least d at which the top-form syzygies modulo the trivial (Koszul and field) syzygies are non-zero.
- *Boolean closure.* Over $\mathbb{F}_2$ with field equations it is convenient to work in $B = \mathbb{F}_2[x]/(x_i^2 + x_i)$. There $W_d(F)$ is the same closure computed on multilinear representatives, and $M_d(F)$ is the Macaulay space spanned by the products mf with $\deg m + \deg f \le d$.

Remark 6.1 (Definitions matter). Three readings of "first fall degree" are in use. They are:

- Petit–Quisquater: a combination $\sum g_i f_i \neq 0$ whose degree drops below $\max \deg(g_i f_i)$;
- Caminata–Gorla: non-trivial syzygies of the top forms;
- operational: the least c with $V_{F,c} \cap R_{\le c-1} \neq V_{F,c-1}$.

They can disagree. Two documented cases follow.

- Nagao's ePrint 2015/984 [29] defines a "fake" first fall degree d'_F computed modulo the field equations, and states $d_F \le d'_F$ as its Lemma 4. Under the equality reading of its condition (1) the lemma fails over $\mathbb{F}_2$:
  - $\{X^2Y,\ XY + X\}$ has $d'_F = 2 < d_F = 3$, and so does $\{X_1X_2 + X_3,\ X_1X_3 + X_2\}$;
  - 94 of 400 random multilinear pairs in three variables violate it.

  The paper's own applications check the unreduced degrees instance by instance, so they are unaffected (`crypto-autoresearcher:papers/nagao-lemma4-note/`, `KN-FIND-936151`; two implementations, 15 tests).
- A finite descent of S_3 over $\mathbb{F}_{2^4}$ ($n = n' = 4$, seed 32201, $x_R = 1$, basis $[9, 3, 2, 12]$) has operational first fall 3. Its closure dimensions are $\dim V_D = 1, 11, 53$ and $\dim(V_D \cap R_{\le D-1}) = 0, 1, 12$ for $D = 1, 2, 3$. Lean finite certificates check the ranks, the semantics on all 256 assignments, and an explicit witness. The literal Petit–Quisquater reading of the same system gives 2, through a field-equation identity.
  - This refutes an internal claim of a uniform degree-2 first fall, since withdrawn.
  - It contradicts no published theorem: Kosters and Yeo say "usually 2" [26].
  - Source: `crypto-autoresearcher:docs/reports/2026-10-22-semaev-first-fall-replay/`.

Every statement below names its convention.

6.2 General closure facts

The following hold for any Boolean system F, with $\tilde{F}$ its lift to R together with the field equations. They are proved in `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/theorem-dossier.md`, §1.

Proposition 6.2 (Bridges). *1. $\pi(V_{\tilde{F},d}) = W_d(F)$, where $\pi : R \to B$ is the quotient map. In particular $1 \in V_{\tilde{F},d}$ if and only if $1 \in W_d(F)$.*
2. If F has no Boolean zero, then $d_{\tilde{F}} = d_{\mathrm{ref}}(F) := \min\{d : 1 \in W_d(F)\}$. This also equals the solving degree for every degree-compatible order (Caminata–Gorla, reduced basis $\{1\}$).
3. If F has no Boolean zero, then every *F4 run on $\tilde{F}$, under any selection strategy, produces a polynomial of degree $\geq d_{\mathrm{ref}}(F)$.*
4. If F has Boolean zero set S and $\operatorname{codim} W_d(F) > |S|$, then $d_{\tilde{F}} \geq d+1$, and every F4 run reaches degree $d+1$.

Lemma 6.3 (Branching). *If both restrictions $F|_{x=0}$ and $F|_{x=1}$ satisfy $1 \in W_d$, then $1 \in W_{d+1}(F)$.*

Proposition 6.4 (Dual certificate). *Let $\pi : B_{\leq d} \to Z$ be linear and onto, with kernel U. Suppose:*
(i) $\pi(f) = 0$ for every $f \in F$ of degree $\leq d$;
(ii) $\pi(xu) = 0$ for every variable x and every $u \in U$ of degree $\leq d-1$.
Then $W_d(F) \subseteq U$, so $\operatorname{codim} W_d(F) \geq \dim Z$. If moreover $\pi(1) \neq 0$, then $1 \notin W_d(F)$. Conversely, the quotient map $B_{\leq d} \to B_{\leq d}/W_d(F)$ is such a certificate whenever $1 \notin W_d(F)$.

Proposition 6.4 is what makes the lower bounds below checkable. A certificate is a matrix. Checking it needs the curve parameters, an independent descent of the system, and three linear-algebra tests: (i), (ii) and $\pi(1) \neq 0$. Nothing about how the certificate was found has to be trusted. With Proposition 6.2(3), one accepted certificate for $1 \notin W_4(F)$ proves that *no* F4 run on that instance stays at degree 4. This is a statement about the system, not about a solver run.

6.3 Binary Weil descent of S_3

Setting.
- The field is $K = \mathbb{F}_{2^n}$ with trace Tr, and $K_0 = \ker \mathrm{Tr}$.
- The curve is $E : y^2 + xy = x^3 + a_2x^2 + a_6$ with $a_6 \neq 0$.
- The target abscissa is $z \in K^\times$.
- $V \subset K$ is an $\mathbb{F}_2$-subspace of dimension k; Semaev's diagonal is $k = \lceil n/2 \rceil$.
- $X_1 = \sum a_j v_j$ and $X_2 = \sum b_j v_j$ are linear in the $N = 2k$ Boolean variables.
- The system F is the Weil descent of $S = S_3(X_1, X_2, z)$: the coordinates $\mathrm{Tr}(\gamma_i S)$. This is Semaev's system (5) for $m = t = 2$.

Fix ϑ with $\mathrm{Tr}\,\vartheta = 1$, and an Artin–Schreier section G (the half-trace for odd n). Put

$$w = \frac{X_1X_2}{z} + G(s^2) + G(a_6/z^2), \qquad s = X_1 + X_2, \qquad \ell_0 = \mathrm{Tr}(s) + \mathrm{Tr}(a_6/z^2).$$

Theorem 6.5 (Exact reformulation). *For every n, V and z:*
1. $S = z^2(w^2 + w + \ell_0\vartheta)$;
2. $\mathrm{span}_{\mathbb{F}_2} F = \{\mathrm{Tr}(\delta w) : \delta \in K_0\} + \mathbb{F}_2\ell_0$;
3. in particular, the affine-linear form $\ell_0 = \mathrm{Tr}(S/z^2)$ always lies in the span of the equations.

Proof. (1) Squaring is a ring endomorphism, so $S = z^2(u^2 + u + s^2 + a_6/z^2)$ with $u = X_1X_2/z$, and $h = G(s^2 + a_6/z^2)$ satisfies $h^2 + h = s^2 + a_6/z^2 + \mathrm{Tr}(s^2 + a_6/z^2)\vartheta$, with $\mathrm{Tr}(s^2) = \mathrm{Tr}(s)$. (2) Put $\gamma' = \gamma z^2$. Then $\mathrm{Tr}(\gamma S) = \mathrm{Tr}((\gamma'^{1/2} + \gamma')w) + \mathrm{Tr}(\gamma'\vartheta)\ell_0$, and $\gamma' \mapsto (\gamma'^{1/2} + \gamma', \mathrm{Tr}(\gamma'\vartheta))$ is a bijection $K \to K_0 \times \mathbb{F}_2$. (3) Take $\gamma' = 1$. □

Part (3) is Kosters and Yeo's Corollary 4.11 [26], seen through a different coordinate. The reformulation makes the falls explicit.

Corollary 6.6 (First fall degree 2, for every n). *Suppose every descended equation has degree exactly 2; this held on every instance computed. Then:*
- *$d_{\mathrm{ff}}(F) = 2$ in the Caminata–Gorla sense, for every $n \geq 3$ and every V;*
- *if moreover $\ell_0 \neq 0$, the Petit–Quisquater first fall degree is also 2.*

The same holds for Semaev's chained system (5) for every $t \geq 2$*: the last equation* $S_3(y, x_t, R_X)$ *always contributes the linear form* $\mathrm{Tr}(Y + X_t) + \mathrm{Tr}(a_6/R_X^2)$.

Semaev states the value 4 for the first fall degree of his systems [34, §4.5]. His argument concerns only the cubic equations of the chain. Under the definition he quotes, the trace relation makes the value 2.

Lemma 6.7 (Explicit degree-3 falls)**.** *Let* $V^\perp$ *be the trace-orthogonal of* V*, and* $\omega = \mathrm{Tr}(\mu w)$ *for any* μ *with* $\mathrm{Tr}\,\mu = 1$.

- *For* $\lambda \in V^\perp$ *and* $i \in \{1, 2\}$, $\mathrm{Tr}(\lambda X_i w)$ *lies in* $M_3(F)$ *and has degree* ≤ 2.
- $\mathrm{Tr}(sw) + \mathrm{Tr}(a_6/z^2)\,\omega$ *lies in* $W_3(F)$ *and has degree* ≤ 2.

These are $2(n-k)+1$ *explicit degree-3 falls, independent of* n *in form. On random subspaces they span the whole degree-*≤ 2 *part of* W_3*, of dimension exactly* $3n$*; this was checked on six instances with* $n \in \{23, 25, 27\}$*. Polynomial subspaces* $\langle 1, t, \ldots, t^{k-1}\rangle$ *have* $2n-1$ *further falls, which is why they resist longer.*

Theorem 6.8 (Upper bound)**.** *If* F *has no Boolean zero, then* $1 \in M_{k+2}(F)$*. So* $d_{\tilde{F}} \leq k+2$*, which is* $\lceil n/2 \rceil + 2$ *on the diagonal. For Semaev's chained system (5) with* t *summands the bound is* $d_{\tilde{F}} \leq (t-1)k + t$*: each auxiliary variable adds 1, not* n.

Proof for $t = 2$. Substitute $c_j = a_j + b_j$. Then $S = X_1^4 + X_1^2 s^2 + z^2 s^2 + z X_1^2 + z X_1 s + a_6$ is at most linear in a and in c, so each $f_i = \sum_j a_j L_{ij}(c) + M_i(c)$. For each $\sigma \in \mathbb{F}_2^k$ the affine system $f_i(a, \sigma)$ in a is inconsistent, so some combination $\sum_i \lambda_i^\sigma f_i(a, \sigma) = 1$. With the indicators $e_\sigma = \prod_j (c_j + \sigma_j + 1)$,

$$1 = \sum_\sigma \sum_i \lambda_i^\sigma e_\sigma f_i \in M_{k+2}(F). \qquad \square$$

Proposition 6.9 (Lower bound $d_{\tilde{F}} \geq 3$)**.** *Under two explicit genericity conditions on* V *and* z*, which a script checks and which held on every random-subspace instance, together with* $\ell_0 \neq 1$*:*

- $W_2(F) = \mathrm{span}\, F + \ell_0 B_{\leq 1}$*, and its degree-*$\leq 1$ *part is* $\mathbb{F}_2 \ell_0$*;*
- *so an unsatisfiable* F *has* $d_{\tilde{F}} \geq 3 = d_{\mathrm{ff}} + 1$.

The bound is attained at $n = 17, 19$.

Computer-verified instances.

- *Implementations.*
  - Implementation A: a C closure over M4RI, `code/lfdclose2.c`.
  - Implementation B: independent Python sharing no code with A; `verify/b_sweep.py` and `verify/indep_verify.py`.
  - Certificate checker: `verify/cert_check.py`, which rebuilds F from the curve parameters by its own Möbius interpolation of S_3 values.
- *Instances.*
  - Every instance on the diagonal $k = \lceil n/2 \rceil$, other than the two satisfiable ones, was shown unsatisfiable by exhaustive count, and the count was repeated by B.
  - B reproduced all 61 instances with $n \leq 29$: the same solution count, the same dimensions of $W_{d-1} \cap B_{\leq e}$ for every e, and $1 \in W_d$.

Figure 6.1 shows the result: as n grows, the first fall degree stays at 2 while the last fall degree climbs.

Theorem 6.10 (Exact last fall degrees; computer-verified on instances)**.** *On the diagonal* $k = \lceil n/2 \rceil$*:*

- $d_{\tilde{F}} = 3$ *on all six random-subspace instances at* $n = 11, 13$ *and on all ten polynomial-subspace instances at* $n = 17, 19$*;*
- $d_{\tilde{F}} = 4$ *on all 29 random-subspace instances with* $15 \leq n \leq 33$*, and on all 25 polynomial-subspace instances with* $23 \leq n \leq 35$ *or* $n = 41, 43$*;*
- $d_{\tilde{F}} = 4$ *on Semaev's own cell (*$n = 40$*,* $k = 20$*, uniform* z*), matching his reported* $d_{F4} = 4$*;*
- $d_{\tilde{F}} = 5$ *exactly on instance R35a (*$n = 35$*,* $k = 18$*, random* V*):* $1 \notin W_4$*, and both restrictions* $a_1 = 0, 1$ *are refuted at* W_4 *(Lemma 6.3).*

Theorem 6.11 (Certified $d_{\tilde{F}} \geq 5$)**.** *For each instance below, a certificate (Proposition 6.4) for* $1 \notin W_4(F)$ *was accepted by the independent checker. Hence* $d_{\tilde{F}} \geq 5$*, and every F4 run reaches degree* ≥ 5.

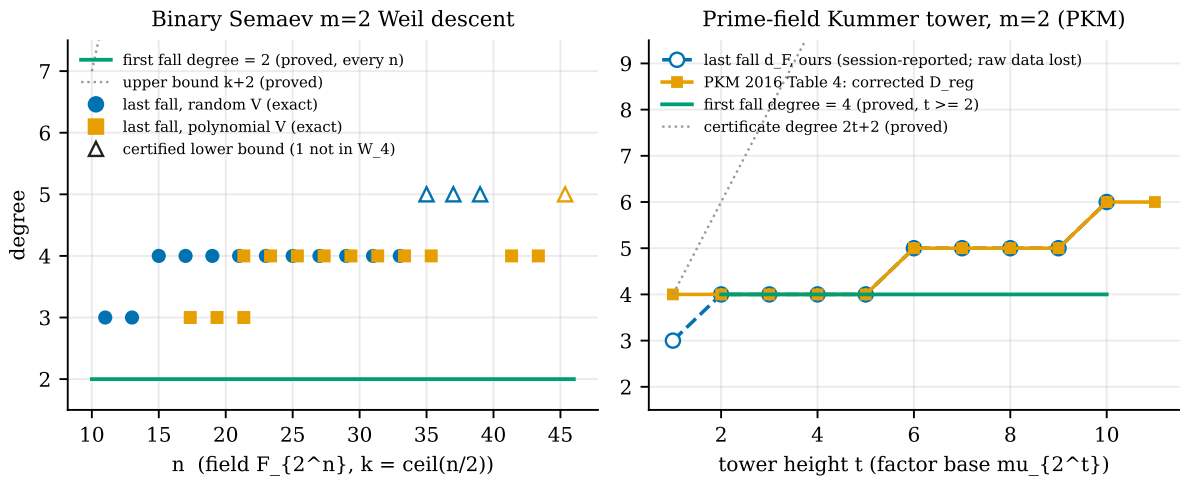


Figure 6.1: *Left:* binary Semaev $m = 2$ descent. The first fall degree is 2 (Corollary 6.6, proved). Filled marks are exact last fall degrees per unsatisfiable instance. Open triangles are certified lower bounds $d_{\bar{F}} \geq 5$ (Theorem 6.11). The dotted line is the proved upper bound $k + 2$. *Right:* prime-field Kummer tower, $m = 2$ (§6.5). The session-reported last fall degree lies on the corrected degree of regularity of PKM 2016, Table 4. The first fall degree is 4 (proved). Values and sources: `figures/data-new.json`.

instance	n	k	*subspace*	*certified* codim $W_4 \geq$
T33a, T33b	*33*	*17*	*random* $\subset K_0$	*4,888*
R35a, R35b, U35a, U35b	*35*	*18*	*random (U: uniform z)*	*5,287*
Q35a, Q35b (satisfiable)	*35*	*18*	*random*	*5,287*
R37a, R37b	*37*	*19*	*random*	*10,066*
P45a	*45*	*23*	*Semaev's default* $\langle 1, t, \ldots, t^{22} \rangle$	*12,742*
C19a, C19b (chain $m = t = 3$*)*	*19*	*7*	*random*	*9,253, 6,817*

Two further instances, R39a and R39b ($n = 39$)*, and C21a and C21b (the chain at* $n = 21$*) have* $1 \notin W_4$ *by implementation A alone.*

For Q35a and Q35b, which have 6 and 2 Boolean solutions, the bound follows from Proposition 6.2(4), since $5{,}287 \gg |S|$. The P45a closure stalls after three rounds at $\dim W_4 \cap B_{\leq d} = (0, 1, 224, 8950, 166705)$, out of 179,447 monomials. Its quadratic part, $224 = 5n - 1$, is exactly the polynomial-subspace count of Lemma 6.7: at degree 4 the closure finds no new falls.

Corollary 6.12 (Semaev's Assumption 1 fails as stated)**.** *Semaev's Assumption 1 [34, §4.5] asserts* $d_{F4} \leq 4$ *for Boolean systems equivalent to his system (5), for* $q = 2^n$, $k = \lceil n/m \rceil$*, any subspace of dimension* k *and any* $2 \leq t \leq m$*. Theorem 6.11 contradicts it in three cells:*

- $(n, m, t, k) = (35, 2, 2, 18)$*, with random subspaces;*
- $(45, 2, 2, 23)$*, with his own default subspace;*
- $(19, 3, 3, 7)$ *and* $(21, 3, 3, 7)$*, inside the parameter range of his tables, where his polynomial subspace gives 4 and random subspaces do not.*

What is new, and what is not. That the degree of regularity reaches 5 has been *observed* before:

- Kosters and Yeo saw Magma step degrees of ≥ 5 at $n = 40$ and $n = 45$ [26];
- Kousidis and Wiemers saw F4 at step degree 5 for $n = 37$–48, $m = 2$, and wrote that a lower bound on the degree of regularity "can potentially answer an open question" [27].

Those are properties of solver runs. Theorem 6.11 is a property of the systems, carried by a certificate any reader can check. With Corollary 6.6 it makes the separation between first and last fall degree rigorous on named instances. What remains open is the same as before: a proof that $d_{\bar{F}} \to \infty$. Semaev's asymptotics survive if the regularity degree is $o(\sqrt{n \ln n})$, so only a growing lower bound bears on the subexponential claim.

Re-verification. For this treatise, the T33a certificate was regenerated from the committed producer (M4RI 20200125, 4 min 40 s on 4 vCPUs) and checked with `cert_check.py`:

- the certificate is byte-identical to the recorded SHA-256 `cf9dcae8...22cf`;
- the checker reports "1 is not in $W_4(F)$, and codim $W_4(F) \geq 4888$" (38.8 s);
- the closure dimensions $(0, 0, 64, 2968, 48068)$ match the record.

Source: `research/paper_opportunities_20261029/REPORT.md`, §6.

6.4 Explicit membership over any field: the guarded chain

A second presentation encodes factor-base membership directly. Let S be a set of B abscissae of rational points, and $g(T) = \prod_{s \in S}(T - s)$. The two-$S_3$ chain with an inverse guard is

$$F_S = \{g(x),\ g(y),\ g(z),\ S_3(x, y, u),\ S_3(u, z, r),\ v(x - y) - 1\},$$

with $r = x(R)$. Degrees are ordinary total degrees with explicit inputs. The first fall degree is the homogeneous-top syzygy degree modulo Koszul syzygies. Source: `research/guarded_fall_degree_20261028/`, which holds written proofs, independent audits, a native exact replay and partial Lean.

Theorem 6.13 (Guarded chain)**.** *For unsatisfiable instances:*

1. *Short Weierstrass curves, $p > 3$, $B = 3$: $(d_{\mathrm{ff}}, d_F) = (4, 5)$, exactly, for every p.*
2. *Same presentation, any $B \geq 3$: $d_{\mathrm{ff}} = B + 1$ and $\max(5, B + 1) \leq d_F \leq 3B - 4$.*
3. *Ordinary binary curves $Y^2 + XY = X^3 + a_2X^2 + a_6$, $B = 3$:*
   - *$d_F = 4$ if and only if $g = T^3 + c$, which is possible only for even n;*
   - *$d_F = 5$ otherwise.*

   So for odd n, which covers every SEC 2 binary size, every unsatisfiable instance has $(4, 5)$.
4. *Binary curves, $B \geq 4$: $B + 1 \leq d_F \leq 3B - 4$.*

Proof sketch of (1). Upper bound. The divided difference $Q = (g(x) - g(y))/(x - y)$ enters V_4 through the guard. The pair is then replaced by the missing root $w = s_1 - x - y$. The two quadratics in u are normalised and their resultant extracted in stages inside V_5. Finally the unit $T(w, z)$ is inverted in $K[w, z]/(g(w), g(z)) \cong K^9$.

Lower bound. The space $W = (I_0 \cap R_{\leq 4}) + Kf_1 + Kf_2$ is closed under the degree-4 operations and excludes 1. This uses characteristic > 3, through $-3 \neq 0$. □

Exact degree-5 refutation DAGs were replayed in native arithmetic on public fixtures:

- $p = 101$ and $p = 5$;
- secp256k1, P-256, P-384 and P-521, with $S = \{x(G), x(2G), x(4G)\}$ and $R = 16G$; the DAGs have 413–415 nodes, and OpenSSL rechecks the points.

Twenty polynomial identities used in the proofs are checked in Lean (`FallDegreeIdentities.lean`, `PrimeBitLocal.lean`; axioms `propext` and `Quot.sound` only). The degree-4 lower bound is now machine-checked: `GuardedModel.no_degree_four_refutation` proves that 1 is not derivable at degree four from the six generators, over every field with $6 \neq 0$ and three distinct base abscissae, and `lastFall_lower` states the last-fall consequence under an explicit unit-ideal hypothesis (`research/guarded_fall_degree_20261028/formal/GuardedModel.lean`; six modules, fresh `leanchecker` replays and a transitive axiom audit, merged in #2814). The degree-5 upper construction, the bridge from rational unsatisfiability to the unit ideal, and the first-fall syzygy statement are *not* formalised, so the exact pair $(4, 5)$ is not yet a Lean theorem. A `lastFallDegree` definition and Caminata–Gorla's Theorem 2.8 in `crypto-autoresearcher:formal/CryptoResearch/FallDegree/LastFall.lean` are the bridge a complete formalisation would build on.

Two Boolean upper bounds accompany the guarded chain:

- $d_F \leq 3\ell + 1$ for the fully Boolean binary encoding, with ℓ bits per base coordinate;
- $d_F \leq 3\ell + 2$ for a prime-field bit-restricted lookup encoding.

Both are instances of the standard brute-force refutation in Polynomial Calculus [6]: fix the base bits, and refute the affine residue. For the binary chain at $t = 3$ they are weaker than Theorem 6.8's $2k + 3$.

6.5 Prime-field Kummer towers

Petit, Kosters and Messeng [31] take the factor base $V = \mu_{2^t}$ over $\mathbb{F}_p$ (with $2^t \mid p - 1$). They write membership as a tower $y_0 = x/g$, $y_{j+1} = y_j^2$, $y_{t-1}^2 = 1$, and argue the complexity from the first fall degree. Source for this section: `research/prime_fall_degree_20261027/DERIVED.md`, written proofs D1–D8.

Theorem 6.14 (Kummer towers, $m = 2$)**.** *Let $R = \mathbb{F}_p[y]/T$ be the tower quotient. Items (3)–(5) concern* inconsistent *systems, those whose reduced S_3 is a unit of R; a satisfiable target has no inverse in R, and the refutation bounds do not apply to it.*

1. *T is a Gröbner basis, and $R \cong \mathbb{F}_p^{V^m}$. So every ideal over R is radical, and an $m = 2$ system is inconsistent if and only if the reduced S_3 is a unit of R.*
2. *$R = \mathbb{F}_p[z]/(z^K - 1)$ with $K = 2^t$. The degree of a monomial z^k is* popcount(k)*, and degree falls are exactly binary carries.*
3. *If $t \geq 2$, $g \neq 0$, $3x_R^2 + a \neq 0$ and the system is inconsistent, then $d_{\mathrm{ff}} = 4$ (operational convention), for every t. The degree-2 part of f is a bilinear form with $\det M = -2g^6(3x_R^2 + a) \neq 0$, so nothing falls at 3; the monomial $q = y_{10}y_{11}$ gives a fall at 4.*
4. *For an inconsistent system, the no-fall certificate has degree $d_F \leq 2 + \mathrm{popdeg}(f^{-1}) \leq 2t + 2$.*
5. *For an inconsistent system in the naive presentation $\{S_3,\ x_i^K - g^K\}$ of the* same *set, $d_F \geq 2^t$. So d_F is not an invariant of the ideal; it depends on the presentation.*
6. *A single summand has $d_F = 2$ for every t. Growth at $m = 2$ must therefore come from the genus-one curve $f(z_1, z_2) = 0$; no specialisation to finitely many points can show it.*

Measured, and prior. The study's session reported a median last fall degree of 3,4,4,4,4,5,5,5,5,6 for $t = 1..10$ at $p = 2{,}013{,}265{,}921$. The raw records were never committed (the study's `RESTART.md`), so these values are session-reported, not re-derivable. They coincide with the corrected degree of regularity in PKM's own Table 4, which reads 4,4,4,4,4,5,5,5,5,6,6 for $n_1 = 1..11$ (Figure 6.1, right). So the *growth* is prior art. What Theorem 6.14 adds is the proved constant first fall degree beside it, and the presentation dependence. A prediction, $d_F = 4 + \lfloor (t-2)/4 \rfloor$, was registered before any $t \geq 11$ row and is untested.

6.6 Measured solving degrees in other presentations

The binary programme measured refutation and closure degrees on many presentations; the full record is in [3, §6].

- *Last fall degree is flat in n at fixed (m, ℓ).* With ℓ the dimension of the base subspace, $d_F = 4$ for $m = 3$, $\ell = 6$ at $n = 16$–24; for $m = 3$, $\ell = 5$ at $n = 10$–15; for $m = 4$, $\ell = 3$ at $n = 9$–14; and for $m = 5$, $\ell = 3$ at $n = 11$–15.
- *It rises with ℓ.* At $m = 3$, $\ell = 7$ every draw is refuted only at degree ≥ 5.
- *Plain refutation tracks the semi-regular degree.* The plain Macaulay refutation degree stays within one of the semi-regular prediction in all 14 exact cells.
- *Closure algorithms refute one or two degrees earlier.* A MutantXL-style closure refutes all 92 unsatisfiable draws at degree 3–4, where XL needs 5–7.
- *Guessing variables restores degree 4.* Fixing one or two variables restores degree 4 at $(m, \ell) = (3, 7)$, at 1/116–1/260 of the projected degree-5 cost.

Every observed growth is growth in the size of the factor base: in ℓ, k, t, B, or the digit count s of a prime digit presentation (where $d_{\mathrm{ff}} = d_F = 4 + \lfloor s/2 \rfloor$ at $m = 2$, reproduced by a curve-free null). None was observed in n at a fixed base.

6.7 What this settles and what it does not

- *Settled.* In the three presentations the programme studied, the first fall degree is a proved constant of the presentation: 2 for binary Weil descent, $B + 1$ for explicit membership, 4 for Kummer towers. The last fall degree is not bounded by it.
  - This is proved for the guarded chain at $B = 3$ (exactly 5 against 4).
  - It is certified on named binary instances: ≥ 5 against 2, at $n = 33$–45 and in the $m = t = 3$ chain.
  - It is measured for Kummer towers (6 against 4 at $t = 10$).

  Any complexity argument that substitutes the first fall degree for the solving degree is unsupported for these systems.
- *Not settled.* No presentation has a proved last-fall lower bound that grows with n. That is the open problem of HKY [22] and Kousidis–Wiemers [27], and the only result that would bear on the subexponential claims. The upper bounds $k + 2$ and $(t-1)k + t$ are linear in n, far above every observed value.
- *Missing before submission.* The certificates are committed only as hashes; they regenerate (above) but need public hosting. The sample is small: two instances per n at $n \geq 33$, one polynomial instance

at $n \geq 41$. The written proofs need expert review. The Kummer-tower data must be regenerated (`research/paper_opportunities_20261029/`).

7 Obstructions: what moving the problem cannot buy

The programme kept 148 leads against ECDLP in a lead atlas (`crypto-autoresearcher:knowledge/leads/ecdlp/`, reviewed 2026-10-24). Of these:

- 73 are closed with a named obstruction;
- 51 are open with a decided next step;
- 16 are blocked;
- 8 are unexamined.

Most obstructions are textbook results applied to the case at hand: the Shoup–Nechaev generic bound, the Corrigan-Gibbs–Kogan preprocessing bound $ST^2 \gtrsim N$ [7], isogeny invariance of the group order, and the Balasubramanian–Koblitz embedding-degree bound. This chapter records the few that the programme *proved* itself and that constrain index calculus specifically. Each states a lever that cannot work, and why.

7.1 Transport conservation

A homomorphic transfer $\varphi : E \to E'$ with rational kernel κ changes the number of ordered m-decompositions of a target by exactly $|\kappa|^{m-1}$ (Lemma 14.1 in Chapter 14). So an isogeny buys only three things:

- the kernel symmetry, which is trivial on prime-order curves;
- the factor-base geometry available on the destination curve;
- the cost of arithmetic there.

Chains of isogenies compose to one homomorphism. A loop is an endomorphism acting as a known scalar. Multi-curve factor bases glued by isogenies are single-curve bases in disguise. Measured across sizes at fixed cofactor, fibre-aware index calculus loses ground to rho as $\#E^{1/6}$; the full measurements are in Chapter 14.

7.2 The summation cover is the same everywhere

Chapter 5 proved:

- the monodromy of the summation cover is $(\mathbb{Z}/2)^{m-2}$ for every curve;
- its fibres split completely on the factor-base locus;
- mean yield at fixed base size is a double count.

Together these close three families of leads: curve selection by anomalous monodromy, fibre censuses as a relation-rate probe, and factor-base geometries claimed to raise yield.

7.3 Oracle-augmented index calculus stays generic

Several leads proposed giving index calculus an extra oracle on the group:

- a jet or p-adic lift;
- an endomorphism;
- an incidence or halving test;
- an elliptic-net evaluation.

The programme proved, at derivation tier, that each of these is simulable in the generic group model, so it cannot beat $\sqrt{r}$ (`KN-FIND-002`, `KN-FIND-b7e091`, `KN-FIND-194294`; halving is equivalent to an x-coordinate oracle).

The minimal non-simulable order-based identifier is a threshold oracle C_t (`KN-FIND-982fdf`). Table-based decomposition then obeys the preprocessing bound $ST^2 \gtrsim N$. Balanced generic index calculus costs about $r^{2/3}/n^{4/3}$, consistent with the measured growth of the Koblitz gap to rho as $r^{0.165\,[0.117,0.213]}$ (`LEAD-ICB-generic-table-oracle`). These are known instances of generic-group reasoning [7]; they are recorded because each retired a concrete proposal.

7.4 Lifting sections does not produce relations

Silverman's xedni calculus lifts points to a curve over a global field, in the hope that rank forces relations. On a frozen family of rational elliptic surfaces (with $\deg b = 6$ and monic quadratic x), the programme derived the exact lift probability

$$P_{\text{lift}}(p) = \frac{1 - 2/p + p^{-3}}{2p^3},$$

so the exponent is $\alpha = 3$, far above the $\alpha < 1/2$ that a relation source would need. This was checked against 596,268,874 enumerated slots with no disagreement, and a parameter count reproduces $\alpha = 3$ (`LEAD-ICP-xedni-function-field-sections`, `EV-XEDN-001`). The scope is one family ($\deg b \leq 10$, $p \leq 31$). A dimension-count theorem over all section shapes would be needed to close the route in general.

7.5 Twist points carry no relations

A curve and its quadratic twist share x-coordinates, and therefore share summation polynomials. Could twist points supply relations for the curve? They cannot. Frobenius acts as $+1$ on $E(\mathbb{F}_q)$ and as -1 on the embedded twist, so the twist part B of any sum satisfies $2B = 0$.

The study `research/twist_obstruction_20261026/` checked this exactly on toy binary curves in the ECC2K-130 model (odd n, twist $a \mapsto a + 1$):

- 0 of 356,530,036 mixed sums are rational;
- 0 violations on 4,954 decomposition targets at $m = 2, 3$;
- the off-parity solution sets equal the predicted ones on 1,154 of 1,154 targets;
- positive controls found all 18,042 and 95 planted solutions.

A heuristic that ignored the obstruction predicted that 74–89% of solutions would contain a twist-lifting coordinate; 0% did at $m = 2$.

A follow-up asked whether a factor-base subspace can *avoid* the twist altogether (`research/lift_dense_factor_base_20261027/`; 11 Lean theorems, standard axioms only).

- *Composite degree $N = dn$, $n \geq 3$: yes.* On suitable $\mathbb{F}_{2^d}$-lines the lift condition collapses to the constant $\mathrm{Tr}(a)$. When $\mathrm{Tr}(a) = 0$ every abscissa lifts, giving 2^n relations per Gaudry–Diem trial. This applies to c2pnb208w1 and c2pnb368w1, but the modelled gain is +0.50 bits, a model output, not a measurement.
- *Prime degree (ECC2K-130): this construction does not exist.* A prime-degree field has no proper intermediate subfield, so there are no $\mathbb{F}_{2^d}$-lines to use. This does not rule out other twist-avoiding subspaces. At the small prime degrees tested, all-lift subspaces of dimension 4–5 exist, though fewer than for density-paired random sets (exact all-lift 4-subspace counts 0.921 ± 0.018 of the null at $N = 11$, permutation $p = 2 \cdot 10^{-5}$). This is toy evidence. Whether useful twist-avoiding subspaces exist at cryptographic prime degree is unresolved.

7.6 What the obstructions leave open

None of the obstructions above concerns the cost of *solving* the decomposition system, nor the linear algebra. Chapter 6 shows that the solving cost is genuinely open. The algorithms of Part III attack the remaining free terms.

Part III

Algorithms and measurements

8 Root-finding decomposition

8.1 Origin: a suggestion in an audited manuscript

The textbook decomposition oracle builds Semaev's polynomial system and solves it with a Gröbner-basis engine. On 12 October a 13-page manuscript, *Point Decomposition by Root Finding*, was supplied for assessment. We froze it (`research/ecbench_rootfinding_assessment_20261012/source-paper.pdf`, SHA-256 `743d3d3e...7c01`) and audited it against its source branch before measuring anything. Its central suggestion is simple: for two summands, fixing one factor-base point turns S_3 into a quadratic in the remaining variable whose roots are already known.

Proposition 8.1 (roots of S_3)**.** *Let $E : y^2 = x^3 + ax + b$ over a field K of characteristic $\neq 2, 3$, and let $R, F \in E(K)$ be affine with $x_R \neq x_F$. Then*

$$S_3(x_R, x_F, X) = (x_R - x_F)^2 \, (X - x_+)(X - x_-), \qquad x_\pm = x(R \pm F),$$

so $S_3(x_R, x_F, X) = 0$ iff $X \in \{x(R+F), x(R-F)\}$.

Proof sketch. With $\lambda_\pm = (\pm y_F - y_R)/(x_F - x_R)$ the chord slopes, $x_\pm = \lambda_\pm^2 - x_R - x_F$. Expanding $(X - x_+)(X - x_-)$ and eliminating y_R^2, y_F^2 with the curve equation gives exactly Semaev's $S_3/(x_R - x_F)^2$. The identity is machine-checked over an arbitrary field in Lean 4 with Mathlib (`papers/rootfinding-decomposition/lean/S3Roots.lean`, theorems `SemaevS3.S3_factor` and `SemaevS3.S3_root_iff`; no `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound`). □

Hence the two-summand system has a parametrised variety: $R = \pm F_i \pm F_j$ iff $x(R \mp F_i) \in x(\mathcal{F})$. No polynomial system is solved; the oracle is one subtraction and one table probe per base point.

Prior art. Neither the identity nor the oracle is new. The identity is a direct consequence of how S_3 is defined: it is the polynomial whose roots are the abscissae of $P_1 \pm P_2$. The oracle mechanism—sum $m-1$ factor-base points in the group and test whether the result lies in the base—is the second algorithm of McGuire and Mueller [28], who also evaluate S_m by iterated resultants instead of solving it, building on the instance-derived bases of Amadori, Pintore and Sala [1]. What this section adds is engineering and measurement: a machine-checked proof, a fail-closed verification protocol, binary pair-sum tables for up to four summands, GPU, Kummer-tower and dense-index variants, and complete-cost measurements in one harness against rho.

What the audit kept and corrected. The identity is exact under the stated generic affine condition, and a returned witness can be re-added in the group for soundness. Completeness of the exhaustive enumeration holds only with an explicit sign and factor-base convention (the binary table stores full point keys over a base containing both signs; the manuscript's schematic x-only table needed that convention stated). The manuscript's NIST baseline was Buchberger, not F4 (§4.5), and its statement that no implementation can use fewer than linear field operations per ideal fails for repeated queries against a reusable pair table. Larger-summand enumeration has a growing exponent in $|\mathcal{F}|$, and the manuscript did not benchmark it end to end. All of this is recorded in `research/ecbench_rootfinding_assessment_20261012/REPORT.md`.

8.2 The algorithms

Prime fields (Algorithm 1, `src/cryptanalysis/prime_fast.rs`): $\Theta(N)$ field operations and $2N$ lookups per target. *Binary fields* (`src/cryptanalysis/koblitz_index_calculus.rs`, `DecompositionStrategy::PairTable`, made the default): a precomputed table of pair sums $F_i + F_j$ turns an m-decomposition into 1, N or N^2 lookups for $m = 2, 3, 4$—the meet-in-the-middle half of a birthday search, exact and complete like exhaustive search, at the price of $\Theta(N^2)$ set-up and memory. Both are verified fail closed: every decomposition is re-added in the group, S_3 is checked to vanish, Gröbner-variety membership and agreement on refutation are checked against F4, and planted positive controls must be found.

Algorithm 1 Prime two-summand oracle by direct subtraction

Require: target R, base $\mathcal{F} = \{F_1, \ldots, F_N\}$, hash set H of abscissae (or full points) of $\mathcal{F}$

1: $d_i \leftarrow x_R - x_{F_i}$ for all i; invert all d_i with one field inversion and $3(N-1)$ multiplications (Montgomery's trick)
2: **for** $i = 1..N$ **do**
3: **for** $\varepsilon \in \{+1, -1\}$ **do**
4: $\lambda \leftarrow (y_R + \varepsilon y_{F_i})\, d_i^{-1}$; $x_3 \leftarrow \lambda^2 - x_R - x_{F_i}$ ▷ $x(R - \varepsilon F_i)$
5: **if** $x_3 \in H$ **then** recover the sign of F_j from y_3; emit $R = \varepsilon F_i + \tau F_j$
6: **end if**
7: **end for**
8: **end for**

Table 8.1: Root-finding generalisations, 12–14 October. Every row is ENGINEERING: no slope fell below its generic floor.

Step	What it adds; artefact
Prime subtraction, binary pair table	Prop. 8.1; batched inversion; $m \leq 4$ tables. `research/semaev_root_vs_groebner_20261012/`
NIST P-192…P-521	standard parameters, 6- and 64-point bases; dedicated P-256 field 11–12× over generic big integers. `research/semaev_rootfind_nist_20261012/`
End to end vs msolve	complete DLPs with only the oracle swapped. `research/semaev_rootfind_e2e_20261012/`
PDP oracle family	`rf-mitm`, `mitm3`, `mitm4` (2-sum table); `rf-pairs`, `pairs3` (pairs then solve by the group law); `subtract2`; binary rungs $\ell = 8..14$, prime Kummer towers $m = 2, 3, 4$. `research/pdp_root_finder_20261012/`
CUDA kernel	one thread per (target, slice); per-thread Montgomery inversion; SoA base; shared-memory hash for $\lvert\mathcal{F}\rvert \leq 2048$; 64-bit and P-256 fields. Validated in host emulation only (no GPU available). `RESEARCH_GPU_ROOTFIND_DECOMPOSITION.md`
ecbench engines	`pdp3-koblitz:m=2|3,engine=rootfind`, `rootfind-subtract`, later `rootfind-window` and `rootfind-negfold`. `research/ecbench_rootfind_engine_20261012/` and successors
Kummer-tower base	PKM's μ_{2^t} base with the root-finding oracles, 20–28-bit primes. `research/pkm_tower_ecbench_20261013/`
Dense root index, $n = 83$	packed five-byte S_3 root table, prefix widths K1271–K1798, 8 GiB of slots. `research/n83_dense_root_20261013/`
Prime tuning	pair-table size, arity, batched table inversions, synchronised lanes. `research/ecbench_rootfinding_\{tuning,arity,batchtable,lanes\}_2026101*/`

8.3 Generalising the observation

Over three days the identity was carried through every layer of the pipeline. Table 8.1 lists the steps; each is its own study with a protocol registered before it ran.

8.4 Results

Oracle stage. Per decomposition, root finding is 383× ($\mathbb{F}_{2^9}$, $\lvert\mathcal{F}\rvert = 55$, 254 targets) and 1,535× ($\mathbb{F}_{2^{13}}$, $\lvert\mathcal{F}\rvert = 4{,}005$, 256 targets) faster than matrix F4; 3.5–$4.7 \cdot 10^4$× faster than `groebner_f4` on 16–24-bit prime curves ($\lvert\mathcal{F}\rvert = 6$, 40 targets each); and 3,021–5,802× faster than Buchberger on the decomposition ideal of the five NIST curves (Figure 8.1). Every binary verdict agreed with F4; every prime solution was in F4's variety and every prime refutation was F4's unit ideal. In the PDP study, `rf-mitm` refutes a binary target in 3.1 μs at $\ell = 8$ and 482 μs at $\ell = 14$, against the PS-0 baseline's 9.3 ms and 55.5 s (3,007–115,160×), costs $5\lvert\mathcal{F}\rvert - 3$ multiplications and $2\lvert\mathcal{F}\rvert$ lookups a target, and sits on the pair-table floor at 1.000 ± 0.027 bits per rung; at prime $m = 3$, $N = 18$ it uses 197 multiplications where F4 uses $8.99 \cdot 10^{13}$ multiply-adds; 707/707 brute-force agreements (`research/pdp_root_finder_20261012/`).

Complete solves. End to end, the gain is two orders of magnitude smaller and not one-sided: 50–105× against msolve on 16–24-bit prime curves; 122× on the one binary instance ($n = 23$) large enough

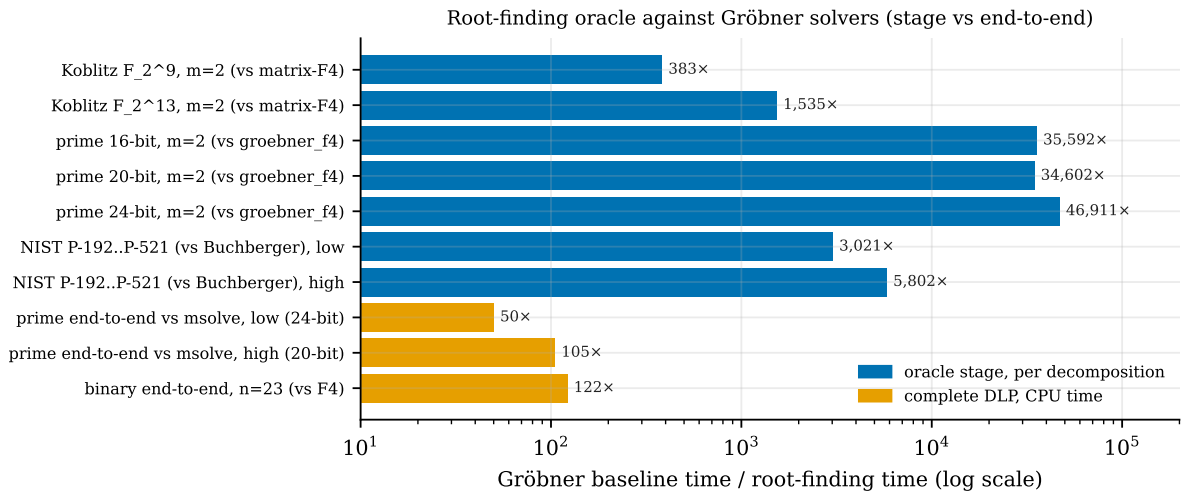


Figure 8.1: Root finding against Gröbner solvers on identical targets. Blue: one decomposition (oracle stage only). Orange: complete discrete-log computations with the oracle swapped and everything else fixed (prime: CPU time against msolve, 20/24-bit totals composed from sampled per-query cost; binary: one cold solve including table build). Sources: `RESEARCH_SEMAEV_ROOTFIND_VS_GROEBNER.md`, `papers/rootfinding-decomposition/README.md`.

to amortise the pair table; and F4 *faster* at $n = 5, 9, 13$, where a solve needs too few decompositions to pay for the table. Inside ecbench, with every group operation charged, the root-finding IC beats inherited F4 in solve wall by about 350× at $n = 13$ (admitted) and about 10^4× at $n = 23$ (descriptive), yet stays 2.9–25,116× the strong rho in S on binary curves and 29–141× rho on prime curves, with the gap widening (`research/pdp_root_finder_ecbench_20261013/`).

The Koblitz loops. Figure 8.2 follows the best complete-count root-finding arm at Koblitz E_0, $n = 13$ through four registered improvement loops (8 public targets × 3 rounds each, L0 operation counts). The first engine had $S = 38.82$ ($m = 3$) and 28.38 ($m = 2$). A windowed third-summand scan cut $m = 3$ by a paired 0.828 $[0.783, 0.863]$; a fold over negation (`rootfind-negfold`) took $m = 2$ to 23.22 and then 22.97. Two further cycles of denser and sparser bases, windowed negfold and plain root finding were dead ends (`research/ecbench_rootfind_vs_rho_20261014/`). With F4's word XORs priced, the paired ratio negfold/F4 is 0.0545 $[0.047, 0.064]$, a complete-against-complete comparison; against strong signed-Frobenius rho ($S = 14.82$) the best complete arm is still 1.55×. A compact-orbit base with a counted Frobenius pair table reaches $S = 15.55$, below the then-unpriced F4 (16.75), but it is itself a lower bound (factor-base construction natives unpriced), and it is 1.08× rho. Amortised over 64 targets that arm reaches $S \approx 11.2$, below rho; by the ledger's rules that is a secondary, multi-target result, not a cold claim.

Prime tuning in ecbench. On a 20-bit prime curve, the pair table's counted S is 0.018861 $[0.018547, 0.019242]$ of direct subtraction (24 pairs; prime native work unpriced, so a bounded ratio), and 0.057047 $[0.041159, 0.092069]$ on a 17-bit Koblitz curve, fully priced, where the table set-up is about 68% of the arm's cold cost. Doubling the table from 32 to 64 abscissae gives 0.830 $[0.697, 0.962]$; three summands instead of two costs 1.431× $[1.225, 1.716]$; batching the table inversions leaves counted GAE unchanged (1.000; 4,096 versus 64 inversions, set-up wall 243 versus 128 μs, descriptive); and a train-set lane selection did not resolve a hold-out winner (0.979 $[0.919, 1.038]$; 1,088/1,088 verified executions) (`benchmarks/README.md`, rows of 12–14 Oct).

Dense root index at $n = 83$. Widening the prefix base from K1500 to K1798 cut mean probes per relation from 300,021.9 to 169,069.8 (paired difference 130,952, 95% [52,058, 209,991]) and gave a one-point online ratio of 18.528 over same-point rho (IC 63.3 ms, rho 1,173.3 ms), with 339.0 s of reusable preparation and an 8 GiB table outside the window. The host was shared, so the result is exploratory; two K1500 runs hit the 900 s timeout and are retained. The probe reduction stands; the online ratio does not: it was taken against rho with an empty table, and rho with a precomputed table matched to the index calculus's set-up is faster in every matched $n = 83$ pair (validation sweep, 20 October; §16.4).

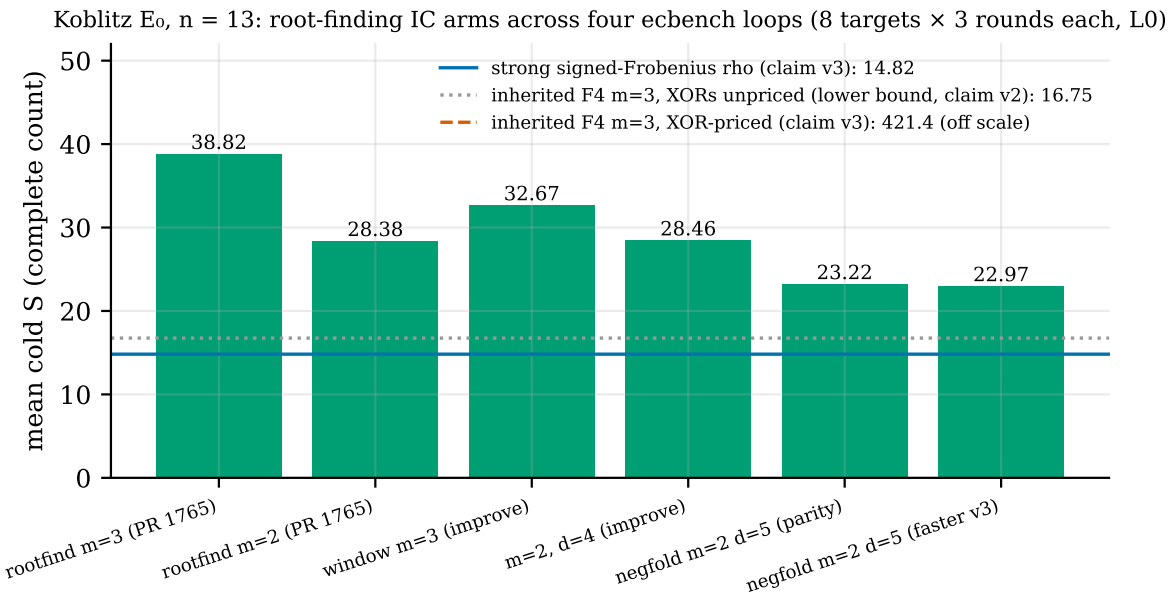


Figure 8.2: Mean cold S of root-finding IC arms at Koblitz E_0, $n = 13$, $d = 5$ unless stated, across the engine, improve, parity and faster loops. All bars are complete counts. Sources: `research/ecbench_rootfind_engine_20261012/`, `research/ecbench_rootfind_improve_20261013/`, `research/ecbench_rootfind_parity_20261013/`, `research/ecbench_rootfind_faster_20261014/`. Each loop had its own seeds and rho arm (rho 14.12–14.82).

Reading. Root finding is the largest constant-factor change of the programme at the oracle stage, and the one with a proof behind it. It does not change the asymptotic count of index calculus: its cost is $\Theta(|\mathcal{F}|^{m-1})$, every measurement is at $m \leq 4$, and it leaves relation yield, factor-base size and linear algebra untouched. Its effect on the cold frontier is to move the decomposition stage from dominant to minor, which exposes the stages that now are.

9 Gröbner-basis engines

Before root finding, the decomposition oracle was a Gröbner solve, and for $m \geq 3$ over binary fields it still is. The repository grew a family of engines (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md`, §1; file names below are under `src/cryptanalysis/`):

Matrix F4 and F5 over $\mathbb{F}_2$
(`koblitz_groebner.rs`, `matrix_f5_f2.rs`): Macaulay matrices of the Weil-restricted Boolean system, reduced by a shared Four-Russians kernel (`gf2_elim.rs`, four 8-bit tables, AVX-512 or AVX2).

Inherited F4
(`inherited_f4.rs`, from 22 September, the default algebraic oracle): a DPLL-style tree over partial assignments in which each child *inherits* its parent's echelon form and re-reduces only what the assignment changed, instead of building a fresh Macaulay matrix per node.

F6-IC
(opt-in): inherited F4 plus a factor-base domain gate (a fully assigned summand whose abscissa is not a base point refutes; partial support pruning refutes earlier), and meet-in-the-middle leaves (with $m-1$ summands fixed the last is looked up; at $m = 3$ and $|\mathcal{F}| \leq 256$ every compatible second point is added and the third looked up).

$\mathbb{F}_p$ F4
(`f4_fp.rs`, `f4_fp_tower.rs`, `sig_fp_tower.rs`): degree-bounded F4 with sparse-by-reducer then dense reduction, a Faugère–Lachartre block for the Kummer-tower oracle, and a signature variant.

Correctness. The audit pinned every engine against every other and against brute force (88+15+9+14+13+12+21+6 passing library tests in the touched modules), and found one soundness bug (§4.5, item 6), two contract defects and several latent holes, all fixed with regression tests.

Engineering results. Figure 9.1 collects the headline ratios; all are ENGINEERING with identical outputs.

- *Inherited versus matrix F4*: matrix work 198.6× (K13) and 222.1× (random-13) the inherited engine's on the same targets (the two can return different decompositions, so their splitting trees may differ; only matrix F4 and F5 share a tree by construction); at $m = 19$ every matrix F4/F5 run timed out at 1,800 s while inherited F4 finished in about 6 s (`research/f4_f5_f6_cross_family_20261011/`).
- *F5 versus F4*: word-XOR ratios 0.6465 / 0.6414 (K13 / random-13), wall 0.79–0.86.
- *F6-IC versus inherited F4*: word XORs about 0.82 at $m = 13$ and 0.77–0.78 at $m = 19$, relation wall 1.7–1.9× faster. But once its batched group additions are charged, F6-IC's S was 64.6–68.8× inherited F4's at $m = 13$ and 1,303–1,353× at $m = 19$: an accounting loss, not a frontier move. Those ratios used inherited F4 before its word XORs were priced (#1774); priced, at K13, F6-IC's S is 1,572.1/634.12 ≈ 2.5× inherited F4's, still a loss, while the R13 and $m = 19$ cells remain lower bounds (`research/f4_pricing_jump_20261018/`). At $m = 31$ both engines hit the 3,600 s cap on the first target.
- *Kernel routing and rewrite*: routing every echelon call to the Four-Russians kernel gave 1.75–3.22× on the oracle's own Macaulay matrices; a word-level `pext`/`pdep` rewrite plan made F4 and F6-IC 10–14% faster at $n = 23$; eager rewrite of kept rows beat lazy rows by 1.40–1.62× (`research/inherited_f4_eager_rewrite_20261011/`).
- *Instruction and allocation work*, measured by paired A/B/A′ instruction counts on a frozen eleven-system suite: fused construction and readback, −5.28% (F4) and −11.02% (F5) instructions; native hashing, −3.715% F5 [3.714, 3.726] and −45.1% for the standalone criterion; bounded direct indexing in the F5 packer, −3.947%; row allocation, 42–50% fewer requested bytes. Elapsed time was not claimed for these.
- *Quotient-curve model*: solving the $m = 2$ system directly on the quotient curve instead of a symmetrised control cut F4 word operations by 1.340–1.642× with identical root digests in five interleaved cells (`research/lit2026_w8_summation_models_20261011/`).
- *The $\mathbb{F}_2$ engine campaigns* (about 90 directories, 22 September–17 October; the programme record [3], clusters "boolean", "gf2-elim", "f5-kernel", "f5-support-cut", "f4-f5-shared"). Each micro-change was gated against the strongest reference, and most failed. Kept as defaults: a branchless AVX2 strip (1.210× one thread; `research/gf2_branchless_strip_20260929/`), a bitmap seen-set in Boolean

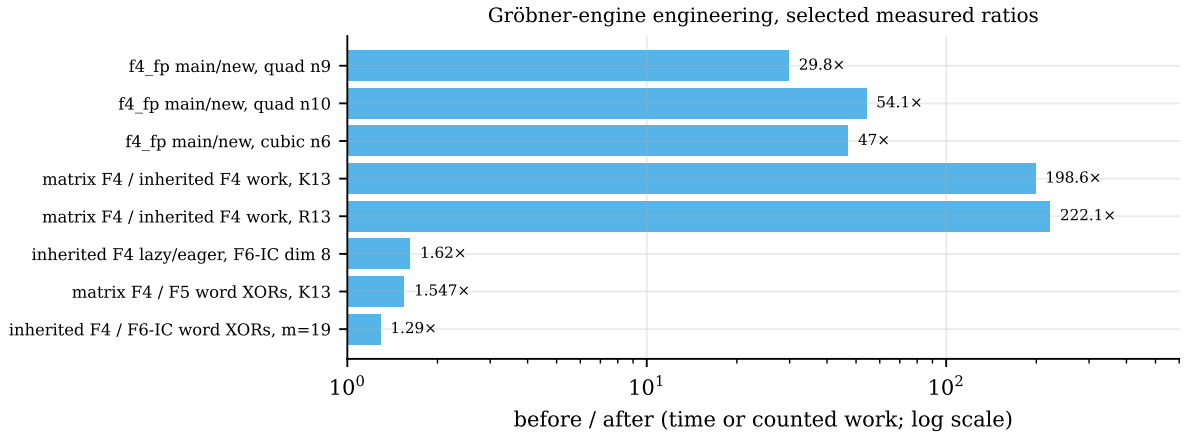


Figure 9.1: Selected engine ratios (before/after or baseline/candidate): `f4_fp` successive versions (single-threaded wall, L0); matrix versus inherited F4 (counted matrix work); lazy versus eager inherited rewrite (median wall); F4 versus F5 and inherited F4 versus F6-IC (word XORs, stage only). Sources in `figures/data-program.json`.

F4 (1.199× the full F4; `research/boolean_f4_bitmap_20260926/`), batched pair inserts, five-column tables, and direct packing in F5 (default since 17 October). The combined F5 options reached 1.530× (`research/f5_combined_2x_20260928/`) and a fixed support cut 3.02–3.07× on one complete call (`research/f5_support_cut19_confirm_20261003/`); about thirty unpack, pivot, layout and packing variants were refuted, and graded reuse in F5 reached 1.78–1.84×, short of the campaign's 2× goal (`research/boolean_f5_graded_batch_20261002/`). These are stage speed-ups of the decision engine; none changes an index-calculus distance to rho.

- $\mathbb{F}_p$ *F4 to msolve parity* (`research/f4_fp_msolve_parity_20261011/`): Gebauer–Möller criteria, packed monomials, interned symbolic preprocessing and a parallel symbolic pass made `f4_fp` 29.8–54.1× faster than its predecessor on square generic systems, with msolve/ours at 0.96–1.22 single-threaded; a later blocked pair reduction made ours 0.75–0.78× msolve's time on the two largest square systems but 1.11–1.49× on four others (`research/f4_fp_blocked_pairs_20261016/`). A day earlier msolve had been 12–24× faster (`research/gb_cas_comparison_20261011/`); every reduced basis agrees with Singular or msolve. On Boolean decomposition systems ($n = 17$, 8 targets) our oracle decides in 7–24 ms per target, while msolve and PolyBoRi spend 40–52 s computing a full basis and Singular exceeds 120 s—a difference of task (decide versus compute the basis), not of engine quality.

What did not work. Degree-2 early-abort probes after a 2025–26 proposal: word-XOR ratios 1.00083–1.00378 (no change) and 0.99714–0.99928 (worse). Gate-first geometry: 1.57–1.78× fewer word XORs but lane-batched F4 roots cost 2.6–4.0× a sequential control. Convolution splitting of the summation polynomial: 1.21–2.07× *more* word operations despite lower degree; Frobenius auxiliary variables: 57–701× more (`research/lit2026_w\{4,5,8\}_*_20261011/`). Symmetry-breaking clauses in SAT did not help (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). A registered law for the $m = 3$ refutation degree resolved all 68 units but ended with the verdict "neither": one variant met its generic degree window on 32/32 draws, the other fell below it on 8, and the heaviest degree-10 Macaulay solves were $2.4 \cdot 10^5 \times 2.0 \cdot 10^5$ matrices at 42–51 minutes each (`research/ic_semiregular_law_20261005/`).

10 Koblitz curves

Koblitz curves $E_a : y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$, $a \in \{0, 1\}$, carry the Frobenius endomorphism τ, which acts on the prime-order subgroup as a known scalar λ; ECC2K-130 is the $a = 0$, $n = 131$ member. They were the programme's main family: the compact-orbit producer, the $n = 83$ campaign, leaderboard Table A and the ECC2K-130 cost model all live here. Random and prime-degree binary curves without τ are in §11, and the Weil-descent degree studies that price ECC2K-130's algebraic route are there too, because their systems use random subspaces rather than Frobenius structure. Every study in this section is listed in the programme record [3].

10.1 Frobenius-invariant and compact-orbit bases

The Koblitz implementation (`src/cryptanalysis/koblitz_index_calculus.rs`) builds GGMP's invariant base from the linearised polynomial of a factor of $x^n - 1$, checks $\pi(\mathcal{F}) = \mathcal{F}$ point by point, rewrites each relation into orbit columns with known λ^k weights, and solves the cofactor-corrected system $ha + hbd \equiv \sum \lambda^{k_i} x_{o_i}$. On small fields the orbit collapse is as predicted: $|\mathcal{F}|/\#\text{orbits}$ of 7.86, 10.89, 12.96 at $n = 9, 11, 13$ for relations and 61.7–168.0 for linear algebra, against rho's 4.24–5.10 (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). Because $\ell = \text{ord}_n(2)$ fixes the invariant base's size, the programme moved to bases sized to the instance: divisor bases, SAT-optimised bases, a rank-guided base at $n = 53$ (220 orbits, a complete implicit S_5 formula of 22,887 clauses), and finally the *compact signed-Frobenius orbit* base, which stores one representative per orbit and no pair table (`docs/ic/COMPACT_ORBIT_PRODUCER.md`). In the ecbench factor-base suite it is the best measured Koblitz construction: $S = 9.114$ at $n = 17$ (272 points, 8 columns), 9.985× matched signed-Frobenius rho, against full orbit 18.7× and subspace bases 47× (`research/ecbench_frontier_20261013/`).

10.2 The oracle sequence

The binary oracle went through SAT (XOR-native CDCL after Trimoska et al. [37]; decides the $n = 19$, $\ell = 6$ corpus but does not scale), matrix F4 (whose refutation range was the real frontier), pairs-and-solve for S_4 (validated to $n = 21$, $\ell = 7$, still $O(2^{2\ell})$), meet-in-the-middle pair tables, inherited F4 and F6-IC (§9), and root-finding engines (§8). The boundary ledger's current Koblitz decomposition record is an $n = 31$, $m = 2$ F4 cell with an exact block-6 M4RI acceleration; at $n = 53$ an exact Frobenius-orbit S_5 compression removes the coordinate ROM but every unrestricted solver arm remains UNKNOWN.

The September toy studies that fed this sequence are catalogued as campaigns. At $n = 13$ and $n = 19$, rotated subspace bases covered 1,591 of 2,003 targets with five spaces against 61 with one, and joint rank recovered 58/64 targets against 29/64, without passing a 10% yield gate at $n = 19$ (`research/notes/ecc2k130/`, clusters "rotated-n13" and "rotated-n19"). Finite-fibre S_3 semantics were proved exact and an O-aware CNF made 252.7× smaller at $n = 13$, but no existing solver interface was admitted as an arm; three SAT engines decided 5 satisfiable and 27 unsatisfiable targets at $n = 13$, with the five negatives DRAT-certified. At $n = 131$ a symbolic full-point DAG replayed a single-edge DIMACS instance but could not be admitted past the 64-bit ceiling, and the exact Frobenius census of the balanced $(m, d) = (10, 13)$ rotated base has 1,048,134 columns, the only shape with as many columns as the field. A frozen test of the Frobenius quotient found that toy fields saturate and so test nothing; an independent SAT reproduction solved $n = 13$ and none of four planted targets at $n = 19$ in 1,200 s (`research/frobenius_quotient_sat_20260930/`). At $n = 13$ the presentation of the curve does not matter: yield spread is explained by cofactor balance (correlation 0.850) and there is no curve effect at $n = 19$ (intraclass correlation ≤ 0.006; `research/koblitz-isogeny/`).

10.3 Against a stronger rho: the September reversals

Three September results that put Koblitz index calculus at or below rho were reversed when rho was strengthened, which is why every later row is quoted against the strongest available rho (§4.5).

- *Candidate tournament.* The 15 September tournament's best candidate used 0.5347 of a worker rho's retired instructions and a triple table 0.409 at $n = 43$, $a = 1$. Giving rho the index calculus's own

canonical form made rho win all five cells, IC at 1.19–2.88× (`research/ic_triple_counted_20260923/`; the tournament README, `research/ic_candidate_tournament_20260915/`, carries no correction note).

- *Matched rho for the orbit DLP.* An IC/rho of 0.2403 in retired instructions ($n = 53$, 1,024 targets) used a rho with software carry-less multiplication on x86-64. Strengthening rho in four rungs moved the ratio $0.240 \to 0.317 \to 1.294 \to 1.923$, and the preregistered sweep then read 0.794, 1.520, 1.923, 2.297 at $n = 37, 41, 53, 61$, rising with n (`research/notes/index-calculus/RESEARCH_STRONG_RHO_LADDER_20260929.md`). Against a strong single-target rho the direct arm costs 44.0× [39.4, 45.4], which withdrew a selected-panel crossover of 0.8392 (`research/notes/index-calculus/RESEARCH_SINGLE_TARGET_STRONG_RHO_N53_20260930.md`).
- *Compact-orbit batch lead.* A point panel had compact orbit at 0.309× a corrected rho at $n = 41$, $L = 1{,}024$ targets. A 32-walk batched-inversion normal-basis rho reversed it (2.081 at $n = 37$, 3.086 at $n = 53$ for $L = 1{,}024$; `research/notes/ecc2k130/compact_orbit_strong_rho_20260929/RESULT.md`), and the orbit-disjoint cold confirmation ended with six eligible no-gos, compact/rho CPU 1.314 ($n = 41$, $L = 1{,}024$) to 22.675 ($n = 53$, $L = 1$) (`research/notes/ecc2k130/disjoint_cold_v2_outcome_20261001/RESULT.md`). Batched compact IC does use fewer instructions than rho (0.5165× at $n = 37$, 0.6221× at $n = 41$, $L = 1{,}024$) while its cold CPU stays above rho, a gap traced to memory behaviour (`research/notes/ecc2k130/compact_ir_ledger_20260930/RESULT.md`). Some sixty notes on S3 batch inversion, swap quotients, Bloom prefilters, window screens and K grids each moved a constant and none crossed rho (cluster "compact-orbit-cold").

The boundary ledger of the same weeks reached the same place from the other side. Re-pricing every row against a matched negation rho raised the vs-rho ratios 2.1–6.5× (`research/ic_rho_reference_20260923/`); the best batch row was 4.86× [4.21, 5.55] at $r = 2^{39}$, and a page $r^{1/6}$ law was falsified at the small end (`research/ic_exponent_20260926/`); single-target online IC beat table-less rho at all six sizes (8.84–17.5×) while cold rho won at all six (1.98–31.6×; `research/ic_single_target_20260930/`). At $n = 37$ in ecbench the cold IC/rho was 3.64× in instructions and 13.6× [10.6, 18.7] in counted S, and the $n = 37$ parameters did not transfer to $n = 41$ or $n = 53$ (`research/ecbench_n37_shared_rank_20261004/`, `research/ecbench_n41_n53_shared_rank_20261005/`). An $m = 4$ exponent audit measured a cost slope of 0.985–1.060 bits per unit of n at $n \leq 19$, against the 0.25 the lead needed (`research/ic_m4_exponent_audit_20260928/`).

Other Koblitz lines. The remaining Koblitz campaigns moved constants inside index calculus and are catalogued in the programme record [3]. Instruction-level rounds on the Gröbner stage left the whole relation log at 964× rho on K_0 over $\mathbb{F}_{2^{13}}$ (`research/groebner_instructions_20260925/`); a September autolab campaign cut cold instructions by 47.7% and left the candidate at 4.98× matched rho's instructions (`research/ic_autolab_next30_20260915/`); at $n = 37$ a full-rank compact-orbit gate passed with IC at 22.6× rho (`research/notes/ecc2k130/full_rank_compact_orbit_ic_20261003/RESULT.md`). Pipeline kernels gave laned canonical keys 1.09–1.31× (`research/canon_laned_keys_20261008/`), u64 root batching 3.26× whole-process wall (rank cell corrected to 2.47×), and aimed gathering 1.037× set-up (`research/ic_aimed_gather_20261023/`). Transporting a base through ECC2K-130's degree-263 and degree-7 isogenies was built and certified, but on toy fields the transported and descendant bases cost 3.6–4.0× the original (`research/ecc2k130_factor_base_pilot_20260924/`); and τ-adic known-scalar arithmetic is 0.413 of the binary cost at $m = 131$ with no measured rho benefit (`research/notes/ecc2k130/tau_adic_arithmetic_20260929/README.md`). On the rho side, GPU iteration redesigns for the ECC2K-130 walk did not beat the shipping distinguished-point walk (`research/ecc2k130_iteration_redesign_20261012/`, `research/dp32_sigma_table_20261012/`).

10.4 The one-target online ladder

The ledger's primary `vs_rho` question is: with reusable IC preparation already done, how long does one unseen public target take, against rho with the automorphism discount on the same point? The ladder (Figure 10.1):

- $n = 53$: guided shared-log precompute with an explicit pair table (9.34 GB peak RSS); earlier, a 1,024-target shared-log batch beat batched rho in all 15 paired blocks (secondary, multi-target).
- $n = 61$: compact orbit, 600 columns, 73,200 points, 2.73 GB; fresh reproduction 124.85× (exploratory).

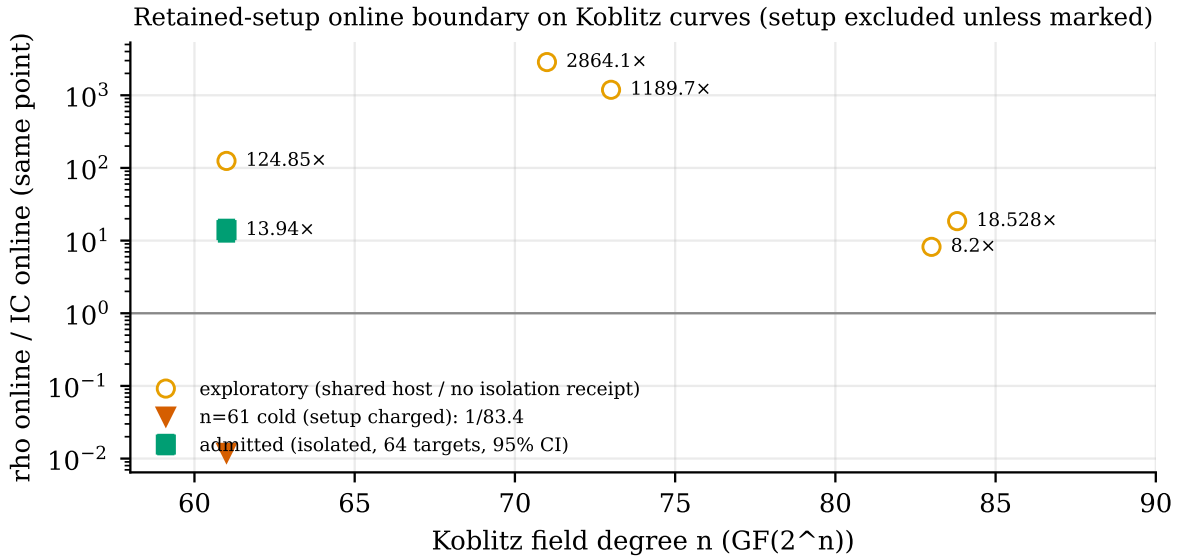


Figure 10.1: Rho online time over IC online time on the identical public point, reusable IC preparation excluded. Open circles: exploratory (verified answers, no qualifying isolation receipt; three repeats of one point or a single point). Filled square: the admitted 64-workload panel with its 95% bootstrap interval. Triangle: the same $n = 61$ candidate with set-up charged (cold). Sources: `docs/ic/boundary_targets.json`, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`, `research/n83_dense_root_20261013/`.

- $n = 71$: first rung past 64-bit packing (u128 words), 85,200 points, 2,864.1× median of three repeats of one point (exploratory).
- $n = 73$: 87,600 points, 1,189.7×; the guided-rank precompute took 5.1 h at 57–70 million probes per relation (exploratory).
- $n = 83$, $a = 1$: IC online 5.6–7.9 s against rho 43–71 s, 8.2× (exploratory). Re-measured on 12 October against a default strong rho on five repeats of the same target, IC was *slower* online, 1.382× $[1.036, 1.882]$, and 218.0× rho in fresh-process wall time (`research/n83_evaluation_20261012/`). The later dense-root reading of 18.5× is in §10.5.
- $n = 61$, **admitted** (12 October): $a = 0$, $K = 500$ orbit columns, 61,000 points, 64 fresh public one-target workloads per host, identical candidate manifest on two hosts. Pinned x86_64: median online speed-up 13.94×, 95% $[9.81, 19.13]$, claim-check 64/64; unpinned arm64: 12.87× $[7.30, 19.08]$. Set-up of about 28 s is outside the window; with it charged, IC costs a median 83.4× rho.

The exploratory ratios are large and irregular because a one-point online window measures the luck of one relation search against one rho walk; the admitted panel's range over 64 workloads is 1.06–648.6×. The admitted point is a constant-factor online result at $n = 61$ (47-bit subgroup) with preparation retained; it is not a cold result, not a scaling law, and not an asymptotic claim.

Retired as an index-calculus win (14 October). Every rung of this ladder compares index calculus with a retained table against rho with none. Given a distinguished-point table of equal or smaller size, Bernstein–Lange preprocessing rho [4] is 53–317× faster online than the compact-orbit arm on the same 64 targets at $n = 41$, 53 and 61 (§16.3). At $n = 83$ (§10.5) rho with a table matched to the index calculus's set-up time or memory won every pair that was run. The ladder therefore measures what a table buys, and generic rho buys more per byte. The validation sweep later found `docs/ic/BOUNDARY_TARGETS.md` still printing the 13.94× as a win; it is superseded (`research/validation_sweep_20261019/B-benchmarks/VERDICT_TABLE.md`).

10.5 The $n = 83$ campaign

Degree 83 was the programme's largest Koblitz target with a working pipeline: 145 study directories between 4 and 21 October (the programme record [3], clusters beginning "n83" and "f6-n83"). In order:

- *F6 micro-studies, 4–5 October* (24 directories, almost all on the $a = 0$ curve). They kept kernel gains (queries up to 2.998× faster, resident memory −52%) and shrank SAT encodings, but every ordinary solve timed out; four-summand coverage was at most $4.662 \cdot 10^{-12}$ of the subgroup and no natural

relation was found. On the 81-bit $a = 0$ gate curve, SAT and XCNF decompositions succeed only with a known witness, and the prebuilt-base question is an open problem record.

- *The $a = 1$ point, 7–12 October.* The archived K600 point read 8.2× online and was re-measured slower (above).
- *Root-index capacity, 13–16 October* (25 directories). Root-index layouts grew the $a = 1$ base from K512 (84,992 points) to K3596 (596,936 points) within 14 GiB. The dense root index cut probes per relation from 300,021.9 to 169,069.8 (K1500 to K1798; paired difference 130,952 [52,058, 209,990]), a within-IC result that stands; its one-point online 18.528× is retired (`research/n83_dense_root_20261013/`).
- *Factor-base selection, 14–16 October* (56 directories). Root4, root11 and root21 layouts, Bloom filters, sparse and canonical variants and K screens from 100 to 20,000 points selected within index calculus; the best means are about 26–29 ms online (K3425, K3450, K3500), and many screens end BOUNDED_UNKNOWN. A three-hash 512 MiB Bloom filter cut a K3300 online mean from 34.77 to 31.32 ms (`research/n83_root11_bloom_results_20261015/`); a K3350 win of 7.18% did not replicate.
- *Tier-1 online readings, 14–16 October* (14 directories): median rho/IC of 54.1, 210.3, 113.0, 104.2, 35.1, 413.3 (one point) and 208.7× (K3072, five fresh points, [38.3, 2,510]; `research/n83_tier1_online_20261016/`), all against a rho with no table and with set-up excluded.
- *The control, 16 October.* Bernstein–Lange rho given a distinguished-point table matched to the index calculus's set-up time or memory was faster online in every pair run: IC/rho 3.74 [2.65, 11.05] (K3300, time-matched), 40.61 (K3300, memory-matched), and 81.5–395.5 (K600) (`research/20261016-n83-precomputed-rho-control/`). Only the K3300 and K600 bases were paired; the K3072 reading was not, so its retirement rests on the same mechanism rather than on its own measurement. The validation sweep and the ledger audit retired all the online readings on 19–20 October.
- *Cold, 16–21 October* (11 directories). The cold one-target process fell from about 150 s (K1024) to 119.8 s (K1120, holdout) and 113.8 s (K1107; `research/n83_cold_base_confirm_20261021/`), against a rho online time of about 2.2–2.6 s on the same hosts. No matched cold rho was run, so no cold ratio is admitted; the hosted single-target benchmark's 133.7× is the nearest cold figure (`benchmarks/20261014-n71-n83-hosted-single-target/`).
- *A second pipeline in `cryptanalysis`* (K600, $a = 1$; [3]). A Tier 1 ratchet of rank-probe, closure, index-build and rank-finder changes brought its latest row to 3.92× a tuned rho end to end, cold, and 0.093× online with set-up excluded (`ca:experiments/koblitz-n83-rankfinder-20261021/README.md`). The study's own limits apply: one run on one host class, a point estimate with no interval, wall time, factor-base construction charged to no arm (so the cold cost is a lower bound), no computable band, and gaps that are not comparable across runs (the same producer read 4.42× on another host). Against Bernstein–Lange rho with equal advice its online time is 22.55× [16.23, 31.44] worse (`ca:experiments/koblitz-n83-bl-rankprobe-20261019/`). It is the closest cold Koblitz figure at $n = 83$ and is still above rho; the two pipelines' numbers are not comparable with each other.

The open pull requests at the time of writing include further $n = 83$ work ($m = 4$ queries on a retained 16,627-column base, AWS replays, shared-binary panels); none is on `main` (the programme record [3]).

10.6 Leaderboard Table A: the Koblitz rules

Table A of `docs/ic/LEADERBOARD.md` prices one unseen public target on Koblitz curves at $n = 41$–61 ($\log_2 r = 36.6$–47.2), cold, against rho on the same point. Its history is a sequence of rules, each a pipeline baseline plus a rho reference (`research/ic_tool_program/`):

- *Rule v3*: IC/rho 5.23–21×, the figure still printed in Table A, which remains unvalidated (pre-gate).
- *Baselines v4–v8* (16–17 October): head-of-main, scan-key (v5, 1.20–1.24× cold), scan-fold (v6, 1.33–1.40×), build streams (v7, 1.07–1.13×) and admitted prefetch (v8, 1.06×), together 1.90–1.98× cheaper cold than v4 (`benchmarks/20261017-ic-r12-admitted-prefetch/`). Seven further candidates (prefetch gates, filter slots, huge pages, a wider base) did not clear their rounds.
- *Rules v6, v7*: cold rho still faster at every size, 3.78–12.06× and then 3.43–11.51×; but their rho references ran at 0.57–0.85 of the strong walk per step, so the distance was understated (`benchmarks/20261016-ic-rule-v7/`).
- *Rule v8* (19 October): with a matched GFNI rho the cold distance is 4.67 [4.34, 5.07] to 18.07 [15.98, 20.58]×, 1.30–1.58× the v7 control, with a fitted slope of 0.168 per $\ln r$ (`benchmarks/20261019-ic-matched-rho-v8/`). The run is archived as incomplete although its row is listed.

- *Rule v4 recipe* (24 October; §16.5): a paired retuning improved one size of six; with rho priced in the same process at the canonical batched step the distance is 9.9–43×. Its validation file is withheld because its rho reference may not be the strongest, so it is not an admitted row either.

Table C prices a 32-target batch on nine Koblitz curves ($\log_2 r = 18$–47.2) against batch rho on the same targets: 3.62–14.9×, a historical diagnostic under the one-target rule. All nine failed the 24 October validation gate, and Table D, the Joux–Vitse historical study, is withheld.

The validation sweep found that Table A understates its gap to rho by 1.7–2.0×, and the 24 October validation gate did not check Table A (`research/ic_validation_gate_20261024/`). Rules v8 and v4 measure the same sizes against differently built rho references and are kept apart (Definition 4.1).

10.7 ECC2K-130

ECC2K-130 is the $a = 0$, $n = 131$ member of the same family, with a 130-bit prime subgroup and rho at about $2^{60.81}$ group operations. The compact producer already solves $n = 41, 53, 61, 71, 73$ of the family; about 300 notes under `research/notes/ecc2k130/` price the routes to $n = 131$ itself. No route prices below rho.

- *Extrapolating the producer.* The fitted probes-per-relation law $C\,r/(n^2K^2)$ with $C = 1$ gives $1.1 \cdot 10^{18}$ core-years at $K = 600$ and $6.8 \cdot 10^8$ at $K = 10^{12}$ columns (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`, with the unexplained factor-of-5 gap in C of §4.5).
- *Decomposition oracles.* Decompositions exist, but no oracle finds one for less than 2^{131} per family; using unions of Frobenius orbits lowers the family's bottom from $2^{132.58}$ to $2^{124.99}$, still $2^{64.18}$ times rho, and needs a materialised base of $2^{44.53}$ points at $m = 3$ (`research/notes/ecc2k130/RESEARCH_ECC2K130_DECOMPOSITION_TARGETS.md`). Relation sweeps over m give a best single-relation row of +4.98 bits at $m = 4$ (with $2^{70.79}$ bytes) and a best honest $m = 8$ row of +13.59 bits with the Frobenius quotient (`research/notes/ecc2k130/RESEARCH_ECC2K130_RELATION_SWEEPS.md`). Crossbred, RR-solver and WDSat oracles are engineering only: Crossbred is 0.023 of F4's bit operations at $m = 3$ but slower end to end, and the RR solvers' advantage over Semaev at $d = 6$ is worthless against brute force.
- *Linearisation.* Plain linearisation leaves a gap of $70.19 - b_{\max}$ bits to rho; the best bilinear XL setting, with each Macaulay solve charged, is 71.82 bits at $\omega = 2$ (78.15 at $\omega = 2.81$), and a trilinear variant is worse (87.16 bits) (`research/linearization_reach_20260930/`, `research/xl_bilinear_core_20261006/`, `research/xl_trilinear_t_unknown_20261006/`).
- *Weil descent.* The descent-degree note's rows are $m = 3$ with a free oracle at $2^{68.58}$; $m = 4$ at degree 4, $2^{68.43}$ (+7.62 bits, now contradicted); and $m = 4$ at degree 5, $2^{71.13}$ (+10.32) at $w = 1$ and $2^{106.52}$ (+45.71) at $w = 2$, in unconverted column touches (`research/notes/ecc2k130/RESEARCH_ECC2K130_DESCENT_DEGREE.md`). The studies behind these rows, and the hybrid guess that puts the $m = 4$ row at +8.43 to +58.27 bits, are in §11.1.
- *Covers.* Over $\mathbb{F}_{2^{131}}$ every ordinary curve has GHS magic number 1 or 131, so GHS descends ECC2K-130 to genus 1 (a zero map on the subgroup) or to genus about 2^{130}, and the Hess–Menezes–Teske walk has nowhere to land. One question is open: $\langle G \rangle$ is the group of a simple 130-dimensional abelian variety, and if that variety were isogenous to a Jacobian with an evaluable isogeny, index calculus on the genus-130 curve would cost $2^{37.17}$, $2^{23.64}$ times *less* than rho; no such curve is known, the Torelli codimension at genus 130 is 8,128, CM constructions with prescribed Frobenius stop at genus 3, and the largest published Weil-descent discrete logarithm is at genus 31 over $\mathbb{F}_{32}$ (§14) (`research/notes/ecc2k130/RESEARCH_ECC2K130_HYPERELLIPTIC.md`, with the magic-number correction of `docs/audits/GHS_MAGIC_NUMBER_MMT_CORRECTION_20261013.md`). The $N = 130$ counterfactual is genus 255 at a modelled $2^{55.60}$ against rho's $2^{60.31}$, and measuring one relation trial turns that margin negative (§16.5). The best explicit curve construction reaches genus $2^{28.92}$ (`research/notes/ecc2k130/RESEARCH_ECC2K130_CURVE_CONSTRUCTION.md`). The isogeny class has 262 curves of conductor gap 263 and no measured level effect (§14); rho on the degree-263 leaves keeps only the negation map, an 11.45× penalty (`research/notes/ecc2k130/isogeny_rho_speed_20260930/`).
- *Rho itself.* The separate GPU rho estimate is about 4.3 RTX PRO 6000 card-years (`paper/ecc2k130/`); on one CPU core the ECC2K-130 client walks 13.2 million steps per second (`research/certicom131_rho_ic_bench_20261022/`). Signed σ steps cut modelled GPU time to 0.961–0.970 (`benchmarks/20261014-ecc2k130-signed-sigma/`).

Index calculus is not a route to ECC2K-130 on this evidence: every route that has been constructed or

priced is at least about 5 bits above rho (the $m = 4$ single-relation row, at $2^{70.79}$ bytes), most by tens of bits, and the one hypothetical route below rho needs a genus-130 curve that no known method can construct.

10.8 Factor bases closed under $\mathbb{Z}[\tau]$

The $\mathbb{Z}[\tau]$ small-norm multiplier base of 23 October (`research/ic_improve_20261023/README.md`) is the latest Koblitz change: 1.9–4.4× child CPU at $n = 41$–61 and 6.2–10.5× at $n = 71$, against the programme's own producer, exponent unchanged; it is discussed with the other known-logarithm bases in §16.5.

11 Random and prime-degree binary curves

This section covers binary curves $y^2 + xy = x^3 + ax^2 + b$ over $\mathbb{F}_{2^n}$ with n prime, attacked with random subspace bases that use no Frobenius structure, whether the curve is random or Koblitz: the Weil-descent route and the degree question behind the subexponential claims (§3), the decomposition oracles, the random-binary leaderboard rows, and the 131-bit ECC2-131 target. Many of the descent studies were priced for ECC2K-130 because it was the programme's named target, but their systems are built on prime-degree fields with random subspaces and use no Frobenius structure (the systems at $n = 19, 21$ say so explicitly), so they belong here. The Gröbner and $\mathbb{F}_2$ engines they run on are in §9; every study is listed in the programme record [3].

11.1 The degree question, measured

The subexponential claims for prime-degree binary curves rest on the first-fall-degree heuristic: that Weil-descended summation systems have a solving degree close to their first fall degree (FFD), which stays small. The programme measured both, on toy fields, in about thirty preregistered studies from late September to 24 October.

First fall degree stays low; the solving degree does not. Every Weil-descended S_3 system for $n = 3$–7 has first fall degree 3 (`research/notes/index-calculus/RESEARCH_FFD_MEASUREMENT.md`), and the HKY last fall degree d_F is 3 or 4 at every reachable size—$m = 2$ to $n = 32$, $m = 3$ at $\ell \le 6$, $m = 4$ at $\ell \le 3$, $m = 5, 6$ to $N = 54$ unknowns—while the plain refutation degree rises from 5 to at least 7 and random controls rise with it (`research/fall_degree_bounds_20261008/`). At $m = 2$ the Semaev d_F sits one below the semi-regular degree at $n \le 18$, and the gap does not grow over $n = 10$–18.

The refutation degree tracks ℓ, within one of semi-regular. For chained $m = 3$ systems at fixed surplus the refutation degree reads 5 at $\ell = 2$, 6 at $\ell = 3, 4$, and exactly 7 at $\ell = 5$ on all four draws of the $(10, 5)$ cell, with FFD 3 throughout (`research/dreg_ell_grid_20260925/`, `research/dreg_surplus_control_20260925/`, `research/dreg_degree7_ell5_20260930/`). At $m = 4$ it reads 5, 6 and at least 7 at $\ell = 1, 2, 3$, flat in n at fixed ℓ (`research/dreg_m4_chain_20261006/`). In 10 of 14 exact cells the semi-regular D_{reg} equals the measured degree and in all 14 it is within one. With a dense in-tree engine at $\ell = 6$ ($n = 19$, both Koblitz curves) the x_4 formulation refutes at 11 and the RR formulation at 8 on every draw, 4 above the sharp first-fall bound of 7, rising one degree per rung from $\ell = 3$ (`research/ic_dense_ladder_20261004/`). A registered semi-regular law then fitted none of seven discriminating cells, with x_4 inside $[d_{\mathrm{reg}}, d_{\mathrm{reg}} + 1]$ on all 32 draws (`research/ic_semiregular_law_20261005/`).

Closure algorithms refute earlier, but rise. A MutantXL-style closure refutes all 92 unsatisfiable draws at degree 3–4 where XL needs 5–7, and none of 345 satisfiable draws (`research/dreg_closure_mutantxl_20261016/`). It stays at degree 4 to $N = 42$ at $m = 4$, $\ell \le 4$, but rises past 4 at $m = 3$, $\ell = 7$ on 8 of 8 draws (`research/dreg_closure_scale_20261016/`); the number of unknowns is not the driver, and of two candidate explanations only "$\ell = 7$" survives a run at $(m, \ell) = (4, 6)$, $N = 54$, which refuted at degree 4 (`research/dreg_closure_driver_20261016/`, `research/dreg_closure_m4l6_20261018/`). Guessing one or two variables restores degree 4 at $(3, 7)$ on 10 of 10 draws at 1/116–1/260 of the projected degree-5 cost (`research/dreg_hybrid_guess_20261024/`).

What the measurements price. Applied to ECC2K-130's $m = 4$ row, the closure results contradict the degree-4 floor (+7.62 bits over rho) and leave degree 5 as the optimistic case, +10.32 bits at $w = 1$ and +45.71 at $w = 2$; the hybrid gives +8.43 bits if the number of guesses stays flat in ℓ and +58.27 if it grows linearly, which only a run at $\ell = 8$ can separate (§16.5). The $m = 4$ cost slope on this engine is 0.985–1.060 bits per unit of n against the 0.25 an exponent lead would need, and a symmetrised refutation degree rises about one per unit of ℓ (`research/ic_m4_exponent_audit_20260928/`, `research/ic_m4_head_engine_20260929/`, `research/ic_symmetry_lever_slope_20260929/`).

Levers that did not move the degree. Decomposition trees cut on the curve's own structure have the same sampled cut-out degree as random sets in all 102 cells (`research/ic_tree_split_structured_v2_20261011/`); nonlinear factor-base labels refute nowhere through degree 6 (`research/nonlinear_factor_labels_20261002/`); Hamming-weight ideals answer correctly but resolve only after 7–18 of n coordinates are assigned (`research/hamming_ideal_pdp_20260930/`); the one measured subspace-shape effect on the degree was a trace artefact (`research/notes/index-calculus/RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md`), and frozen shape cells were killed at a gap of zero (`research/shape_search_g1_g6_20261015/`). The offensive first-fall-degree thread found no lever within 9 bits of the 2^N boundary (`research/notes/index-calculus/RESEARCH_DEGREE_REDUCTION.md`). A census over $\mathbb{F}_{2^7}$ found non-subfield quasi-subfield polynomials below the HKPYY Lemma 4.1 bound, which answers a tightness question but moves no cost (`research/notes/index-calculus/RESEARCH_QUASI_SUBFIELD.md`).

The same question in crypto-autoresearcher. The companion repository measured it independently ([3]). msolve's solving degree of the descended $m = 3$ system is 7 at every n from 11 to 23, a lower bound from $n = 17$ on, where the run walls on matrix size (`ar:experiments/EXP-ALBIN-008/analysis.md`); the study reads this as evidence on the bounded side in a narrow range that cannot distinguish bounded from slowly growing. A degree-4 reuse closure refutes all 400 unsatisfiable S_3 descents at $n = 19$ that plain degree-4 Macaulay refutes none of (`ar:experiments/EXP-CERTBIN-060020/`), the same effect as the MutantXL closure above. A crossbred solver's total exponent is 0.891 $[0.869, 0.914]$ against rho's 0.498 over $n = 10$–24 (`ar:experiments/EXP-ICI-001/analysis.md`); on its older range, $n \leq 20$, the fit reproduces the repository's earlier 0.863 as 0.864.

Reading. On toy fields, the evidence says the solving degree of Weil-descended summation systems grows with ℓ and stays within one of the semi-regular prediction, while the first and last fall degrees stay at 3–4. That is the behaviour of a generic system, and it supports no subexponential claim. It does not refute one: the sizes are small, the closure algorithms bend the curve by a degree or two, and the $(4, 6)$ and $\ell = 8$ cells that would decide the $m = 4$ question have not run. The PKM analogue over prime fields (§12) is in the same state.

11.2 Decomposition oracles

The pair-table floor. On the non-Koblitz curve $y^2 + xy = x^3 + zx + 1$ the meet-in-the-middle pair table is the floor every oracle is measured against. Over Gaudry's pairs-and-solve it is 94–277× at $\ell = 8$, 2,474× and 4,244× at $\ell = 9, 10$, and 928× and 2,321× at $\ell = 11, 12$, with no disagreement on 128 targets per rung (`research/binary_mitm_oracle_l8_20261015/`, `research/binary_mitm_oracle_l9_l10_20261015/`, `research/binary_mitm_oracle_l11_l12_20261016/`; the $\ell = 12$ figure is the validation sweep's recomputation from the committed per-target medians, the study quotes 2,187×). These are wall times on a shared host and unvalidated; the counted evidence—8,068 and 16,092 point additions against 2.1 and 8.4 million quartic solves—carries the result. An earlier pairs-and-root-find oracle was 926× faster than triple enumeration at $\ell = 8$ and decided $\ell = 12$ (11.5 billion triples) in 15 s (`research/notes/index-calculus/RESEARCH_SEMAEV_DECOMPOSITION.md`).

Algebraic oracles against the floor. No algebraic arm beats the pair table: the cheapest is 303× the floor at $n = 13$ and $3.8 \cdot 10^4$× at $n = 31$ for $m = 3$, and the slopes in n are steeper (F4 0.534, F6-IC 0.583, floor 0.207; at $m = 4$ a wide chained solver 0.805 against 0.333; `research/ic_m3_algebraic_vs_mitm_20261014/`).

SAT, SMT and WDSat are enumerators. Every SAT, SMT and WDSat back end spends about one conflict per candidate tuple (CDCL 0.97–1.02 per triple), so the best enumerate-and-root-find oracle beats every solver by 10^2–10^4 (`research/notes/index-calculus/RESEARCH_PDP_SAT_SMT_NAGAO_SURVEY.md`). The SAT pipeline decides all 11 satisfiable $n = 19$, $\ell = 6$ corpus instances in 17.8 s; SMT runs at 0.44–2.63× the native CDCL+XOR solver (`research/pdp_smt_oracle_20261008/`); WDSat solves $n = 5$ (47 against 51,425 conflicts with and without XOR-Gauss) and crashes at $n = 9$ on a static allocation limit, where CryptoMiniSat takes 0.4–2.6 s (`research/wdsat_rr_20260920/`). Run unmodified on the 60 published S_4 instances of Trimoska et al., WDSat is the fastest solver, 2.37 s on the unsatisfiable $n = 19$, $\ell = 6$ cell against 124.6 s for CryptoMiniSat with XOR clauses (`ar:experiments/EXP-ICPERF-66fd51/`; [3]); that is a solver comparison on toy instances, with no relation, linear-algebra or rho column. The

SAT route is closed as an exponent lead. Isogenous and random-b neighbours give identical Macaulay ranks and FFD 3 (`research/pdp_isogeny_probe_20261008/`).

The PDP oracle programme. The point-decomposition speed-up programme (`research/pdp_oracle_speedup_20261011/`), run over $\mathbb{F}_{2^{3\ell}}$, measured the oracle in steps before root finding replaced it: a baseline (PS-0; 338–452 ns per pair across five rungs, A/A spread $\leq 8.8\%$), a gate-count model of the group-law form (PS-1; 7,410–14,608 fused ternary-logic issues against 70,695–190,275, 9.5–13.0×), a scalar group-law kernel (PS-2; 29.8–40.8 ns per pair, 0.067–0.123 of baseline, 400 corpus root sets agreeing), and a bit-sliced kernel (50–62× the baseline on two hosts, about equal to the scalar kernel). The root-finding oracle then ran 3,007–115,160× the PS-0 baseline at $\ell = 8$–14, at the generic floor's slope (`benchmarks/20261012-pdp-root-finder/`). A split/hybrid solving-degree study (Koblitz $b = 1$) projected its closest relation phase at 22.8× strong rho, 391× signed-Frobenius rho (`research/pdp_split_dreg_20261011/`); with Frobenius closure the priced S fell to 0.35–0.54 of the standard base but the best arm stayed at 36× strong rho at $n = 11$ and 889× at $n = 19$ (`research/pdp_split_slots_20261012/`). A shape-replay switch cost more on all 9 cells and stays off (`research/pdp_shape_replay_20261015/`).

Norm and circuit searches at $n = 83$. The 17–19 October autolab cycles built a native norm predicate over GF(2^{83}): a cutoff-8 circuit cuts AND gates 60.5% and clauses 32.9%, an affine support limit of 2 cuts internal coordinates from 58,176 to 45,427, and 1,237 of 1,800 positive-dimensional cells were certified empty. Every hard search returned UNKNOWN with zero verified witnesses, and a primary-source check found the hybrid ANF/CNF approach already established in Bosphorus, SMFF and PolyBoRi (`research/autolab_cycle_20261019_v1/`, `research/autolab_cycle_20261019_v3/`). The companion draft `papers/norm-rootfinding/` reports the same: no measured speed-up.

Toy end-to-end. On toy fields the fastest index calculus, a cached Nagao arm, was 2.33× plain rho, and S_4/S_3 with CryptoMiniSat 14.5–14.7× (`research/toy_ic_e2e_candidates_20260922/`). Frobenius transport preserves zero and lift sets; a constant term c changes zero counts only; neighbouring curves show no consistent F5 advantage; symmetry quotients help three summands (0.26–0.31 of the ordered cost) but not two (1.06–1.32) (`research/toy_binary_elimination_20260924/` and the toy cluster).

11.3 Leaderboard rows

Table B of `docs/ic/LEADERBOARD.md` has five random-binary rows, $\mathbb{F}_{2^{15}}$ to $\mathbb{F}_{2^{27}}$ ($\log_2 r = 14$–24.4), one target, $m = 2$, folded pair table with walked targets: IC is 13.1, 10.6, 18, 23 and 38.2× matched rho. All five passed the validation gate (commit `a8d8e3cd38`; `research/ic_validation_gate_20261024/`), which the ledger paper, written a day earlier, records as no row passing. The cofactor-projected base of LEAD-19 cuts three of these rows by 1.29, 1.58 and 3.36× in cold S, leaving 6.1–14.4× to matched rho (`research/lead19_projected_base/`); it is not yet a leaderboard row. The integer-multiple closure of 23 October cuts S 3–4× against plain seeds at degree 26–28 and remains 6–12× the framework's rho reference (§16.5).

11.4 ECC2-131

The Certicom ECC2-131 target ($\log_2 r = 130.0$, rho $2^{64.83}$ GAE with the negation map) has a cost model, gates and two falsifiers (`docs/ic/CERTICOM131_IC_PROGRAM.md`), all on toy instances. With a free oracle the $m = 3$ floor is $2^{65.2}$–$2^{67.2}$, above rho from linear algebra alone; $m = 4$ is the first usable arity and needs a solving degree of at most 3 (generous pricing) or 1 (plain). On random-b curves at $m = 3$, $n = 5$–11, the measured degree is $D = 4, 5, 6$ at $\ell = 1, 2, 3$, equal to the semi-regular reference with zero spread over 60 draws, and equal to the Koblitz $b = 1$ value on 52 of 52 paired draws; extrapolated to $n = 131$ it is between about 6–7 (if it plateaus) and 39 (semi-regular). One CPU core walks ECC2-131 rho at 8.50 million steps per second with a batched pclmul walk, and 11.46 million bit-sliced (`research/certicom131_rho_ic_bench_20261022/`, `research/certicom131_rho_parity_20261023/`); a CUDA port compiles but has not run on a GPU. The bench's IC ladders on smaller random-binary curves read 11–24× matched rho; it is not an admitted row.

sect113r1. The 113-bit SECG curve is weak by size alone ($2^{56.33}$ generic work); its two degree-5 isogeny routes transfer the problem without an established speed-up, and the GHS and Semaev routes reached no conclusion (`research/sect113r1-weak-curve-20261006/`).

12 Prime fields

Over $\mathbb{F}_p$ there is no subfield to descend to and no Frobenius to fold, so index calculus must find structure in the factor base itself (§1). The programme tried known-answer ladders, root-finding and pair-table oracles, the Petit–Kosters–Messeng Kummer tower, structured bases on the P-256 field, isogeny-class walks on the NIST curves, and lattice reformulations. No prime-field arm comes within a single-digit factor of rho except on the smallest Table B rows, and the gap grows with size. The section ends with Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$, the one odd-characteristic extension field the programme measured end to end. Every study is listed in the programme record [3].

12.1 Known-answer ladders and leaderboard rows

Over prime fields the ledger tracks staged known-answer ladders (`docs/ic/boundary_targets.json`, regime `prime`). On $j = 0$ curves the staged driver recovers a 20-bit logarithm ($p = 1{,}048{,}609$) in 56.8–64.9 s of IC wall against 150–153 ms of rho on the identical public point, with split timers and three seeds; running the relation search on 14 threads made that stage 3–4.4× faster in wall time with counted work unchanged (`research/j0_parallel_relations_20261016/`). On the generic $a = -3$ shape (P-256 and CryptoPro-B class), a Semaev-S_4 three-decomposition family with Cantor–Zassenhaus root finding yields one relation per trial at $p = 65{,}519$ with a 120-point base; at fixed base size, median trials per relation grow from 2 to 28 to 379–506 at 16, 20 and 24 bits, about 16× per four bits. The prime `vs_rho` row is *not achieved* at any measured size.

Table B of `docs/ic/LEADERBOARD.md` has eight prime rows, $\log_2 r = 9.7$–23.4, one target, $m = 2$, folded pair table with walked targets: IC is 6.34, 5.47, 6.43, 6.71, 5.82, 8.3, 10.5 and 13.5× matched rho, the ratio rising from about 18 bits. All eight passed the validation gate on 24 October (`research/ic_validation_gate_20261024/`). In ecbench the root-finding pair table is 16–20× matched rho in counted GAE, and 7.46× [5.82, 9.63] with a lane collector, all lower bounds (`research/ecbench_rootfinding_lanes_20261013/`); a planner that picks the base, stop rule and oracle beat every one-change alternative on six curves at $w = 0$, the root-find-and-subtract variant by 171–439× (`research/ic_planner_validation_20261020/`). The frontier campaign's two-summand prime arm goes from 10.5× rho at 20 bits to 102× at 40 bits (§16.1).

Root finding against Buchberger on NIST shapes. Per query, root finding of Semaev's S_3 was 3,021× (P-256 shape) to 5,802× (P-224) cheaper than the comparison arm at a six-point base (`research/semaev_rootfind_nist_20261012/`); the arm was `groebner_f4::buchberger`, not F4, so the ratio is against Buchberger (§4.5). It is an oracle ratio with no complete solve. A host emulation of a GPU decomposition kernel passed six configurations on 40–61-bit P-256-class curves (`research/gpu_rootfind_decomposition_20261012/`); no device ran it. Three frozen SMT formulations of a 20-bit decomposition all returned UNKNOWN at 120 s, where the native reference needed 146 probes (`research/prime_field_smt_20261005/`).

Earlier, in crypto-autoresearcher. The first end-to-end prime-field index calculus of the laboratory predates this repository's: an imported Autolab round of 1 June 2026 solved logarithms with an $m = 3$ Semaev index calculus over a factor base of the smallest valid abscissae, at a fitted cost exponent of 0.889 against rho's 0.5, IC/rho growing from 21.5 to 205.7 over 10–18 bits (`ar:experiments/EXP-ALPF-032/analysis.md`). Structured bases there weakened rather than supported a relation excess (`ar:ledger/evidence/EV-PFDR-f6b76c.yaml`), and on one 13-bit curve an apparent early fall of the degree fired identically when the Semaev form was replaced by a random polynomial but not when the factor-base constraints were, so it came from the factor base, not from the summation polynomial (`ar:experiments/EXP-PFDR-4bfc6f/analysis.md`). A separate repository asked the underlying question directly—is there any coordinate-defined subset of $E(\mathbb{F}_p)$ that interacts with addition better than a random subset?—and found none on toy curves (`ecdlp-prime-field-smallness`, `README.md` at `4f20be9b36`).

12.2 The Kummer tower

Petit, Kosters and Messeng's base $V = \mu_{2^t}$ (the roots of $x^{2^t} - 1$) gives prime-field summation systems extra structure for an algebraic solver. Its claim rests on the solving degree of those systems staying bounded as $N = m\ell$ grows; the programme measured that degree for four weeks.

The solving degree. The pilot found F4's solving degree at 4–5 for $m = 2$ through $N = 18$ and read the preregistered "grows like N" as failing at toy size (`research/pkm_tower_pilot_20260924/`). Round 2 showed it is not bounded: at $m = 2$ it is 5 to $N = 18$ and 6 at $N = 20, 22$, at $m = 3$ it reaches 7 at $N = 15$, and the null family behaves the same (`research/pkm_tower_round2_20260925/`). At $m = 4$, $N = 16$ finished at $D = 7$, as at $N = 12$ (`research/pkm_tower_round3_20260925/`); at $m = 3$ the line reads $6, 6, 7, 7$ over $N = 9$–18 (`research/pkm_tower_round7_20261006/`). Signature engines agreed with F4 but needed one or two degrees more and 1.7–28× the memory, so F4 stayed (rounds 4–5); a full-echelon exit cut multiply-adds 39.8% at $m = 4$, $N = 12$ without moving D (round 6). The torus and split Dickson presentations give the same D in every cell, and the Kummer D does not depend on the 2-adic valuation (`research/pkm_dickson_torus_20261015/`). At $m = 3$, $N = 21$ two engines finished every degree-7 step without reaching the unit ideal and stopped inside degree 8 on memory—the native `f4_fp_tower` at a 317 GB matrix cap after a 67 GB peak, msolve at 99.6 GiB on 16 threads (`research/pkm_lead1_m3n21_20261016/`, `research/pkm_m3_n21_msolve_20261018/`). That implies $D \geq 8$ at $N = 21$, which both studies read as inconclusive because no degree-8 step completed. The Certicom-131 programme's gate needs $D \leq 3$ at $m \geq 4$; the toy plateau at 7 is above it, at 2^{89}–2^{98} (generous) and 2^{129}–2^{139} (plain) at 131 bits (`docs/ic/CERTICOM131_IC_PROGRAM.md`). The exponent sum of decompositions on the tower is indistinguishable from random at $k = 1$–6 (`research/ic_exponent_falsifiers_20261013/`).

In ecbench. We ran the tower end to end with the root-finding oracles on five curves of 20–28 bits (896 runs, all verified, all replay audits OK; `research/pkm_tower_ecbench_20261013/`). Index calculus costs 34.0, 41.9, 29.4, 96.6 and 140.6× rho-negation in S (95% intervals in Figure 12.1), growing by 0.26 ± 0.28 bits of S_{IC}/S_ρ per bit of r, consistent with the $p^{1/6}$ the floors predict. The tower base costs the same as a plain base of the same size to within 1% (ratios 0.999–1.012): PKM's structure helps an algebraic solver, not a table. An isogeny-tower variant costs 1.00–1.33× the plain base once its auxiliary-curve search is charged, and every arm is 17–137× rho (`research/pkm_isogeny_tower_ecbench_20261015/`); for a P-192-sized tower that search would cost about 2^{74} GAE, or 11.2 core-years, while a supersingular auxiliary curve gives the needed point in 7 ms with the same solving degree (`research/pkm_isogeny_tower_p192_20261015/`, `research/pkm_supersingular_aux_20261015/`). A 30-bit row of 24 October is 158.9× rho (§16.5). Swapping in an A11 presentation could not be priced: ecbench has no method that builds a tower summation system (`research/a11_pipeline_pricing_20261019/`).

12.3 Structured bases on the P-256 field

Twenty-four rounds between 4 and 6 October (`research/icv1-fp256-t89188191154553853111372247798585809583-f188c491_affine_bitbox_degree_round1_20261004/` and its 23 successors; cluster "icv1-fp256") searched for factor bases on the P-256 field whose summation systems have lower degree. They found degree reductions on split systems: Dickson-torus branching lowered the solving degree of components from 4 to 3, image trees for S_9 and S_{17} reached local degree 2, and an indexed quadratic image cut one stage 498×. At the P-256 modulus the images hit their generic size ceilings (width ratio 1.000000), so the degree reduction transferred and the compression did not (round 14). No round found a relation for a target that was not planted: planted length-17 relations were recovered, hash-public targets were not, and a log-transport round found zero relations among 14.4 million images (round 21). Projected end to end, the best schedule is 16.2–19.4 bits above rho in the same unit, after a symmetric projection of $2^{144.625}$ was withdrawn for exceeding its domain (round 20); the final geometric bases are 392–404× rho, and a known-log constant of 0.964× rho in round 24 is a lower bound that the study itself says is not an attack.

12.4 Small-abscissa bases and isogeny classes

Two-point small-x bases have an exact pair yield of $2B^2/(n-1)$ with no pair-sum collisions: on P-192 a base of 8,192 points needs 2^{178} trials for enough relations against rho's 2^{96}, and on all 33 SECG curves the route loses to rho by at least $2^{40.9}$ in trials (`research/prime_small_x_p192_20261014/`, `research/secg_small_x_20261014/`). At $p = 2^{61} - 1$, 1,228,800,000 target trials over 1,024 curves of a horizontal isogeny class found zero relations, while a 20-bit positive control found 33 and recovered the scalar (`research/prime61_factor_base_20261016/`, `research/prime61_disk_extend_20261018/`). Native walkers enumerated 1,064,000 distinct curves isogenous to P-256, 20,000-curve classes of P-224 and P-192, and verified isogenies of degree up to 10,453 (P-192) and 1,471 (P-224); no preregistered structural screen fired on any walked curve, so no factor-base test was ever selected (`research/p256_isogeny_million_20261006/`, `research/p192_p224_large_degree_isogenies_20261009/`). By Lemma 14.1 a homomorphic transfer could change only bounded constants in any case. The class of a 61-bit curve has a certified order of 358,094,304 with adjacency still open (`research/p61_terminal_closure_20261023/`).

12.5 Lattices

Every lattice intrinsic to $(E, P, [k]P)$ has rank 2 (4 over $\mathbb{Z}[\tau]$), and relation lattices have a derived ceiling of $p^{5/6}$ (`research/notes/ecdlp-general/RESEARCH_LATTICE_ECDLP_ROUTES.md`). A Coppersmith reformulation of the $m = 4$ oracle via S_5 reaches $\delta^* = 1/80$ against about $1/5$ needed and is closed (`research/lit2026_w10_s5_coppersmith_20261011/`); on S_3, a frozen falsifier recovered no planted decomposition in any of 96 cells with $\beta \geq 0.20$ at 24–56 bits, so the $\beta = 1/4$ ceiling stands at multiplicity ≤ 3 (`research/coppersmith_s3_theta2_20261023/`).

12.6 ECCp-131

ECCp-131 ($\log_2 r = 130.19$, rho $2^{64.92}$ GAE) has exact parameters wired into the BSGS and IC command paths; a three-point small-x probe found zero natural relations and rank zero at a 128-point base, which blocks any full-parameter IC solution and is an open problem record (`research/eccp131_bsgs_ic_20261020/`, `problems/eccp131-full-parameter-index-calculus.md`). Structured prime bases for the Certicom programme were a null result (N/E 0.969–1.000 against a random control of 0.974). One CPU core walks ECCp-131 rho at 1.23 million steps per second with a 32-bit Montgomery batched walk, and 15.06 million with an AVX-512 eight-lane walk (`research/certicom131_rho_ic_bench_20261022/`, `research/certicom131_rho_parity_20261023/`).

12.7 Audits of standard curves

Several studies examined the NIST, SECG and CryptoPro curves themselves. None found an ECDLP weakness. The NIST prime-curve seeds were regenerated as 1997–98 `drand48` output, with the skip rule and base points still unexplained (`research/nist_seed_provenance_20261016/`, `research/nist_basepoint_provenance_20261017/`); a claimed P-256 $n - 1$ outlier was not established after controls, and the published b sits at the median of every statistic over a million seed samples (`research/nist_cyclotomic_controls_20261011/`, `research/p256_trapdoor_round2_20261012/`). The $j = 0$ SECG curves have sextic twists of weak order (rho $2^{65.5}$ for secp256k1's), a fact about twists, not about the curves (`research/safecurves_trapdoor_slots_20261017/`). Special prime shapes change neither the distribution of $\#E$ nor the embedding degree, so only small-embedding-degree pairing curves open an SNFS-type route (`research/snfs_prime_families_20261023/`, `research/solinas_distribution_20261023/`). A P-192 "BREAKS" verdict applies to a singular nodal cubic, an implementation weakness (`research/p192-weak-curve-20261006/`), and the P-224 directory is an invalid-curve attack on an unvalidated ECDH oracle (`research/p224_attack/`). The CryptoPro-B and CP6-782 work is constructive: GLV-2 scalar multiplication 1.25–1.50× faster via isogeny-chain endomorphisms (`research/cryptopro_b_chain_sweep_20261006/`, `research/endo_native_bench_20261016/`). Cheon's attack with auxiliary inputs beats rho from 24 bits up, but only for the DLP with auxiliary inputs (`research/torsion_auxiliary_inputs_20260918/`).

12.8 Literature claims that did not survive

The lit2026 programme (`research/lit2026_speedup_program/`) ran each finding of a 2025–26 search as a measured change against an interleaved baseline binary. Workstream W12 implemented the zero-minor guess-and-determine attack (Mahalanobis, arXiv 2607.09814v2): it solved 48/48 planted targets on 16–26-bit curves, but its search phases scale with slope 0.905 $[0.769, 1.021]$ in $\log_2 r$ and its cost over rho grows from 260× to 6,715×—exhaustive in r, as the source's own table says. A second flagged ePrint (Hu, 2025/015) did not specify an implementable algorithm and is recorded as blocked. On the generic side, W11 cut BSGS from 16.75 to 3.71 multiplication equivalents per visited point with paired steps (wall 1.15–1.67×), which moves rho's and BSGS's constants, not index calculus.

12.9 Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$

Over $\mathbb{F}_{p^k}$ with a base of points whose abscissa lies in $\mathbb{F}_p$, a relation needs a k-decomposition solved by a Macaulay system. We made the `gaudry_quartic` solve generic in the number of unknowns and faster (median whole solve 1.96/4.19/6.08× at $d = 4, 5, 6$, 37.3 → 6.1 s at $d = 6$, as recomputed by the validation sweep from the committed records on a shared host; the S_5 solve from about 30 minutes to 164–203 s; every count and solution identical). Then we built FGHR's D_4 variant (`research/gaudry_quartic_d4_20261013/`): on a curve with rational 2-torsion T, the coordinate $Y = (x - x_0 - c)/(x - x_0 + c)$ satisfies $Y(P + T) = -Y(P)$, so S_5 in Y is invariant under the Weyl group D_4 (order 192). In the invariants $\sigma_k = e_k(Y^2)$, $\pi = \prod Y_i$, the system has Bézout number 512 instead of 4,096, Macaulay degree 14 instead of 29, and a 13,412 × 11,628 matrix instead of 41,780 × 40,920.

At $p = 269$, the cost C_4 of one decomposition falls from $1.21–1.23 \cdot 10^{12}$ to $2.54–2.55 \cdot 10^{10}$ $\mathbb{F}_p$ multiplications, 47.5–48.4×; a solve takes 4.2–5.6 s instead of 164–203 s. Decompositions agree with an independent meet-in-the-middle enumeration on 19 of 19 solved targets; one target in 20 is degenerate (quotient dimension 517) and is discarded, adding about 5% to the relation phase. The Y-line base decomposes about 1/8 as often (pooled ratio 0.133 over 8,000 targets; prediction $1/8 \pm 30\%$ held on one of two curves), but each column covers $\{\pm P, \pm(P + T)\}$, halving the unknowns.

Inserted into the repository's crossover model (`research/notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md` §11.16), the measured C_4 and rate move the $k = 4$ crossover with rho from $2^{149.4}$–$2^{152.7}$ to $2^{135.1}$–$2^{135.8}$ (cofactor $h = 2$, forced by the 2-torsion), and the asymptotic ratio past it from 0.34–0.63 to 0.12–0.22 of rho. This is the one place in the programme where a crossover moves; it moves in an *extrapolated model* ($n^{1/4}$ relations, $n^{1/2}$ linear algebra and rho) fed by a measured constant, so it is classed as ENGINEERING, not an advance. The validation sweep notes that this crossover was not re-derived against a rho with the negation map; against the negation walk, a later note puts the plain prime-order $k = 4$ handover at $2^{156.1}$ and a ψ-folded one at $2^{146.0}$ (`research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`). The measured characteristic-two $k = 4$ ladder is in §16.3. Extension fields of degree four are not the fields deployed curves use.

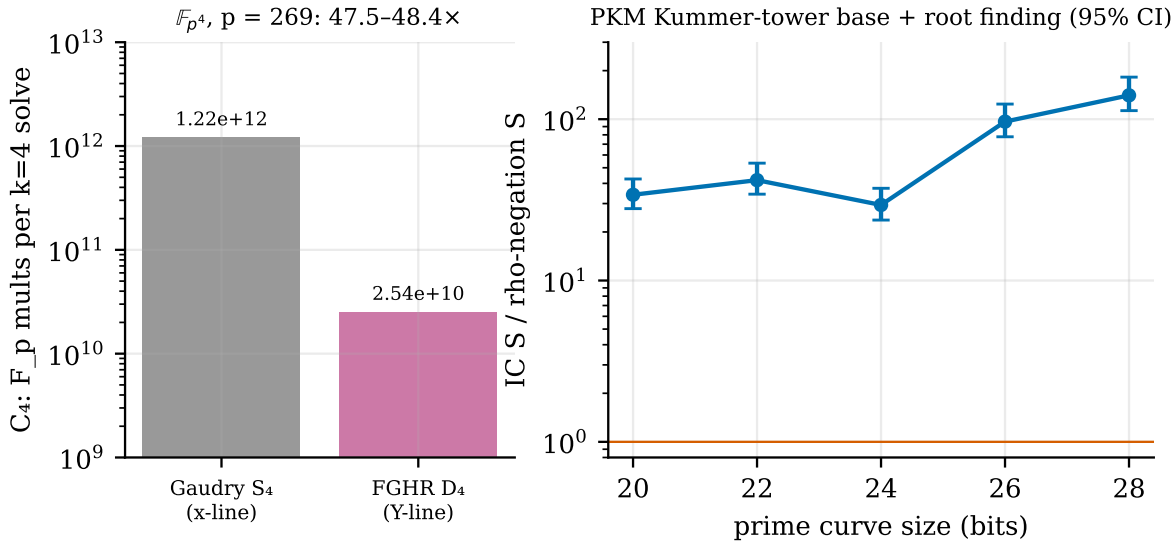


Figure 12.1: Left: cost C_4 of one $k = 4$ decomposition over $\mathbb{F}_{p^4}$, $p = 269$, Gaudry's S_4 system on the x-line against FGHR's D_4 system on the Y-line (`research/gaudry_quartic_d4_20261013/`). Right: index calculus with the PKM tower base and root finding over rho with negation, eight one-target workloads × three rounds per size, paired 95% cluster-bootstrap intervals (`research/pkm_tower_ecbench_20261013/`).

13 Extension fields: $k = 4$ and the D_4 symmetry

Over $\mathbb{F}_{q^k}$ with a factor base of points whose abscissa lies in $\mathbb{F}_q$, Gaudry's index calculus [19] decomposes each target into k summands by Weil restriction of S_{k+1}. It costs $\tilde{O}(q^{2-2/k})$, below rho's $q^{k/2}$ for $k \geq 3$, but with a Gröbner constant that grows very fast in k. Faugère, Gaudry, Huot and Renault [13] show that on curves with rational 2-torsion the system has the larger symmetry group D_k, the Weyl group of type D, in a well-chosen coordinate. At $k = 4$ this chapter reports how far that symmetry moves the measured gap to rho. Chapter 12 (§12.9) gives the odd-characteristic engineering in detail.

13.1 The D_4 system

On E with rational 2-torsion point T, the coordinate $Y = (x - x_0 - c)/(x - x_0 + c)$ satisfies $Y(P+T) = -Y(P)$. In Y, S_5 is invariant under D_4, of order 192. Written in the invariants $\sigma_k = e_k(Y_i^2)$ and $\pi = \prod Y_i$, the system shrinks:

- the Bézout number falls from 4,096 to 512;
- the Macaulay degree falls from 29 to 14;
- the matrix falls from 41,780 × 40,920 to 13,412 × 11,628.

The engineering ladder, each step against the previous incumbent, was:

1. *D_4 invariants.* One decomposition costs $C_4 = 2.54 \cdot 10^{10}$ $\mathbb{F}_p$ multiplications instead of $1.21 \cdot 10^{12}$ at $p = 269$, a factor of 47.5–48.4. All 19 solved targets agree with an independent meet-in-the-middle enumeration (`research/gaudry_quartic_d4_20261013/`).
2. *The Y-line on a complete ladder.* On $p = 101$–1033 (21 curves) the cost is 0.0548 $[0.048, 0.061]$ of plain S_4, at the same exponent.
3. *Sparsest pivots, the weighted grading $\sigma_4 = \pi^2$, and shape replay.* C_4 falls to $6.89 \cdot 10^7$, by 369.8× $[367.9, 371.6]$; all 648 of 648 solves are exact. Ecbench S falls 398–406× (`research/gaudry_quartic_d4_cost_20261014/`).
4. *First-dependency stop.* The cost is 0.979 $[0.973, 0.986]$ of the previous step. Double large primes cut the linear algebra to 0.827× but regress the whole solve, to 1.025×.
5. *A Frobenius-reduced form in characteristic two.* Weights $(2, 2, 1, 1)$ give Macaulay degree 11 and a 384×448 matrix. One decomposition is 481× $[479, 483]$ below the plain form at $m = 8$ (`research/gaudry_quartic_d4_char2_20261015/`).

The D_n symmetry is FGHR's [13]. Invariant rings in characteristic two, including even-power substitutions, were studied by Galbraith and Gebregiyorgis [16]. Whether the Frobenius-reduced weighting is in that work has not been checked.

13.2 Distance to rho

Odd characteristic. Over $\mathbb{F}_{p^4}$, index calculus is still $1.39 \cdot 10^6$× rho at the measured sizes. The modelled handover moves from 2^{150} to about 2^{103}–2^{119}, depending on the frame, but that is an extrapolation through a measured constant.

Characteristic two. Over $\mathbb{F}_{(2^m)^4}$ the gap closes with size. IC over the negation-map rho model is:

m	6	8	10	12	14
$\log_2 n$	24	32	40	48	56
IC / rho model	5,080	648	80.6	49.7	6.06
IC / measured rho	4,427	919	98.9	47.2	4.27

These are first targets; $m = 14$ is a single target. The fitted slope is -0.248 in $\log_2 n$, which is Gaudry's $n^{-1/4}$. The extrapolated crossover was $n \approx 2^{68}$, where rho takes about 2^{34} steps, so it could be tested directly.

The red-team verdict. An adversarial review tested that crossover (`research/gaudry_k4_char2_redteam_20261026/REPORT.md`). Its verdict: **not survived (unsupported), not refuted**.

- *Data.* The only clean point is at $m = 15$, with two targets per arm. The recorded IC/rho there is 6.30. After the corrections that favour index calculus it is 5.2, with range 2.4–15.0 over unit-price variants. The $m = 17$ ratio of 0.72 came from a killed run and is not evidence. No interval is possible with $n = 2$.
- *Rho reference survives.* These are generic ordinary curves with Aut $= \{\pm 1\}$, and the floor 0.886 is about 3.5% generous to rho.
- *Pricing partly survives.* The unit price moves the ratio by 0.64–0.89, with a plausible range of 0.31–3.1. Unpriced rho work moves it by about 0.93.
- *Fitted exponents* (cost in $q = 2^m$): IC relations 1.07 $[1.01, 1.13]$, IC linear algebra 2.03, rho 2.00 $[1.71, 2.29]$. A prime-field cross-check agrees.
- *Corrected crossover.* The IC/rho ratio falls by about $2^{-0.93}$ per unit of m, which puts the crossover at $m^* \approx 17.6$, with range 16.3–19.2: at or above the edge of what was measured.

Linear algebra is 0.4% of the cost at $m = 15$. Basic Gaudry linear algebra grows as q^2, like rho, so the $n^{-1/4}$ trend lasts only while the relation phase dominates. Beyond that point double large primes are required.

The red team froze a plan to settle the question: $m = 14$–17, eight fresh targets per arm, a strong rho with batched inversion and charged canonicalisation, counted and hardware units, about 30 h of wall time. It has not run. This is the cheapest experiment in the programme that could yield a measured crossover of index calculus below rho on an elliptic curve. It would be a crossover in a weak family: $\mathbb{F}_{(2^m)^4}$ is not a field deployed curves use.

13.3 Other extension-field routes

- *Joux–Vitse $(n-1)$-decomposition.* The $(n-1)$-decomposition [25] at $n = 5$ over $E(\mathbb{F}_{p^5})$ completed verified logarithms at $4.5 \cdot 10^7$–$1.8 \cdot 10^8\times$ rho, with a crossover extrapolated to groups near 2^{291} (`research/joux_vitse_n1_20261015/`).
- *Joux–Vitse cover.* The cover of $E(\mathbb{F}_{p^6})$ is in Chapter 14.

14 Covers, isogenies and other transfers

A recurring hope was that some other representation of the same instance—an isogenous curve, a curve at another level of the isogeny volcano, a Jacobian, a Weil descent—would make index calculus cheaper. These routes produced the programme's clearest negatives, and one lemma that explains most of them.

Lemma 14.1 (transport conservation, `docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md` §1)**.** *Let $\varphi : E \to E'$ be an $\mathbb{F}_q$-rational isogeny (or any homomorphism of rational points), $\kappa = \ker\varphi \cap E(\mathbb{F}_q)$, $R \in E(\mathbb{F}_q)$, $\mathcal{F}' \subset \varphi(E(\mathbb{F}_q))$ a factor base on the codomain and $\mathcal{F} = \varphi^{-1}(\mathcal{F}') \cap E(\mathbb{F}_q)$. Then the number of ordered m-decompositions of R over $\mathcal{F}$ is $|\kappa|^{m-1}$ times the number of ordered m-decompositions of $\varphi(R)$ over $\mathcal{F}'$.*

Proof. Images of decompositions of R are decompositions of $\varphi(R)$. Conversely, for $\sum Q_i = \varphi(R)$ choose rational preimages P_i of the Q_i; then $\sum P_i - R \in \kappa$, and of the $|\kappa|^m$ choices exactly $|\kappa|^{m-1}$ sum to R (translate P_1 by the kernel element). □

So a homomorphic transfer changes nothing but three curve-specific inputs: the rational kernel symmetry $|\kappa|^{m-1}$ (bounded by the rational torsion; trivial on prime-order curves), the factor-base geometry available on the destination, and the cost of arithmetic there. Sequences of isogenies compose to one homomorphism; a loop is an endomorphism and acts on the subgroup as a known scalar, useful only as a known-log multiplier. Multi-curve factor bases glued by isogenies are single-curve bases in disguise.

Conductor gaps in the ECC2K-130 class. Jao, Miller and Venkatesan [24] prove random self-reducibility among curves with the same endomorphism ring, leaving open whether curves whose conductors differ by a large prime could differ in hardness. The ECC2K-130 class has conductor gap 263; we computed its structure (`docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md`) and ran several rounds of experiments. Native counted-S index calculus on two fixed CM pairs gave floor/crater ratios 0.975 (99% $[0.603, 1.451]$) at $\ell = 8209$ and 1.127 $[0.922, 1.342]$ at $\ell = 131101$: no level effect (192 verified records, 96 exact replays; `research/ecbench_conductor_gap_20261012/`). Two-torsion symmetrisation is a quotient isogeny: at a matched base of abscissae it cut F4 row-reduction work 4.1–4.3× ($|H| = 2$) and 17.6–19.6× ($|H| = 4$) over 1,128 systems—but a correction the next day showed that at a matched number of *columns* the saving disappears; it was the cost of a redundant orbit base. A combined Weil descent used 234–529× the F4 operations of the blind system (`research/isogeny_conductor_gap_round3_20261011/`).

Fibre-aware index calculus. Isogeny fibres are torsors whose arithmetic is controlled by one cohomology class (`RESEARCH_ISOGENY_FIBERS_ECDLP.md`); relation generation is isogeny-equivariant at zero lifting cost, and a fibre-closed pair table was the best lever of the conductor-gap rounds (12.8× rho in its best cell). Measured against matched rho across sizes at fixed cofactor, the envelope IC/rho ratio grows from 8.4–8.8 to 35.1–37.0 ($h = 4$) and from 13.4–20.1 to 60.4–63.1 ($h = 72$), as $\#E^{1/6}$: the gap widens, it does not close (816 audited records; `research/fiber_ic_scaling_20261011/`).

Other structure. Symmetrised u-frame bases saved 0.84× at $n = 17$ and 0.76–0.77× at $n = 23$, but cost 4.01× at $n = 31$ (`research/koblitz_symmetrised_e2e_20260927/`). GHS and hyperelliptic covers: ECC2K-130 does map to hyperelliptic curves, but the genus that would pay is missing. In odd characteristic the cubic GHS transfer from $E_0(\mathbb{F}_{q^3})$ to a genus-3 Jacobian now runs end to end with double-large-prime index calculus, at about $q^{1.4}$ against rho's $q^{3/2}$, within 1.15× of a measured negation-map rho at $q = 2^{20}$ (§16.3). The isogeny-loop GLV/GLS work produced a measured 1.09–1.15× faster four-dimensional scalar multiplication (`research/isogeny_loop_glv_20261012/`) and a $\sqrt{\text{élu}}$ kernel cost exponent 0.822 $[0.805, 0.838]$; these speed up curve arithmetic, not the discrete logarithm.

Covers, 21–24 October. The Joux–Vitse genus-3 cover of $E(\mathbb{F}_{p^6})$ is now within 1.6–2.0× of a pooled rho constant at $p = 1009$; measuring the per-trial cost on a genus-255 binary Jacobian turned the $N = 130$ GHS counterfactual's modelled margin negative (-1.83 bits); and two censuses bound what covers exist at toy sizes (genus floor 13 for odd-order $E/\mathbb{F}_{q^4}$; a 2^{-l} genus-3 fraction over $\mathbb{F}_{2^{3l}}$). All are in §16.5. None touches a prime field or a prime-degree binary field, where the GHS genus is out of reach.

The published scale of high-genus index calculus. The largest Weil-descent hyperelliptic discrete logarithms in the literature are at genus 31. Jacobson, Menezes and Stein [23] applied GHS to elliptic curves over $\mathbb{F}_{2^{62}}$, $\mathbb{F}_{2^{93}}$, $\mathbb{F}_{2^{124}}$ and $\mathbb{F}_{2^{155}}$ and obtained genus-31 curves C_{62}, C_{93}, C_{124} and C_{155} over $\mathbb{F}_{2^2}$ to $\mathbb{F}_{2^5}$, each with $\#J = 2r$, r prime. They solved C_{62}, C_{93} and C_{124} with Enge–Gaudry (relation collection for C_{124} took 379 CPU-days on a mixed cluster) and only estimated C_{155}, at 26,290 days on one 1 GHz Pentium III. Velichka [40] solved C_{155} ($\#J \approx 2^{155}$) with Vollmer's algorithm and sieving: 136,533 relations, about 1,735 CPU-days in total, 21 days wall-clock, of which 14 days were linear algebra. No genus above 31 appears in that thesis; the journal version [39] was not available to us and is unchecked. Our genus-255 per-trial measurement (§16.5) is a cost sample, not a solved instance, at about eight times the largest published genus. The cover ECC2K-130 would need, of genus about 2^{130} (§10.7), lies about 2^{125} times beyond it.

Genus three against the strongest rho. Rounds three and four of the genus-3 work ran without a rho arm; a 23 October study supplied one (`research/ghs3_vs_rho_20261023/`; it has a frozen protocol and tables but no README, and no review is recorded in it). On round 4's instances, $q = 2^{14}$ to 2^{20}, three curves per size, in round 4's unit and binary, the reference was an ideal negation rho priced with Karatsuba products and 32 walks per inversion (R_{ideal}, the primary reference, the most favourable to rho) and a measured negation walk (R_{meas}). The frozen rule declares "IC below the strong rho" at a size only if every curve is below both references at every weight of the band. At $q = 2^{16}$ it is: IC/R_{ideal} is 0.70, 0.71, 0.70 at $w = 0$ and 0.64 and 0.38 (geometric means) at $w = 0.1$ and 1. At the other six sizes the decision is *unresolved*, with geometric means 0.97–1.52 (1.27 at 2^{20}). Three things qualify the result. The registered unit prices one subgroup multiplication as one $\mathbb{F}_q$ multiplication ($c_\ell = 1$); the protocol's own sensitivity, $c_\ell = 9$, puts every instance at 2.7–12.4× rho. The ratio carries $\sqrt{c}$ of the cofactor, because IC depends on q alone and rho on $\ell = \#E_0/c$: the three curves at 2^{16} all have cofactor 4, and a post-hoc fit gives exponent 0.474 in c (theory 0.5) and -0.081 per bit of q at fixed c (theory -0.167). And the pooled trend over all 21 instances is flat (0.034 $[-0.047, 0.105]$ per bit), so no crossover size is determined. This is the programme's only preregistered decision of index calculus below the strongest rho. It is a measurement of a known pipeline in a structured, weak class ($E_0/\mathbb{F}_{q^3}$ with $4 \mid \#E_0$, never prime order), where theory predicts $q^{4/3}$ against $q^{3/2}$; it depends on the pricing of one operation; and it is not yet reviewed. It updates the ledger paper's "1.15× at 2^{20}" and bears on that paper's prediction of a crossover at $q \le 2^{22}$ once rho is measured: below rho at one size, unresolved at the others, with no measurable pooled trend.

The same family through a plane quartic. The $m = 2$ type of the cubic GHS cover gives a smooth plane quartic over $\mathbb{F}_q$ directly. On a plane quartic, Diem's index calculus is $\tilde{O}(q)$ in theory, against $q^{4/3}$ for the hyperelliptic genus-3 route above. Before 18 October the repository's plane-quartic "pure IC" toys could not be priced: their logarithms came from enumerating the subgroup, and the degree-3 class keys merged distinct classes. A canonical Pic^0 arithmetic now gives every class a unique key (checked against brute-force class counts on 21 curves and full Cayley tables at $q = 5$). A generic addition costs 3,523 multiplications and 44 inversions, flat in q up to $2^{31} - 1$ (`research/quartic_pic_canonical_20261018/`).

On that arithmetic, a pipeline with relations on lines, a reduced factor base with one large prime, Lanczos modulo ℓ and canonical descent solved and verified 51 of 51 units. It ran on the GHS-weak family $E_0/\mathbb{F}_{q^3}$ with $4 \mid \#E_0$, for $q = 223$ to 262,111 (`research/quartic_n4_20261019/`).

- Its reference is a matched *plain* rho, without the negation map, on the same subgroup.
- At $q = 262{,}111$, index calculus costs 0.122 $[0.10, 0.16]$ of that rho in $\mathbb{F}_q$-multiplication equivalents. The order holds across the pricing band for every $q \ge 4051$.
- The fitted exponents on $q \ge 4051$ are 1.083 $[1.040, 1.129]$ for index calculus and 1.485 $[1.437, 1.525]$ for rho.
- In wall time, index calculus is cheaper only above $q \approx 3 \times 10^4$.

This was a preregistered decision of index calculus below a matched *plain* rho. Its successor, N4 v2 (`research/quartic_n4v2_20261020/`, 20–25 October), took the same family and instances from $q = 2^8$ to 2^{22} (63 instances, 501 index-calculus units and 2,442 rho runs, all verified) and added three changes: double large primes (D), structured Gaussian elimination with Lanczos (L), and a negation-map rho. Against the frozen pipeline B0 at $q = 2{,}097{,}091$:

- D costs 0.081 of B0 and the combined arm DL 0.042 $[0.042, 0.042]$; a post-hoc arm DL1 reaches 0.022.
- DL's fitted exponent is 0.983 $[0.975, 0.993]$ over the full range and 0.988 on $k = 17..22$, which is the $\tilde{O}(q)$ of Diem's analysis [8, 10]; negation rho fits 1.543 $[1.494, 1.583]$.

- DL over negation rho is 0.005 $[0.004, 0.008]$ at the top size, with an interpolated crossover at $q^* = 555$ $[389, 752]$ in counted units ($q^* \approx 1{,}889$ in CPU time). At $q = 4{,}194{,}271$ one solve took 12.3 CPU-seconds against 616 for negation rho.

The qualifications are substantial. The host was L0 (load average 22–60); the intervals resample only three units per size at $k \geq 17$; the conversion constants move D's high-range exponent from 1.009 to 1.048; the frozen B0 records do not replay at any of 11 replayed sizes; the negation rho fails the calibration rule at three sizes; and there is no validation file. The program's closing note calls the result "our measurements of a known result" (`research/plane_quartic_program_20261018/CLOSING.md`). It covers the GHS-weak family only; nothing here is about a generic curve, a prime-order curve or a standard curve.

Smith's isogeny and the walk to it. The other genus-3 type of the census is hyperelliptic. Smith's $(2, 2, 2)$-isogeny takes it to a plane quartic when his Propositions 2 and 4 pass. The walk study looked for curves on which Smith's construction succeeds by walking 2-isogenies on the elliptic side (`research/smith_elliptic_walk_20261018/`).

- Exhaustively at $p = 7$ to 31, 0.53–0.61 of the descended curves pass Smith's full test, against 0.1857 for random curves. The preregistered expectation of about 0.186 is falsified, upward.
- A failing curve has a passing one in its 2-isogeny component in 0.74–0.98 of cases, at a mean distance of 1.5–3.6 steps.
- On sampled walks at $p = 2^{10}$ to 2^{20}, the frozen fit labels the walk cost $o(p)$, with exponent -0.14 $[-0.80, 0.95]$.
- A labelled post hoc analysis attributes that label to censored walks at the small primes. Successful walks at $p \equiv 3 \pmod 4$ scale as $p^{1.13}$ $[1.00, 1.26]$ in cost, which reads as $\tilde{\Theta}(p)$.
- At $p \equiv 1 \pmod 4$ most walks stay inside small trapped sets.

A second study, also open and not yet merged, maps divisor classes through Smith's isogeny and runs the whole route at toy sizes. The program's closing note lists what stays open (`research/plane_quartic_program_20261018/CLOSING.md`).

Joux–Vitse: two arms of one cover. The Joux–Vitse cover of $E(\mathbb{F}_{p^6})$ (full rational 2-torsion, a weak class) has been measured with two solvers, and their comparisons with rho do not agree. The *plain Nagao arm* with a sparse-FGLM linear stage is 1.59–1.98× a pooled rho constant at $p = 1009$ and closes as about p^{-2} (`research/jv_cover_p_scaling_20261024/`; §16.5); it is a lower-bound comparison and current. The *sieve arm* (`jv_cover -exp sieve`) measured 0.024 $[0.019, 0.031]$ of a strong rho at $p = 1009$ and 0.0042 at $p = 1823$ on 19 October, but its 20 October review withheld every comparison (no numeric acceptance for the controls, uncharged rho hashes, incomplete accounting; `research/jv_sieve_strong_rho_20261019/`); a replication with fixed thresholds failed its acceptance rule and is in an open pull request, and a prospective re-run is registered and has not run. A 9 October whole-route replay from a non-weak challenge curve, through an isogeny walk to a weak curve and its cover, read 0.058 of a pooled, unmatched rho at $p = 1009$ on one seed per prime, with an earlier revision of the solver whose source includes the sieve module (`research/jv_cover_e2e_30bit_20261009/RESULT.md`). The roughly thirtyfold gap between the arms at $p = 1009$ is unexplained on `main`. The cover-decomposition note still states a measured sieve crossover near $p \approx 430$ (`research/notes/index-calculus/RESEARCH_COVER_DECOMPOSITION_LEDGER.md` §11) with no erratum; it should be read through the review. Leaderboard Table D, the historical study, is withheld. On the method side, the sparse-FGLM stage made the plain arm 1.28× cheaper end to end (`research/jv_cover_sparse_fglm_20261018/`); relation collection is 99.99% of S (`research/ic_cover_leads_20261018/`); depth-1 isogeny classes hold no weak curve over 4,871,062 curves, while 86–95% of deeper classes hold one (`research/jv_weak_class_law_20261007/`); and the $(n-1)$-decomposition variant is $4.5 \cdot 10^7$–$1.8 \cdot 10^8$× rho (`research/joux_vitse_n1_20261015/`).

Cover censuses. The minimal cubic GHS genus over an isogeny class follows $\#E(K) \bmod 4$: 13 for odd order, 5 for 2 mod 4, 3 for 0 mod 4, and genus 3 is the only value below rho's $q^{3/2}$ (`research/ghs_cubic_cover_census_20261012/`); at $n = 5$ the floors are 33, 33, 25 and 5 by 2-torsion type (`research/ghs_cover_census_n5_20261013/`). No deployed prime-field or prime-degree binary curve has a usable subfield; only the RFC 2409 Oakley groups 3 and 4 are reachable, at genus 15–16, where exponent models disagree in sign and the measured genus-16 relation phase alone is 5.2 bits slower than rho (`research/cover_route_prime_binary_20261008/`, `research/ghs_oakley_extended_20261023/`). All 15,791 cover maps in the registry were assessed: 165 binary maps have genus-3 arithmetic and the 15,626 odd-characteristic genus-2 covers lack a Jacobian implementation (`research/cover_trans`

fer_assessment_20261013/). An inventory of about 90 cover and decomposition studies found every route on prime, Koblitz and prime-degree binary curves closed, negative or constant-factor (research/cover_decomposition_gap_search_20261018/). Smith's $(2, 2, 2)$-isogeny route was reproduced natively (full success 0.18638 against a theoretical 0.18574 over 10^6 curves) and its sign bias shrinks like $1/\sqrt{p}$ (research/smith_prop24_native_20261014/, research/smith_sign_scaling_20261016/).

Volcano levels and generators. A 403,168-run census found the level effect equivalent in 33 of 35 comparisons and different in none (research/volcano_ecdlp_census_20261017/), and a large-sample rho study found the floor/crater ratio 0.998 $[0.986, 1.010]$ (research/conductor_gap_rho_large_sample_20261018/). The one real level effect is the automorphism group: $\sqrt{3}$ on the $j = 0$ crater, confirmed at 1.86–1.92× (research/drastic_prime_conductor_gap_round3_20261012/). At $\ell = 2$, floors give 5.8% fewer relations than craters, and 99% of the contrast is explained by the base's cofactor classes (research/isogeny-volcano-cost-v2_20261017/, research/generator_effect_ic_20261019/).

Other transfer lines. Index calculus on genus-2 to -4 Jacobians over $\mathbb{F}_p$ (the ADH–Gaudry attack, measured against rho on the same Jacobian) read below a distinguished-point rho in early rounds, but the reference was later found slow and the ratios are lower bounds, unranked under the band (research/notes/index-calculus/RESEARCH_HYPERELLIPTIC_IC_RHO.md); later rounds are in review. A weak-curve atlas of 922 exact models and 36 published cases ranks covering-model and pairing signals but leaves the selected-subgroup advantage unmeasured (research/weak_curve_leaderboard_20261012/); three complete isogeny volcano graphs (24,823 vertices) were built and their graph-wide IC surveys halted at checkpoints (research/full_volcano_graphwide_ic_20261017/). Weil restriction of ECC2K-130 to extension fields costs at least 2^{131}, $2^{70.19}$ times rho (cluster "transfer-boundaries").

Named obstructions. Blocked requirements are kept open as problem records (problems/README.md), among them the complete-cost calibration of root-finding IC (problems/rootfinding-complete-cost-calibration.md), with exit criteria rather than abandoned.

Part IV

The frontier

15 The summer record: July–September 2026

The October campaign measured in the rest of this part did not begin from nothing. Between 19 July and 30 September the crypto-autoresearcher ledger recorded 1,402 coordinator decisions on ECDLP and index calculus:

- July: 74 decisions on 20 experiments, 977 run directories;
- August: 364 decisions on 92 experiments, 321 run directories;
- September: 964 decisions on 449 experiments, 776 run directories.

Each decision names a verdict, the hypotheses and experiments it rules on, and the evidence it cites. The dated index (`research/ecdlp_jul_sep_index_20261029/`: `INDEX.json`, `INDEX.tsv`, and the generator `build_index.py`) lays these out together with the run-data path of each experiment. A decision is counted as ECDLP work if its goal is one of 19 ECDLP goals or, when it names no goal, if a target or evidence identifier carries an ECDLP family token. That scope rule is a choice and is stated in the generator. Seven decisions filed under late-September names carry October dates and are left out of the counts above. Twenty-three decision files did not parse and are not indexed.

Most of this activity was process (Figure 15.1). Of the 1,402 decisions, 561 approve a contract, 267 revise one and 124 expand one. About 157 carry an outcome on a hypothesis (support, reject, weaken, inconclusive, supersede, pause or replicate). This chapter reports those outcomes by theme, with decision identifiers so that each can be traced. All results are at toy or derivation tier unless a scale is given. None moved an exponent on a prime field.

15.1 July: the first baseline and two retracted wins

The ledger's first ECDLP decision, `DEC-20260719-001`, expanded a toy Semaev S_3 Gröbner study (`EV-SEMAEV-001`). It had twelve runs at 8–12 bits with a factor base of 14 points. The degree proxy stayed at most 4 and the wall time stayed flat near 0.16 s. The decision itself noted that three bit sizes and two seeds cannot establish a trend.

Two supports followed on 26 July, and both were overturned the next day:

- *Index calculus beats enumeration* (`H-IC-001`, 748 runs at 8–36 bits) was weakened by `DEC-20260727-008`. The criterion had been revised after the outcome. There were no decompositions at 20 bits or above, so the measured descent time was the cost of a failed attempt. The relation cost was 2287× the group order at 36 bits. The decision records: "no asymptotic exponent moves".
- *A structured factor base induces sparsity* (`H-STR-002`) was weakened by `DEC-20260727-009`. Three headline instances had no run records. The one-variable ablation gave $\alpha = 27,52,197,386$ against $27,52,199,387$ for a random base, so the factor base induced nothing.

In the same week a generic-group classifier (`EXP-GGM-001`) had its support revoked (`DEC-20260727-007`). It was a lookup table, and its nine runs were byte-identical. Fixed-curve preprocessing showed no robust gain over 72 runs (`DEC-20260726-001`, paused). A subresultant degree exponent was measured at $\beta_{\deg} = 0.5985$ against a gate of 0.3 (`DEC-20260728-003`, weakened).

Xedni-style lifting was closed in exact form (`DEC-20260724-021`, reject_scoped): $P_{\text{lift}} = (1 - 2/p + p^{-3})/(2p^3)$, so $\alpha = 3$. This is the result restated in Chapter 7.

15.2 August: yield, monodromy, oracles and transfers

Yield matches the heuristic from below. Semaev decomposition yield at $m = 2$ reached 0.80, 0.98 and 1.00 of the $B^m/(m!N)$ heuristic at $p = 1009$, 4001 and 9001 (`EV-YIELD-e1adbf`). `DEC-20260804-9da614` weakened super-heuristic yield at $m = 2$–4. Structured bases gave small-x 0.845, arithmetic progression 0.658 and random 0.563, with no super-heuristic yield (`DEC-20260804-2bfc30`). These measurements came before the conservation proposition of Chapter 5, which explains them: at matched base size the mean is fixed. A complete $m = 2$ index calculus solved a discrete logarithm at $p = 1009$ (`DEC-20260804-db75bb`). A Fourier-flatness constant for the factor base was fitted as

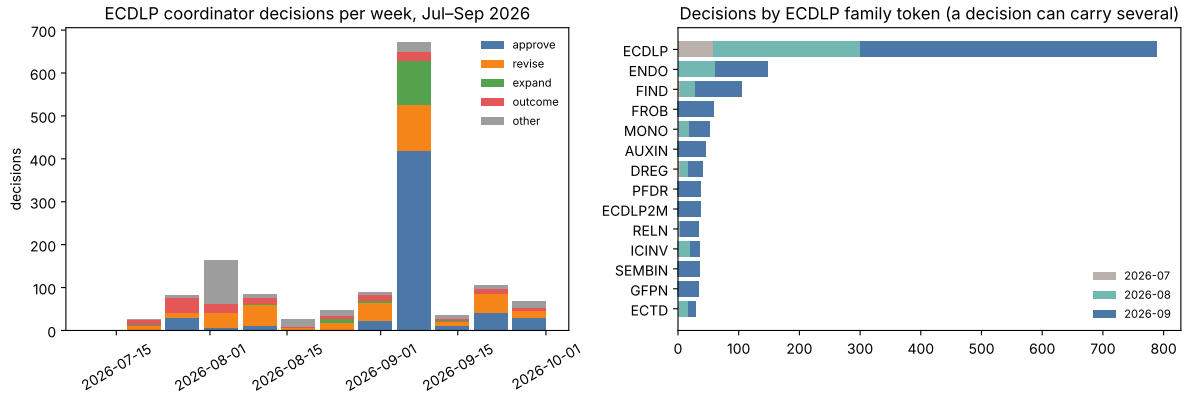


Figure 15.1: ECDLP coordinator decisions, July–September 2026. Left: decisions per week by verdict class. The class *outcome* groups support, reject_scoped, weaken, inconclusive, supersede and pause. Right: decisions per family token, by month; a decision can carry several tokens. The week of 7 September holds 672 decisions, most of them approvals of frozen stage-0 contracts.

$C(p) \sim p^{0.055}$–$p^{0.079}$ and recorded as an open problem (`H-PSEUDO`). Eight proof routes were closed as non-proving.

BKK and meet in the middle: constant factors. A BKK sparsity argument promised a speedup of at least $(m+1)/2$ (`DEC-20260804-1825ea`). It measured 1.99988 against 2.0 at $m = 3$ on a smoke test. `DEC-20260806-26c0e8` then found the multi-target window empty at the planned sizes and the gain about 1.0–1.8×. Meet-in-the-middle structure gave 18×, which rho dominates (`DEC-20260808-030b1f`). A degree-split claw lost its advantage to a matched null, $R = 0.0472$ against 0.0311 (`DEC-20260801-005`). The decision reads: "the instrument, not the hypothesis, has been refuted".

Monodromy. The work that became Chapter 5 was done in August:

- the $m = 3$ derivation (`DEC-20260802-a51c82`);
- the all-m theorem, supported as scoped and "not closed" (`DEC-20260807-64df6a`, `KN-FIND-a8990a`);
- containment in $(\mathbb{Z}/2)^{m-2}$ at $m = 4, 5$ on eight curves, with zero exceptions in an exhaustive census of 2,915,514 triples at $p = 103$ (`DEC-20260830-0f8def`);
- monodromy D_4 for the $m = 3$ symmetric-base cover (`DEC-20260830-1fb2e3`).

The fibre census was ruled blind to relation rate as an instrument (`DEC-20260810-2f86db`, reject_scoped), two months before the census corollary proved why.

Oracles and transfers. Two analysis-only results bounded the oracle leads. Oracles C and D are simulable in the generic group model, and A and B give no sub-birthday advantage (`DEC-20260804-3b4258`, support_scoped). The doubling oracle was then shown equivalent to an x-coordinate oracle (`DEC-20260806-08b9ed`). Three transfer and invariant routes were closed:

- Isogeny transfer: the anomalous, MOV and Weil-descent families are invariants of the isogeny class (`DEC-20260804-2fae6a`).
- Elkies augmentation: with $\ell \leq 23$ it matched a random-bijection null to within 0.004 in all eight cells at 20–28 bits (`DEC-20260804-d0c452`).
- CM: complex multiplication changes the lifted-height obstruction "by exactly nothing" (`DEC-20260810-2c7e3c`).

Within one isogeny class the over-dispersion of ECDLP-relevant statistics traced to the 2-torsion type (`DEC-20260811-a3398a`, `KN-FIND-ddd982`).

Degree of regularity. The boolean chained $m = 3$ system was run at $n = 21$, $D = 5$, and stopped at rank 265,950 of a predicted 268,674 (`EV-DREG-4500cd`). The record calls this "not evidence about d_{reg} in either direction". All six $n = 15$ runs then exceeded memory (`DEC-20260828-653923`). The certified fall-degree results of Chapter 6 later replaced this line.

Late August was spent on governance: budget ceilings, pauses and zero-run successors. `DEC-20260820-0c8ffb` paused at the budget ceiling. The "fund every goal" decisions of 28 August

lifted the limits.

15.3 September: tables, lifts, audits and the first binary costing

Distinguished-point tables re-selected from the batch. The claim that a table of $T/2$ entries re-selected from solved walks matches a static Bernstein–Lange table of T was contradicted at $a = 1/4$. It reached 0.65–$0.69\,T$ at steady state, with intervals excluding 1, while paying 2.6–$8.9\,T$ of working storage (`DEC-20260906-1f5b71`, weaken: "a tradeoff point, not an improvement"). At $N = 2^{24}$ the feasibility crossing was then bracketed between $a = 1/8$ and $a = 3/16$ (`DEC-20260907-9b1b02`, `DEC-20260907-6d9427`). The result is a ceiling only, with no online arm (`DEC-20260910-34e654`).

The canonical lift carries no logarithm. The p-adic lift statistic D_1 took exactly the values $\{0, 8\}$ on 1,035 pairs at $p = 1009$, and the same in a third independent implementation and on a second instance. It tracked precision $r = 4, 8, 12$ exactly (`DEC-20260903-256c7c`). The digit identity held in 1360/1360 cases, and the Teichmüller contrast showed no inflation: all three ratio intervals contain 1 (`DEC-20260911-74c9dc`, `DEC-20260912-2a626b`). Both results were recorded as "no attack".

Natural statistics are not biased enough. An L^1 Fourier registry over 406 cells and primes 4099–16,777,259 falsified two of its four predictions (`DEC-20260921-4516a0`, weaken):

- interval statistics: $\lambda = 0.0806 \pm 0.0020$, against a prediction of at most 0.05;
- cosets and quadratic residues: 0.50, against a prediction near 0.

Semaev degree and the Semaev 2015 / Nagao audits. The successful-subset loophole in the Semaev degree law was closed: $\beta_{\text{work}} = 0.797$ worst case over 36 cells and 1,152 queries (`DEC-20260929-780751`, reject_scoped, `EV-SDEG-4a1c3d`). The audits of Semaev's 2015 paper and Nagao's programme (Chapter 6) recorded these verdicts:

- eq. (11)'s λ is right to 1% at $t = 2$, but $P = 1 - e^{-\lambda}$ fails by 18–22% in the low-degree window (`DEC-20260913-3b51c8`);
- the typed-image ratio is not $m!$ at any of seven cells (`DEC-20260913-2386b5`);
- trace-parity confinement explains the whole deficit at $t = 2$ (`DEC-20260913-92aa48`);
- Nagao's memory claim fails at $n = 571$, and 81 of the 120 flagged cells were an artifact of an incoherent memory pairing (`DEC-20260914-537000`);
- a published measurement claim, P3, fails under the corrected XOR product: the Semaev median is 2 in all eight cells, with the null at least one above it (`DEC-20260929-e37785`, reject_scoped).

Prime-field degree versus yield. A closed form for the first fall degree of a digit-map presentation, $d_{\text{ff}} = m2^{m-1} + \lfloor (s - 2^{m-1})/2 \rfloor + 1$, survived 26,000 instances with zero deviations. Its companion fall-dimension clause was refuted by a certificate at $p = 13$ (`DEC-20260904-1e27a2`). The generator degree is $m2^{m-1}$, not $2m$, and the "cells below rho" of a cost table had zero rows at D_0, so they were rejected (`DEC-20260904-d47cd2`).

Frobenius-stable factor bases. The stable subspaces are exactly the ideals of $\mathbb{F}_q[T]/(T^n - 1)$, and the orbit count is exact on 14 of 14 cells (`DEC-20260922-5796b8`, `EV-FROB-d336b0`). The orbit gain is n, a constant factor. At $n = 131$ there are only four stable subspaces, and the orbit trick *loses* 57.0–96.0 bits for $m = 2$–5; at $n = 163$ it loses 72.7–121.7 bits. Whether WDSat is blind to basis choice was never measured, because 19 stage-0 launches failed.

The first costing of descended systems against rho. The ICPERF lane priced the descended algebraic normal form against the whole rho budget on standard binary curves (`DEC-20260922-925da8`, `EV-ICPERF-784b25`). Forming the system alone exceeds rho by +6.3 bits at $n = 233$ and by +40.8 bits at $n = 131$ (ECC2K-130). Below $n \approx 300$ the obstacle is building the system, not solving it. The excess turns negative at $n = 409$ and 571 for $m \geq 4$, and $m = 2$ loses at every degree. This is the first September result priced against rho. It is a model with measured inputs, not a solve.

Lanes without a result. Several September lanes produced approvals and redesigns but no measured outcome:

- SAT on summation systems, trapdoor detection, invariant calibrators and endomorphism decomposition, mostly frozen stage-0 contracts approved on 13 September;
- the auxiliary-input census, whose run was ruled invalid as evidence because its controls were inadmissible and its escalation was generic to 256-bit primes (`DEC-20260923-ce06a6`);
- EcGFp5 symmetrisation, where no $m = 5$ ideal degree was measured (`DEC-20260923-d86ff2`), although the audit found that the curve's self-reported 323 bits recomputes to 322;
- characteristic-2 quasi-subfield work, which found zero violations of its bound in about 340,000 candidates and produced the derivation $\beta > 1/2$ for every completely splitting L (`KN-FIND-617d78`).

The certified binary refutations of Chapter 6 begin here: W_4 refutes 62/62 at $n = 17$ and 400/400 at $n = 19$ (`DEC-20260924-e61f3b`, `DEC-20260926-901ea1`).

15.4 What the summer established

Four patterns run through the 1,402 decisions:

1. *No measured improvement over rho survived review.* Every July and August support that claimed a gain was weakened or revoked within days. The surviving gains are constant factors at toy scale.
2. *The theory of Chapter 5 was found by measurement first.* The yield ratios at or below 1, the census ruled blind, and the exhaustive monodromy checks all preceded the proofs that explain them.
3. *Most closures were instrument repairs.* A large share of the weaken and reject decisions say that a control, a metric or a criterion was wrong, not the hypothesis. This is the origin of the matched-null and replay rules of Chapter 4.
4. *The binary line reached a priced verdict only in late September.* The ICPERF costing and the W_4 certificates opened the fall-degree and Koblitz work that the October campaign measured.

These four patterns are the author's reading. The decisions cited above are the record.

16 What improved, and where the frontier stands

16.1 The measured frontier: one campaign, every cost priced

The frontier synthesis of 13 October assembled ratios from studies that used different curves, sizes and accounting. To replace it we ran one campaign (`research/ic_frontier_campaign_20261015/`, protocol frozen before any measured run). Every method family ran on the same fixed curve ladders: eight Koblitz curves chosen by subgroup size ($\log_2 r \approx 11$–39, $n \in \{13, 17, 19, 23, 37, 41, 43, 47\}$), six NIST-shaped $a = -3$ prime-order curves of 20–40 bits, and the five 2-adic curves of the Kummer-tower ladder. Each curve had 16 public hash-to-subgroup targets nobody knows the logarithm of, 2 measured rounds and a warm-up, and every answer was checked outside the solver. That is 4,848 executions, 3,232 of them measured: 3,178 verified and 54 exhausted, all of the latter in the shared-rank arm at $n = 41$ and $n = 47$. Every session is audited by re-executing every measured run and requiring identical counts. Index-calculus parameters (compact-orbit columns, negfold subspace dimension, prime base size) were chosen per size on a disjoint set of training targets, with grids widened until the optimum was interior. Every cost is reported across the pricing band of §4.3.

What the campaign shows.

- *No index-calculus arm beats generic rho on any curve, at any weight.* Signed-Frobenius rho is below every index-calculus arm on all eight Koblitz curves and across the whole band; rho with negation is below every prime-field arm on all eleven prime curves (Table 16.1). The one unresolved pair in the campaign is compact orbit against the *strong* rho reference at $n = 13$, whose scalar-multiplication set-up dominates on an 11-bit subgroup; the cheaper signed-Frobenius walk is below both.
- *Compact orbit is the closest index calculus.* It sits 3.8–15× the best rho with native work free and 4.9–17.8× at $w = 0.1$; the ratio does not shrink with size.
- *Prime-field index calculus falls further behind with size.* The two-summand pair-table arm goes from 10.5× to 101.8× rho (at $w = 0.1$) between 20 and 40 bits, the $r^{1/6}$ widening its $r^{2/3}$ cost against rho's $r^{1/2}$ predicts; the Kummer-tower arm is 28–159× rho.
- *Negfold root finding stops scaling.* It is 12–45× rho up to $n = 23$, about 1,900× at $n = 37$, and does not complete within $5 \cdot 10^7$ trials from $n = 43$ on (PROTOCOL A4b).
- *Inherited F4 is not competitive once its work is priced*: S rises from 16 (word XORs free) to about $2 \cdot 10^4$ at $w = 0.1$ at $n = 13$. The band weighs one 64-bit word XOR like a table lookup, which overstates F4 (its pinned calibration is about 0.002 GAE per XOR); even with XORs free it is above rho.
- *BSGS has the lowest time on several curves* because it spends $\sqrt{r}$ memory; it is reported on the time–memory Pareto set, not ranked against constant-memory methods.

Cost exponents. Fitting total cost $\propto r^{\alpha}$ per arm over each size tier with `ecbench bound fit -weight` (two-stage bootstrap, 95% intervals; a tier needs at least four sizes for an exponent):

- prime pair table $\alpha = 0.620$ $[0.557, 0.663]$ at $w = 0$ against rho with negation 0.435 $[0.331, 0.501]$; separated at $w = 0$ and $w = 0.1$, overlapping at $w = 1$ (0.594 $[0.482, 0.665]$ against 0.444 $[0.334, 0.517]$). Index calculus grows faster than rho (the predicted 2/3 against 1/2), but under the band rule the difference is not established at every weight;
- Koblitz compact orbit, $n = 37$–47: 0.589 $[0.518, 0.743]$ against signed-Frobenius rho 0.436 $[0.358, 0.508]$ at $w = 0$, overlapping at $w = 0.1$ and $w = 1$; on $n = 13$–23 the two overlap at every weight (0.25 against 0.29 at $w = 0$, both with wide intervals because the automorphism count $2n$ changes with the size);
- Kummer tower 0.717 $[0.358, 0.940]$ against 0.438 $[0.332, 0.599]$, overlapping.

No index-calculus arm has a fitted exponent below rho's at any weight, so no crossover with rho is supported by these data, extrapolated or otherwise. The fitted rho exponents (0.435–0.444) lie below the theoretical 1/2, which reads as a toy-size effect (fixed per-run overheads and the size-dependent automorphism count) rather than a property of rho; so the measured widening of the gap is a fit on

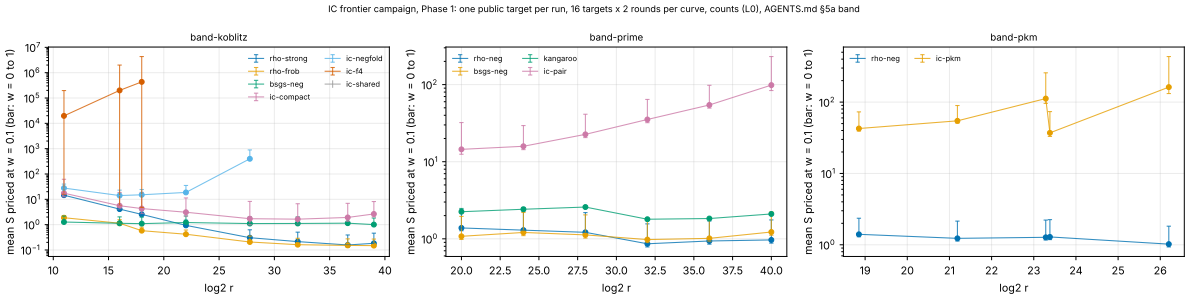


Figure 16.1: The campaign frontier: mean S per arm and curve at $w = 0.1$ GAE per native operation, the bar spanning $w = 0$ to $w = 1$; one public target per run, 16 targets × 2 rounds per curve, operation counts. Sources: `research/ic_frontier_campaign_20261015/analysis/band-*.json` (`ecbench band`).

Table 16.1: Index calculus over the cheapest constant-memory generic arm on the same curve (signed-Frobenius rho on Koblitz curves, rho with negation on prime curves), mean S ratio at three band weights. Every run verified (32/32 per cell). Source: `research/ic_frontier_campaign_20261015/analysis/ratios_to_rho.tsv`.

Curve	$\log_2 r$	Arm	$w = 0$	$w = 0.1$	$w = 1$
Koblitz $n = 13$	11.0	compact orbit	6.9	9.3	27.1
		negfold	14.4	14.8	17.5
Koblitz $n = 17$	16.0	compact orbit	3.8	4.9	11.8
Koblitz $n = 19$	18.0	compact orbit	4.9	7.5	23.6
Koblitz $n = 23$	22.0	compact orbit	5.5	7.4	18.4
Koblitz $n = 37$	27.8	compact orbit	5.2	8.3	23.3
		negfold	1806	1927	2512
Koblitz $n = 43$	32.1	compact orbit	7.4	10.2	23.1
Koblitz $n = 47$	36.6	compact orbit	9.9	12.7	25.3
Koblitz $n = 41$	39.0	compact orbit	15.0	17.8	30.3
prime 20 bit	20	pair table	9.8	10.5	13.5
prime 24 bit	24	pair table	12.1	12.2	12.9
prime 28 bit	28	pair table	18.6	18.6	18.9
prime 32 bit	32	pair table	40.9	40.9	41.1
prime 36 bit	36	pair table	58.1	58.0	57.5
prime 40 bit	40	pair table	95.2	101.8	131.4
PKM 20 bit	18.9	Kummer tower	30.3	30.4	31.0
PKM 28 bit	26.2	Kummer tower	141.3	158.7	238.7

small sizes, and the conclusion that it widens as $r^{1/6}$ rests on the generic-oracle bound, not on these fits alone. The sixty-three fits are in `research/ic_frontier_campaign_20261015/analysis/bounds/`.

Limits. Operation counts only (isolation L0); wall time and a second-host replay certificate need the planned RunPod phase. Parameters were selected on four training targets per size. The shared-rank arm leaves three categories of work uncounted and is reported as a lower bound only. Fitted cost exponents per arm and tier are in `research/ic_frontier_campaign_20261015/analysis/bounds/`.

16.2 The engineering record

Table 16.2 collects the measured improvements of the programme on the boundary each was measured on. Ratios are baseline/candidate; "stage" means one oracle call or one solver run; "complete" means a verified discrete logarithm with the stated phases.
Derived entries: $53.0 = 1/0.018861$; $17.5 = 1/0.057047$; $1.21 = 1/0.828$ with interval from $[0.783, 0.863]$; $18.3 = 1/0.0545$ with interval from $[0.047, 0.064]$; $1.77 = 300{,}021.9/169{,}069.8$; $1.20 = 1/0.830$ with interval from $[0.697, 0.962]$; $1.87 = 18.714/9.985$ (one session; the compact arm's S is a lower bound, so the ratio is an upper bound); 1.69 and 25.2 compare means from different sessions with different seeds and are descriptive; F5 and F6-IC ratios are reciprocals of the recorded candidate/baseline word-XOR ratios.

Table 16.2: Measured improvements, July–October 2026. Class: E = ENGINEERING, A = ACCOUNTING. Boundary: St = stage, Cx = complete solve outside ecbench, Cold = ecbench cold one-target S, On = online window after retained set-up. Sources: Appendix A.

Change	Bnd.	Cl.	Ratio	Scope
Decomposition oracle: root finding				
S_3 roots, binary pair table vs matrix F4	St	E	383–1,535	$\mathbb{F}_{2^9}$, $\mathbb{F}_{2^{13}}$, $m = 2$
S_3 roots, prime vs `groebner_f4`	St	E	3.5–4.7·10^4	16–24-bit, $\lvert\mathcal{F}\rvert = 6$
S_3 roots vs Buchberger, NIST curves	St	E	3,021–5,802	P-192…P-521
`rf-mitm` vs PS-0 oracle	St	E	3,007–115,160	binary $\ell = 8..14$
Root-finding DLP vs msolve DLP	Cx	E	50–105	16–24-bit prime
Pair-table DLP vs F4 DLP	Cx	E	122	binary $n = 23$ (F4 wins $n \leq 13$)
Pair table vs subtraction	Cold	E	53.0	20-bit prime (bounded)
Frobenius pair table vs subtraction	Cold	E	17.5	Koblitz $n = 17$, priced
Root-finding K13 arm, first to best	Cold	E	1.69	38.82 → 22.97, across sessions
Windowed $m = 3$ scan	Cold	E	1.21	$[1.16, 1.28]$
Negfold root finding vs XOR-priced F4	Cold	E	18.3	$[15.6, 21.3]$, K13
Dense root index K1798 vs K1500 (probes)	On	E	1.77	$n = 83$, one point
Pair-table size 64 vs 32 abscissae	Cold	E	1.20	$[1.04, 1.43]$
Gröbner engines				
Inherited vs matrix F4 (matrix work)	St	E	198.6–222.1	K13, R13
F5 vs matrix F4 (word XORs)	St	E	1.55–1.56	K13, R13
F6-IC vs inherited F4 (word XORs)	St	E	1.22–1.30	$m = 13, 19$
`f4_fp` new vs previous	St	E	29.8–54.1	square systems, = msolve
Four-Russians routing	St	E	1.75–3.22	oracle matrices
Eager vs lazy inherited rewrite	St	E	1.40–1.62	dim 8–9
Quotient-curve $m = 2$ model	St	E	1.34–1.64	five cells
Factor bases and symmetry				
Compact orbit vs full orbit	Cold	E	≤ 1.87	Koblitz $n = 17$; compact is LB
FGHR D_4 vs Gaudry S_4, C_4	St	E	47.5–48.4	$\mathbb{F}_{269^4}$
`gaudry_quartic` solve	St	E	1.96–6.08	$d = 4..6$ (sweep recompute)
One-target online				
Compact-orbit IC vs strong rho	On	E	13.94	$n = 61$; rho without a table; retired (§16.3)
Accounting (costs that went up when priced)				
F4 word XORs priced	Cold	A	1/25.2	16.75 → 421.4, K13
F6-IC batched geometry priced	Cold	A	1/64.6–1/1,353	$m = 13, 19$; F4 unpriced (about 1/2.5 priced)

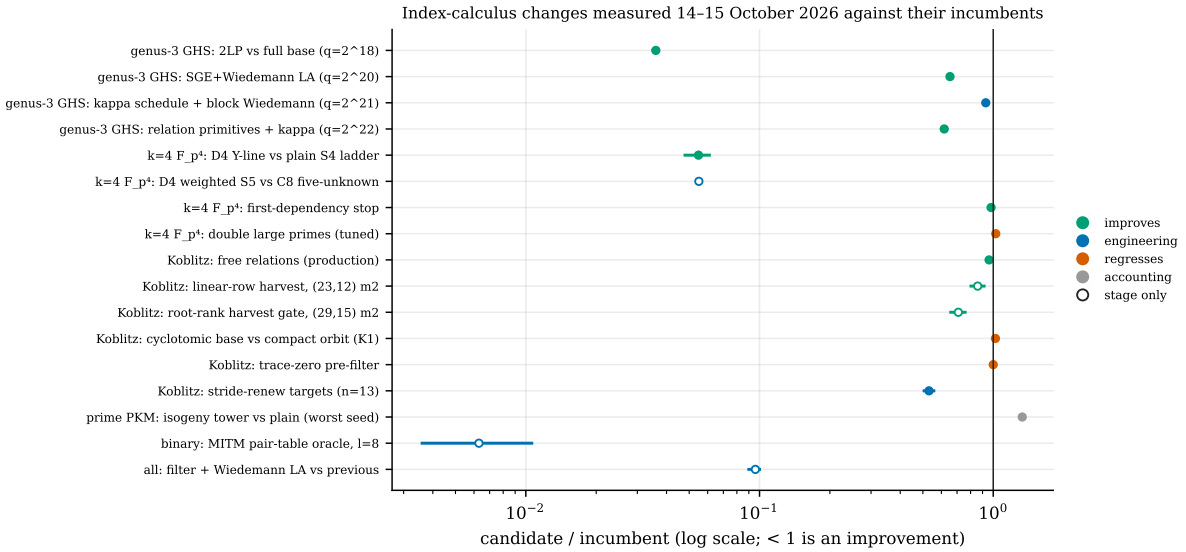


Figure 16.2: Candidate over incumbent for the changes of 14–15 October, with 95% intervals where the study reports one (the binary oracle's range is the spread over three runs). Filled: whole solve; open: one stage. Values: `papers/ecdlp-index-calculus-program/figures/ladder.tsv`; sources in Table 16.3.

16.3 The improvement ladder, 14–15 October

The repository's owner fixed the criterion for success on 15 October: a verified, measured improvement over the current index-calculus baseline is a success; each one becomes the new baseline; the gap to the matched rho is reported as the distance remaining, not as a pass/fail gate; and relabelling or accounting changes do not count. In the two days before that, two improvement rounds and a next-steps campaign landed about twenty measured changes, each paired against its own incumbent under a protocol frozen before data. Figure 16.2 and Table 16.3 list them, with the distance to rho that remains after each one. None is in the frontier campaign of §16.1, which predates them, and none was priced across the band; the boundaries are the ones the studies used, which differ from row to row.

Genus three: an exponent below rho's. The cubic GHS transfer [20] maps the DLP in $E_0(\mathbb{F}_{q^3})$ to the Jacobian of a genus-3 hyperelliptic curve over $\mathbb{F}_q$, where double-large-prime index calculus [21] costs $\tilde{O}(q^{2-2/g}) = \tilde{O}(q^{4/3})$ against rho's $q^{3/2}$. That is known; what is new is a priced, end-to-end, verified run of it on a family the repository can build ($4 \mid \#E_0(\mathbb{F}_{q^3})$, split arm, $q = 2^8$–2^{22}, every logarithm checked against the planted scalar). Adding double large primes changed the exponent from $q^{1.910}$ to $q^{1.385}$ $[1.379, 1.391]$, and keeping the large primes as columns, merging them with structured Gaussian elimination and finishing with Wiedemann moved it to $q^{1.380}$ and the constant to 0.653×. This was the first index-calculus family in the programme whose fitted exponent lies below rho's (the $k = 4$ binary family of §16.4 is the second), and the only one where measured IC beats a rho arm: the item-7 rho, which walks at $1.75\sqrt{\ell}$ steps, at $q = 2^{15}$ (0.69 $[0.55, 0.83]$). Two further rounds kept going. A frozen schedule for the factor-base fraction $\kappa(q)$ plus block Wiedemann brought the whole solve to 0.605× the first double-large-prime arm at $q = 2^{21}$. Cheaper relation primitives (one-inversion Cantor addition, a Stickelberger pre-filter, reused exponentiations) then cut the relation phase to 0.47× without changing a single relation, and with κ re-tuned the whole solve to 0.617 $[0.616, 0.619]$ at $q = 2^{22}$, in a unit that also charges inversions. The distance to rho shrank from 2.58× the negation-map rho model at $q = 2^{20}$ (first double-large-prime arm) to 1.15× a measured negation-map rho there after the second round ($2.31 \cdot 10^{10}$ against $2.01 \cdot 10^{10}$ $\mathbb{F}_q$ multiplications; a reference on overlapping but not identical curves). The third round's 0.62× has no matched rho in its unit yet, so whether it crosses is not established. The local exponents sit at 1.38–1.41, still above 4/3, because the linear algebra grows faster than $q^{4/3}$ and the tuned κ falls with q. The family is narrow: the transfer needs the right 2-power torsion, and $\mathbb{F}_{q^3}$ curves are not deployed. *Next improvement*: a negation-map rho measured in the third round's unit on the same instances, then the run past $q = 2^{22}$ that would show a crossover or rule it out.

Table 16.3: Changes measured 14–15 October 2026 against their own incumbents. Ratio: candidate/incumbent, < 1 better. Class, against the index-calculus incumbent (not the paper's ADVANCE, which is against rho): Ie = improves the exponent, Ic = improves the constant, E = ENGINEERING on one stage, R = regresses, T = trades, A = ACCOUNTING. "To rho": the distance left to the matched rho on the study's own boundary; "d" marks a derived or extrapolated value. Sources under `research/`: `ic_next10_20261014/`, `ic_improve_20261014/`, `ic_improve_20261015/`, `ic_improve3_20261015/` and the named study directories.

Change	Cl.	Ratio [95%]	To rho
Genus 3 after the cubic GHS transfer, $E_0(\mathbb{F}_{q^3})$, cost in $\mathbb{F}_q$ multiplications			
Double large primes vs full base (04)	Ie	0.036 $[0.035, 0.036]$ at 2^{18}; $q^{1.910} \to q^{1.385}$	2.58× negation rho model at $q = 2^{20}$
Large primes as columns, SGE, Wiedemann (01)	Ie	0.653 $[0.647, 0.660]$ at 2^{20}; $q^{1.398} \to q^{1.380}$	—
$\kappa(q)$ schedule + block Wiedemann (06)	T	0.929 at 2^{21}	1.15× measured negation rho at 2^{20} (ref.)
Cheaper relation primitives + re-tuned κ (r3-01)	Ic	0.617 $[0.616, 0.619]$ at 2^{22}	no matched rho in this unit
$k = 4$ *over* $\mathbb{F}_{p^4}$*, operation counts*			
D_4 Y-line vs plain S_4, $p = 101$–1033	Ic	0.0548 $[0.048, 0.061]$	handover 2^{126}–2^{137} (d)
D_4-weighted S_5, four unknowns, vs five	E	1/18.5–1/17.8	handover $2^{135.6} \to 2^{118.9}$ (d)
First-dependency stop (`d4-ed`)	Ic	0.979 $[0.973, 0.986]$	no $k = 4$ run below rho
Double large primes, tuned	R	1.025 $[1.016, 1.037]$ (LA 0.827)	—
Koblitz, compact orbit and Gröbner oracle			
Free relations in production, $n = 41, 53, 61$	Ic	0.960 $[0.946, 0.976]$	3.8–15× rho-frob ($w = 0$, §16.1)
Linear-row harvest, $(23, 12)$, $m = 2$	Ic	0.858 $[0.803, 0.916]$	oracle stage; default on
Root-rank harvest gate, $(29, 15)$, $m = 2$	Ic	1/1.41 $[1/1.52, 1/1.32]$; end to end 1/1.20	oracle stage; default on
Gated fall-row augmentation	Ic	1/11.9 on barrel $n = 13$; 1 in production	production no-op
Equivariant Macaulay oracle	T	4.0 ($n = 5$) … 1/15.5 ($n = 13$)	not applicable to production
`stride-renew` targets, $n = 13$	E	0.531 $[0.506, 0.557]$	4.7× rho-strong at $n = 19$
Cyclotomic γ-weight base, K1	R	1.022 $[1.021, 1.023]$	yield-neutral
Trace-zero pre-filter	R	1.00003	vacuous on odd order
Prime fields and shared stages			
PKM isogeny tower, search charged	A	1.002–1.331	17–137× rho
Binary MITM pair-table oracle, $\ell = 8$	E	1/277–1/94 per target	no exponent claim
Filter + Wiedemann LA, $D = 2^{18}$	E	1/10.40; 1/1.93 vs CADO-NFS	LA is 0.1–1.9% of runs

The $k = 4$ chain. The D_4 Macaulay system of §12.9 entered a complete ladder ($p = 101$–1033, 21 curves): 18.2× cheaper than plain S_4 at the same exponent, with the handover with negation rho extrapolated to 2^{126}–2^{137}. Writing the D_4-symmetrised S_5 in four weighted unknowns cut the per-solve cost another 17.8–18.5×, to $1.42 \cdot 10^9$ $\mathbb{F}_p$ multiplications, 854× below the plain form; in the 2-torsion frame that moves the derived handover from $2^{157.9}$ to $2^{118.9}$. Double large primes cut the linear algebra (0.827×) but regress the whole solve; stopping at the first dependency gains 2.1%. No $k = 4$ run below rho exists; every handover is an extrapolation of fitted exponents through a measured constant. *Next improvement*: put the four-unknown solve into the complete ladder, then keep double large primes' linear-algebra gain without its relation-phase cost.

Koblitz. Free relations—root-table self-collisions used as homogeneous rows—are now the production default (0.960× counted, 0.941× wall); the linear rows the Gröbner oracle used to discard are harvested (0.86–0.93× where the gate fires, and a root-rank gate then made it fire on more cells: 1/1.41 at $(29, 15)$, 1/1.20 end to end); the gated fall-row and equivariant passes pay only on overdetermined systems the production pipeline does not build. The cyclotomic base and the trace-zero filter were refuted. A census closes the free-relation lever at the rungs run: at $n = 61$, $K = 600$ the self-collisions span 49 independent relations of the 599 needed, and full rank would need $K \approx 2{,}050$. The distance to signed-Frobenius rho is still 3.8–15× cold at $w = 0$, and the fitted Koblitz ratio grows as about $r^{1/6}$ (slope 0.165 $[0.117, 0.213]$ per $\ln r$ on eight public curves, every native priced). *Next improvement*: relation yield per probe, which none of these changes touched.

The online result is retired. An equal-advice audit gave Bernstein–Lange preprocessing rho [4] a distinguished-point table no larger than the index-calculus pair table and ran both on the same 64 unseen targets at $n = 41$, 53 and 61. Rho with the table is 53–317× faster online (at $n = 41$, with exactly equal advice, 317 $[137, 739]$), and even at 1/1,024 of the bytes it stays 7.7–32× faster. The 13.9× online win of §10 and the later 7.3× and 12.3× hosted single-target points at $n = 71$ and $n = 83$ compared index calculus *with* a table against rho *without* one. They are measurements of the preprocessing model, not index-calculus improvements, and the online frontier belongs to preprocessing rho.

The reference moved too. Two rho studies shortened the distances above. On prime curves, choosing the better of $P + R$ and $P - R$ at each step takes 1.129× fewer steps under distinguished points (1.081–1.184); each step costs more, so the predicted net gain is 1.03–1.12× with 8–64 walks per inversion and a net loss (0.95) at 1,024. On ECC2K-130, signed σ steps cut GPU time per solve to 0.961–0.970 (derived; not rolled out). Neither is yet in a campaign arm. A comparison against a rho that has not adopted them would overstate index calculus's position by that much.

16.4 The ladder continued, 16–20 October

Work that reached `main` between 16 and 20 October, after the programme record was first assembled (some of it measured on 14–15 October), added a genus-3 round, two $k = 4$ oracles, a long binary-oracle ladder, Gröbner-engine work and a repository-wide validation sweep (Figure 16.3, Table 16.4). The reading rules of §16.3 apply: each row is against its own incumbent, on its study's boundary, unpriced across the band.

Genus three, round four. A field-level splitting test (a Lucas ladder on the Cardano resolvent's cubic character, with the same relations) removes most of the per-trial log factor (9.75 to 2.71 units per bit) and moves the whole solve to 0.756× at $q = 2^{22}$; with 32 walks sharing one inversion, to 0.715×, with local exponent 1.356 (fitted 1.368 over 2^{16}–2^{22}), 0.023 above 4/3. Chained with round three, the whole solve is about 0.44× the second round's at $q = 2^{22}$ (derived, in the inversion-charging unit). Rounds three and four ran no rho arm; a 23 October study then measured round 4 against the strongest rho at $q = 2^{14}$–2^{20}: under the registered pricing it is below rho at 2^{16} at every band weight and unresolved at the other sizes (geometric means 0.70–1.52), and above rho under the protocol's heavier sensitivity pricing; §14). Re-tuning κ on training again steepened the schedule and raised the exponent. A separate rebalance study measured IC at 0.16× (0.25× renormalised) a rho on $\mathrm{Jac}(C)$ at $q = 2^{14}$: that rho walks on the Jacobian (four walks per size, at 1.6–2.8 steps per $\sqrt{\ell}$ against an ideal 1.25), not on $E_0(\mathbb{F}_{q^3})$, so it is not the matched reference and the ratio is not a distance to rho. Memory now limits the ladder: $q = 2^{22}$ peaks at 3.9 GB of a 4.5 GB cap.

Table 16.4: Changes that landed 16–20 October 2026, against their own incumbents; columns and classes as in Table 16.3. Sources under `research/`: `ic_improve4_20261015/`, `ic_improve_20261014/04b-genus3-2lp-rebalance/`, `gaudry_quartic_d4_cost_20261014/`, `gaudry_quartic_d4_char2_20261015/`, `joux_vitse_n1_20261015/`, `binary_mitm_oracle_l9_l10_20261015/`, `binary_mitm_oracle_l11_l12_20261016/`, `ic_m3_algebraic_vs_mitm_20261014/`, `secg_small_x_20261014/`, `greedy_factor_base_yield_20261014/`, `f6_gate_first_x86_20261016/`, `pdp_shape_replay_20261015/`, `f4_fp_blocked_pairs_20261016/`, `ic_linalg_pipeline_20261019/`, `k_conversion_20261018/`.

Change	Cl.	Ratio [95%]	To rho
Genus 3 after the cubic GHS transfer			
Lucas splitting test (round 4, C1)	Ie	0.756 [0.755, 0.756] at 2^{22}; local 1.408 → 1.364	no rho arm since round 2
C1 + 32 walks per inversion	Ic	0.715 at 2^{22}; local 1.356	as above
Factor base retuned per size (04b)	Ic	0.961 at 2^{20}	0.16× rho on Jac(C) at 2^{14}, not on E_0
$k = 4$ over $\mathbb{F}_{p^4}$ and $\mathbb{F}_{(2^m)^4}$			
D_4 oracle: pivots, weighted form, replay, $p = 269$	E	1/370 [1/371.6, 1/367.9]	—
Same oracle, ecbench S	Ic	1/402 [1/406, 1/398]	$1.39 \cdot 10^6$× rho
Frobenius-reduced D_4 form, characteristic 2, $m = 8$	E	1/481 [1/483, 1/479]	—
Same, ecbench S ($m = 6$)	Ic	0.0057	6.1× rho model, 4.3× measured, at $n \approx 2^{56}$
Koblitz and binary oracles			
F6-IC gate-first order	E	1/1.73–1/1.34 word XORs	default on
Greedy coverage base, $K_0/2^{13}$	Ic	0.843 (expected trials)	gain 1.000 at $r = 2^{39}$
Inherited-F4 root-shape replay	R	1.015–1.328	switch stays off
Binary MITM oracle, $\ell = 9..12$	E	1/4,244–1/928 per target	relation collection $\Theta(2^n)$
Algebraic $m = 3, 4$ oracles vs MITM floor	R	slope 0.53–0.81 vs 0.21–0.36	$3 \cdot 10^2$–$3 \cdot 10^4$× floor
Prime fields and shared stages			
`f4_fp` blocked pair reduction	E	0.78–0.89 (wall, quad $n10, 11$)	beats msolve there
Sparse-CADO LA inside ecbench	E	S 1.000; LA wall 0.25–0.72	149× rho
S_3 solve on SECG small-x bases	—	table 2,600–313,000× cheaper	52–280 bits over rho
Column-touch conversion K (ECC2K-130 $m = 4$)	A	+7.62/ + 10.32 → +9.78/ + 12.48 bits ($w = 1$ floor; +45.71 at $w = 2$, unconverted)	accounting

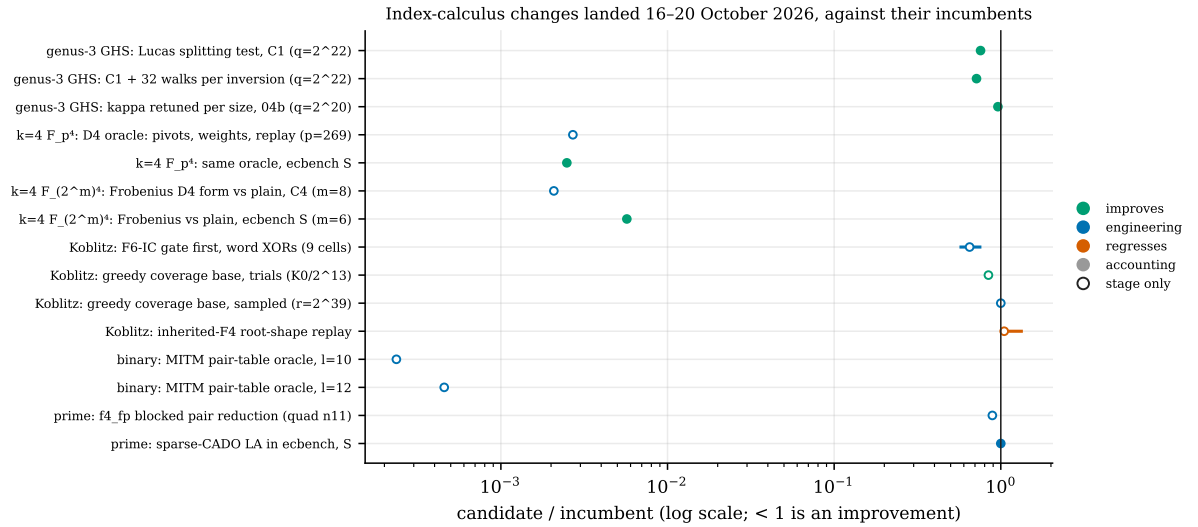


Figure 16.3: Candidate over incumbent for the changes that landed 16–20 October, as in Figure 16.2. Values: `papers/ecdlp-index-calculus-program/figures/ladder2.tsv`; sources in Table 16.4.

$k = 4$: two oracles and a closing gap. Over $\mathbb{F}_{p^4}$, sparsest pivots, the weighted grading $\sigma_4 = \pi^2$ and shape replay cut one D_4 decomposition from $2.55 \cdot 10^{10}$ to $6.89 \cdot 10^7$ $\mathbb{F}_p$ multiplications (370×; 648/648 solves exact) and ecbench S by 398–406×; index calculus is still $1.39 \cdot 10^6\times$ rho at the measured sizes and the model handover moves from 2^{135} to 2^{103}. In characteristic two, a Frobenius-reduced form of the same system (weights $(2, 2, 1, 1)$, Macaulay degree 11, a 384×448 matrix) is 481× below the plain form. End to end on $E(\mathbb{F}_{(2^m)^4})$ the ratio of IC to the negation-map rho model falls from 5,080 at $m = 6$ to 6.06 at $m = 14$ ($n \approx 2^{56}$; 4.3× a measured rho), with fitted slope -0.248 in $\log_2 n$—Gaudry's $n^{-1/4}$—and an extrapolated crossover near $n \approx 2^{68}$ (2^{63}–2^{69} under the model). After genus three, this is the nearest any index calculus in the programme has come to rho, and its extrapolated crossover lies where rho itself takes about 2^{34} steps, so the extrapolation can be tested directly. A Joux–Vitse $(n-1)$-decomposition at $n = 5$ on $E(\mathbb{F}_{p^5})$ completed verified DLPs at $4.5 \cdot 10^7$–$1.8 \cdot 10^8\times$ rho, with a crossover extrapolated to groups near 2^{291}.

Binary oracles and their floor. The MITM pair-table oracle on a non-Koblitz curve reached $\ell = 12$ over $\mathbb{F}_{2^{37}}$ (2,474–4,244× over pairs-and-solve at $\ell = 9, 10$; 928–2,321× at $\ell = 11, 12$, the last as recomputed by the validation sweep; 0 disagreements on 256 targets). Against that floor, no algebraic oracle (inherited F4, F6-IC, SAT, split S_4 hybrids, a wide chained S_3 solver) grows more slowly in n: the cheapest is $3 \cdot 10^2\times$ the floor at $n = 13$ and $3.8 \cdot 10^4\times$ at $n = 31$ for $m = 3$, and the $m = 4$ chain grows at slope 0.805 against the floor's 0.333. A greedy coverage base needs 1.08–1.18× fewer trials on small Koblitz curves, but its gain falls to 1.0000 at $r = 2^{39}$. On the 33 SECG curves an S_3 solve over a small-x base, with no table, is 52–280 bits over rho.

The reference moved again. The strong signed-Frobenius rho of every Koblitz comparison executed 718 instructions per walk step after three rounds, down from 1,196; a fourth round of batched inversion took main from 707.2 to 679.7 Callgrind instructions per step at $n = 53$, with no established native-time gain (outputs identical; `research/rho_batchinv_20261022/`; steps per ideal 0.92–0.99 on 400 fixtures), and ecbench's standing `rho.signed_frobenius_strong` runs at 0.802× its former instruction count with identical GAE. On hyperelliptic Jacobians the programme's rho was slow at large N until full-width branch coefficients brought S_{walk} from 2.10 to 1.48; the negation fold saves only 0.97× at $N \geq 10^5$. Counted-unit comparisons (S) are unaffected by the instruction counts; wall-time comparisons against the older rho overstate index calculus by up to the same factor.

Corrections from the validation sweep. A repository-wide validation sweep (19–20 October) contradicted five recorded results, all now struck through in their sources with dated notes. Three bear on this paper. (i) The one-point $n = 83$ online ratio (18.5×, §8) and the hosted $n = 71/n = 83$ panels (7.3×, 12.3×) compared against rho with an empty table; with a Bernstein–Lange table matched to the

index calculus's set-up time or memory, precomputed rho is faster in every matched $n = 83$ pair (3.7× time-matched and 40.6× memory-matched against $K = 3300$; 82–396× against $K = 600$). They join the $n = 61$ point as retired. (ii) The root-finder session's replay-all, recorded as 120/168 identical with the 48 mismatches confined to inherited F4, replays 72/168 on current `main`, with mismatches in all four IC arms, root finding included. The sweep later isolated the cause: an x86-only BMI2 rewrite in inherited F4 charges a different word-XOR count from the portable path, and with it disabled x86 and emulated aarch64 both replay 168/168 (`research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`). F4 counts are therefore comparable across instruction sets only when the path is stated, and until the method identity records it the root-finder rows rest on the recorded receipt. (iii) The equal-bytes preprocessing control at $n = 73$ and $n = 83$ is built and frozen but not run. *Next improvement* (from the 18 October leads review): a large-memory host. The two exponent-level questions named then—the Weil-descent cell $(m, \ell) = (4, 6)$ and the PKM tower at $m = 3$, $N = 21$—have since run on rented hosts: the $(4, 6)$ cell refuted at degree 4 at $N = 54$, and the PKM cell finished degree 7 and stopped on memory inside degree 8, implying $D \geq 8$ but read as inconclusive (§§11.1,12.2). The $(3, 7)$, $\ell = 8$ and $m = 4$, $N = 20$ cells remain.

16.5 The ladder continued, 21–24 October

Work that reached `main` between 21 and 24 October falls into four groups: factor bases closed under cheap known-logarithm maps, a paired re-run of the Koblitz leaderboard recipe, a guess-and-determine attack on the Weil-descent degree question, and transfer measurements, among them a third family whose measured gap to rho is now single-digit (Table 16.5). The reading rules of §16.3 apply: each row is against its own incumbent and on its study's boundary. The companion ledger (`papers/ecdlp-ic-improvement-ledger/`) records the same period as a ratchet of incumbents; this section keeps to what moves the frontier or corrects it.

One mechanism, four factor bases. The week's measured gains on the index-calculus side are all the same idea: put a point and its images under maps of known logarithm—the Frobenius τ on a Koblitz curve, the CM endomorphism ϕ, small integer multiples, the cofactor $[h]$—into one column with a known coefficient, so the linear-algebra stage needs fewer relations. On Koblitz curves the $\mathbb{Z}[\tau]$ base needs $K+1$ relations instead of about one per orbit: child CPU of the complete process falls 1.9–4.4× at $n = 41$–61 and 6.2–10.5× at $n = 71$. These are measured against the repository's own tuned producer, on a host at load about 20 (child CPU is the less affected number), with three targets up to $n = 61$ and one at $n = 71$. The prime CM base ($j = 0$, $j = 1728$, $D = -7, -8$) cuts S 10–20×, and the integer-multiple base on random binary curves 3–4×; both at 36–40-bit or degree-26–28 sizes, where the best bases have 3–10 columns over 512–3,480 points and, as their sources note, tend to a baby-step giant-step table over multiples as the column count falls. The cofactor-projected base (LEAD-19) is the only one measured on leaderboard rows, with 20 repeats and the order stable across the band: 1.29–3.36× in cold S, leaving 6.1–14.4× to matched rho. Every source states that the exponent is unchanged; the gain is bounded by the size of the multiplier set and shrinks as linear algebra's share of the cost shrinks. The known-coefficient columns are those of GGMP's Frobenius-orbit bases [17] (§10); what is new here is extending them from automorphisms to small-norm and integer multipliers. Whether that extension is in the literature has not been checked. All four are ENGINEERING, and none is measured against a published index calculus; the $\mathbb{Z}[\tau]$ source notes that the summation-polynomial methods have no table-based counterpart at these parameters.

Koblitz: the recipe is near its optimum, the distance is larger. Rule v4 retuned columns, collection window and unit size on disjoint training seeds and re-ran Table A paired against rule v3 (32 pairs per size, 16 seeds, A/A and a positive control that detects a 10–15% slowdown). One size improves (0.953 $[0.904, 0.998]$ at $n = 57$) and five are not separated, so it is not a Table A improvement under its own rule. The free-relation lever costs 5–9% in this pipeline. More consequential is the reference: with rho priced in the same process at the canonical batched step, the cold distance is 9.9–43×, 1.76–1.94× the as-built figure that rule v3 used. The ledger's rule v8 reads 4.67–18.07× on the same sizes in its own unit; the two are different domains in the sense of Definition 4.1 and are not combined here.

Weil descent: a hybrid closes one cell, the degree question stays open. The $(m, \ell) = (3, 7)$ cell had no refutation at degree 4. Fixing one or two variables restores degree 4: all branches refute

Table 16.5: Changes that landed 21–24 October 2026, against their own incumbents; columns and classes as in Table 16.3 (N = null or diagnostic, no cost row). Sources under `research/`: `ic_improve_20261023/` (`README.md`, `PRIME_README.md`, `BINARY_README.md`), `lead19_projected_base/`, `ic_tool_program/rule/v4/`, `dreg_hybrid_guess_20261024/`, `jv_cover_p_scaling_20261024/`, `hcdlp_trial_cost_20261024/`, `ghs_cover_census_n4_20261024/`, `coppersmith_s3_theta2_20261023/`; and `docs/ic/CERTICOM131_IC_PROGRAM.md`.

Change	Cl.	Ratio [95%]	To rho
Factor bases closed under known-log maps			
$\mathbb{Z}[\tau]$ small-norm multipliers, Koblitz $n = 41$–61	Ic	1/1.9–1/4.4 child CPU (3 targets, loaded host)	not measured
Same, wide path, $n = 71$	Ic	1/6.2 equal points; 1/10.5 retuned (1 target)	not measured
$\mathbb{Z}[\phi]$ multipliers, prime CM curves, 36–40 bit	Ic	1/10–1/20 in S	7–12×
Integer-multiple closure, random binary, degree 26–28	Ic	1/3.1–1/3.9 vs plain	6–12×
Cofactor-projected base (LEAD-19), $\mathbb{F}_{2^{215}}, \mathbb{F}_{2^{221}}, \mathbb{F}_{2^{227}}$	Ic	1/1.29 [1/1.52, 1/1.12], 1/1.58 [1/1.82, 1/1.35], 1/3.36 [1/4.02, 1/2.80]	6.1, 11.5, 14.4×
Koblitz leaderboard recipe (Table A, $n = 41$–61)			
Rule v4 recipe vs v3, paired, six sizes	Ic	0.953 $[0.904, 0.998]$ at $n = 57$; five sizes not separated	9.9–43× at the canonical step
Weil descent, degree question			
Hybrid guess, $(m, \ell) = (3, 7)$, $k^* \le 2$ guesses	E	1/116–1/260 of projected degree-5 closure	—
Implied ECC2K-130 $m = 4$ row (derived)	N	+8.43 (flat k) to +58.27 bits (linear k)	vs +10.32 ($w = 1$), +45.71 ($w = 2$)
Prime fields			
Kummer tower, 30 bit (new row)	N	—	158.9× [121.6, 221.4]
Coppersmith lattice on S_3, $\ell \le 3$	N	0/10 recovered in all 96 cells with $\beta \ge 0.20$	ceiling $\beta = 1/4$ stands
Transfer and covers			
Joux–Vitse cover of $E(\mathbb{F}_{p^6})$, $p = 53$–1009	N	slope −2.05 $[-2.28, -1.83]$ in $\ln p$	1.59–1.98× at $p = 1009$ (LB)
Measured trial cost, genus-255 binary Jacobian	A	103 curve additions per trial ($2^{6.69}$)	$N = 130$ margin $+4.71 \to -1.83$ bits
GHS covers of $E/\mathbb{F}_{q^4}$, odd q, census	N	genus floor 13 on odd-order curves	model exponent gap 0.154

with $k^* \leq 2$ on 10 of 10 unsatisfiable draws, at 1/116–1/260 of the projected degree-5 cost, in 3–6 GB (planted-solution and placebo controls passed). For the ECC2K-130 $m = 4$ row this changes less. Its often-quoted "+10.32 bits over rho if degree 5 holds" prices one column touch per Macaulay column ($w = 1$), a floor no elimination beats; at $w = 2$, the exponent the measured closure tracks, the same row is +45.71 bits, and the measured conversion $K = 4$–16 rho steps per touch raises both (§16.4). With the hybrid, the row is +8.43 bits if the number of guesses stays flat in ℓ and +58.27 bits if it grows linearly; one value of ℓ cannot separate them, and the run at $\ell = 8$ that would did not complete. In no scenario does the row go below the degree-4 floor of +7.62 bits, so the route does not reach rho.

Prime fields: further behind. A new 30-bit Kummer-tower row in ecbench (160/160 verified, loaded host) is 158.9× [121.6, 221.4] rho at $w = 0$, against 141× at 28 bits in the campaign; its source calls it "a measurement, not an improvement". Two falsifiers for the Certicom 131-bit programme (random-b binary degrees of regularity; structured bases) were null. A frozen-protocol test of a Coppersmith lattice on S_3 recovered no planted decomposition in any of 96 cells with $\beta \geq 0.20$ at 24–56 bits, with positive controls passing below, so the $\beta = 1/4$ ceiling stands at multiplicity ≤ 3.

A third family approaches rho: the Joux–Vitse cover. The genus-3 cover of $E(\mathbb{F}_{p^6})$ over $\mathbb{F}_{p^2}$ with Nagao decompositions (full rational 2-torsion; a weak class, not prime-order curves) ran at $p = 53$–1009, three seeds per prime, every logarithm checked. The gap to rho falls from a geometric mean of 620× at $p = 53$ to 1.73× at $p = 1009$ (1.59–1.98), with fitted slope -2.05 $[-2.28, -1.83]$ in $\ln p$ against the cost model's -2; the line meets rho near $p \approx 1330$ ($[1170, 1540]$, an extrapolation). Two limits keep it a lower-bound comparison: the cover arm does not emit its unpriced F4 and Wiedemann work, so it cannot be ranked under the band; and rho cannot be run at $p \geq 503$ (subgroups of 2^{52}–2^{58}), so those ratios use the pooled rho constant $S = 1.344$ measured at $p \leq 251$. If that constant kept falling at its (insignificant) point trend, the ratio at $p = 1009$ would be about 20% higher. A separate sieve arm on the same cover read far lower ratios, but its comparisons are withheld by review (§14).

Covers: censuses and a repriced counterfactual. Measuring the cost of one relation trial (a Cantor addition plus a smoothness test) on a genus-255 binary Jacobian gives 103 additions of a point over $\mathbb{F}_{2^{131}}$ ($2^{6.69}$), which turns the $N = 130$ GHS counterfactual's modelled margin over rho from +4.71 bits to −1.83 bits; only a trial below about 29 additions would preserve it, and the smoothness test alone costs 16.5. Two censuses fix what covers exist at toy sizes: for $E/\mathbb{F}_{q^4}$ in odd characteristic the genus floor is 13 on odd-order curves ($2 - 2/g$ against rho's 2 leaves an exponent gap of 0.154, constants excluded), and for binary curves over $\mathbb{F}_{2^{3l}}$ the genus-3 class is a 2^{-l} fraction (0.78% at $l = 7$).

The validation gate. On 24 October the leaderboard's gate ran for the first time: all 13 Table B rows (eight prime, five random binary) passed—every counted field replays identically, every scalar checks in an independent verifier, and IC stays above rho at every weight of the band (4.05–54.76×)—and all nine Table C rows failed; Table A was not in scope (`research/ic_validation_gate_20261024/`). The study staged its manifest rather than installing it, and the leaderboard now marks the Table B rows validated. The ledger paper, current to 23 October, records that no row had passed; for Table B that is now out of date. Validation says the numbers are what the archive says and the reference is fair, not that index calculus approaches rho.

What is still not measured. No arm in the programme has yet been compared with a published index calculus at the same (q, n, m): FGHR-style symmetrised summation polynomials over the standard base $\{P : x(P) \in \mathbb{F}_q\}$, solved by msolve, or WDSat for binary decompositions. (WDSat has been run as a solver on its authors' published instances, §11, but not inside an index calculus with relations, linear algebra and a rho comparison.) Every ratio in Tables 16.3–16.5 is against rho or against the programme's own previous arm. Defining and running that reference arm (open problem 10) is the precondition for any claim of improving on the state of the art.

16.6 Three frontiers

Cold, one target, set-up included. Unchanged: generic rho. Every measured index-calculus arm is above its matched rho (Figure 16.4); the closest complete count is root-finding negfold at 1.55× strong signed-Frobenius rho at $n = 13$; the closest lower-bound arms are 1.08× (compact orbit with a counted

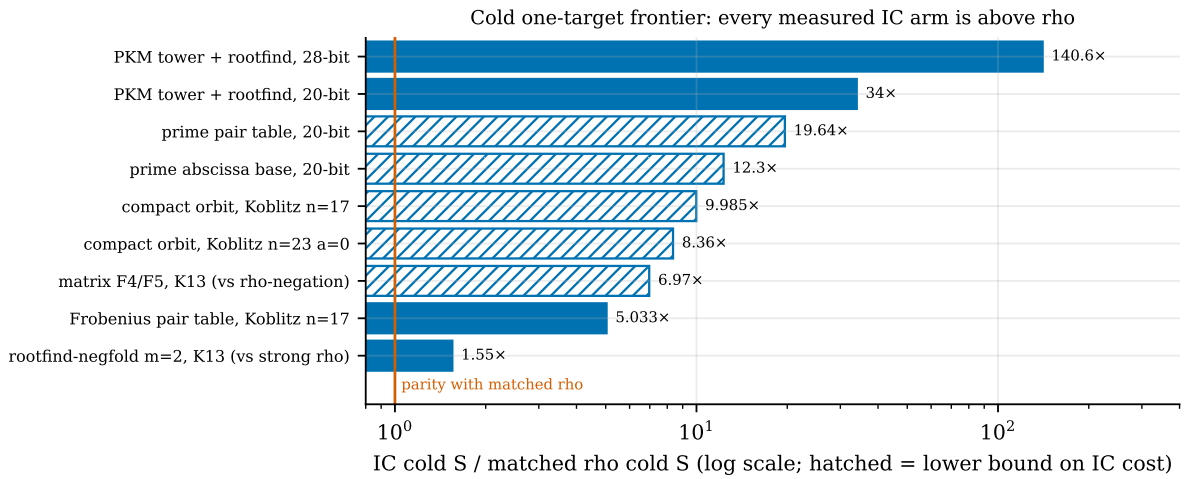


Figure 16.4: Cold one-target S of index-calculus candidates over their matched rho arm (each study's own estimator; curves differ, so bars are not ranked against each other). Hatched: lower bound on IC cost (unpriced native or solver work). Sources: `research/ecbench_frontier_20261013/`, `research/ecbench_rootfinding_assessment_20261012/`, `research/pkm_tower_ecbench_20261013/`, `research/ecbench_rootfind_faster_20261014/`.

Frobenius table) and 6.97× (matrix F4/F5 against rho-negation at K13). Prime-field arms are 12–239× rho and diverge with size (the 30-bit Kummer-tower row of §16.5 is 159×). Off the prime and prime-degree binary fields, three families are within a single-digit factor, all in weak classes: genus three after cubic GHS (below the strongest rho at $q = 2^{16}$ under the registered pricing, unresolved at the other sizes to 2^{20}, above it under a heavier sensitivity pricing), $k = 4$ over $\mathbb{F}_{(2^m)^4}$ (4.3× a measured rho at $n \approx 2^{56}$, closing as $n^{-1/4}$), and the Joux–Vitse cover of $E(\mathbb{F}_{p^6})$ (plain Nagao arm; 1.59–1.98× a pooled rho constant at $p = 1009$, closing as about p^{-2}; a lower-bound comparison).

Online, after retained set-up. Generic again: Bernstein–Lange preprocessing rho. The admitted compact-orbit point at Koblitz $n = 61$, 13.9× $[9.8, 19.1]$ over strong rho without a table (cold median 83.4× rho), is 53–221× slower online than preprocessing rho given equal or less advice (§16.3), and the $n = 83$ points lose every pair run with rho given a table matched on set-up time or memory (§§16.4,10.5).

Stage. Moved substantially: the decomposition oracle is two to five orders of magnitude cheaper than at the start of the programme on the workloads where root finding applies, the $\mathbb{F}_p$ F4 is at msolve parity, the Boolean F4 family is one to two orders of magnitude cheaper in counted matrix work, and the $k = 4$ extension-field solve is 48× and then a further 370× cheaper over $\mathbb{F}_{p^4}$ (481× below the plain form in characteristic two). None of this, by itself, moves the cold frontier: the stages it leaves untouched—relation yield, factor-base size, linear algebra, table set-up—now dominate.

16.7 What would move the cold frontier

From the frontier synthesis and the problem records: (i) register the torsion-symmetrised bases and the $n = 83$ dense-root engine in ecbench under immutable method identities and emit cold and online records; (ii) resolve the inherited-F4 cross-host counter drift before using that counter in a bound; (iii) run a four-size native challenge against the standing rho bound—two sizes cannot establish an exponent; (iv) obtain an isolated cold session at $n = 83$ with every set-up phase and retained byte charged; (v) price F6-IC's batched geometry with a pinned conversion. Since then three items have moved to the front: (vi) run $k = 4$ over $\mathbb{F}_{(2^m)^4}$ at $m = 15$–17 against a measured negation-map rho, where the fitted trend predicts parity near $n \approx 2^{68}$; (vii) measure the matched rho on $E_0(\mathbb{F}_{q^3})$ in the genus-3 rounds' unit (done on 23 October: below rho at 2^{16} under the registered pricing, unresolved elsewhere, §14); (viii) obtain a large-memory host for the $(4, 6)$ Weil-descent cell and the PKM tower. The 21–24 October work adds two more and re-weights one: (ix) measure rho on the Joux–Vitse cover family at $p \geq 503$ (or a size where it can be run) and emit the cover arm's unpriced counters, so that the p^{-2} trend can be ranked

under the band and tested past $p \approx 1330$; (x) define and run a published-index-calculus reference arm (open problem 10); and (viii) is worth less than it looked for the binary question, since the $m = 4$ row at $w = 2$ is about +46 bits over rho even if degree 5 holds (§16.5). A result in the cold domain that beats rho at four sizes would be the first ADVANCE of the programme.

17 The distance to rho, family by family

This chapter gives, for each curve family, the current distance between index calculus and a matched rho. Each figure carries its reference, its boundary and its validation status. Here a ratio is IC over rho, so greater than 1 means rho is cheaper. Chapter 16 gives the frontier campaign, the improvement ladders of 14–24 October, and the three frontiers (cold, online, stage). This chapter adds the work of 25–29 October and states the position as it stands at the end of the programme.

17.1 Koblitz curves

The leaderboard's Table A measures a Koblitz descent: aimed $m = 3$ collection, then an $m = 2$ or $m = 3$ descent. It is measured cold on one unseen public point per process, with 64 points per size, at six sizes from $2^{36.6}$ to $2^{47.2}$. Figure 17.2 shows the rule at three baselines.

- *v3 to v7 (same rho).* Index calculus became 1.52–1.89× faster, and the cold ratio fell from 5.23–20.97 to 3.43–11.5. The board carries the v7 figures, validated (§18.2).
- *v8 (matched rho).* In the same binary, rho's signed-Frobenius canonicalisation batched 16 lanes at a time raised the gap by 1.30–1.58× at every size, to

$$4.67\,[4.34, 5.07],\ 5.70,\ 7.07,\ 11.75,\ 10.65,\ 18.07\,[15.98, 20.58].$$

  All six values lie inside ranges declared before the rows were taken. The 408 rows of each run all pass the `vs_rho` check (`research/ic_tool_program/rule/v8/README.md`).
- *Canonical-step rho (a model).* Pricing rho at the canonical batched step gives 6.45–23.32× in v8's unit, and 9.9–43× at rule v4 (§16.5).

The slope of the gap in $\ln r$ is 0.168 [0.128, 0.208] at v8 against 0.176 at v3. The v3→v7 improvement lowered the constant, not the growth.

The priced campaign. The closest Koblitz arm, compact Frobenius orbits, is 3.8–15× signed-Frobenius rho at $w = 0$. Fully priced against fully priced rho, its best cell is 1.566 [0.992, 2.347]; the ratio is never below 1 and grows as $r^{0.165\,[0.117, 0.213]}$.

$n = 83$. The cold pipeline uses the base K1107/root11, which confirmed a saving of 1.195 s [0.957, 1.433] on a 115 s process (16 public points, GitHub Tier 1 host). The last cold IC/rho ratio is the K1120 holdout's 55.4, from five points. A matched precomputed-rho reproduction on 28 October gave median online IC over rho as follows (`research/n83_k1107_bl_reproduction_20261028/RESULT.md`):

- 40.9 [7.58, 141.9] against 650 s table walks;
- 4.10 [1.58, 24.8] with walks matched to IC set-up;
- 33.1 [6.60, 74.4] with a table matched to IC memory.

Cold, IC took 155.5 s against 36.96 s for plain rho over all 16 targets. In operation counts the order flips across the pricing band, so no ranking is made.

17.2 Prime curves

Generic track. The prime-field programme's owner decisions of 29 October split prime-field index calculus into two counted tracks (`research/prime_ic_program_20261029/DECISIONS.md`):

- a *generic* track, with table, pair-table and root-finding oracles, labelled "bounded by $\sqrt{r}$; cannot beat rho";
- an *algebraic* track, whose prime baseline is empty.

On the generic track:

- *Table B* (validated). The prime rows are 5.47–13.5× matched negation rho at 10–24 bits.
- *Campaign.* The two-summand pair table runs from 10.5× rho at 20 bits to 101.8× at 40 bits ($w = 0.1$). Its fitted exponent is 0.620 [0.557, 0.663] against rho's 0.435, which is the $r^{2/3}$ against $r^{1/2}$ of Chapter 2.
- *A k-sum table oracle* (28 October, `research/prime_ksum_table_20261028/`). A negation-folded table of every signed 4-sum of a smaller base lowers mean S at $w = 0$ by 1.9× at 20 bits, rising to 7.2× at 40 bits (83.7 → 11.6). The fitted exponent falls from 0.620 [0.586, 0.650] to 0.521 [0.506, 0.538],

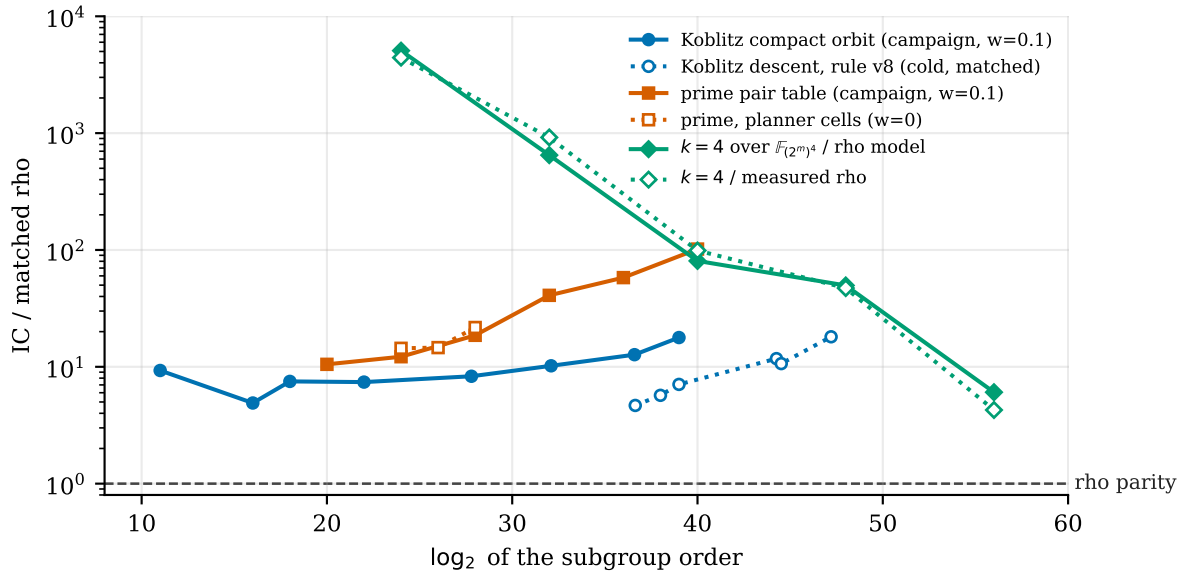


Figure 17.1: Distance to matched rho by family and size. Koblitz and prime curves: the frontier campaign's priced arms ($w = 0.1$, 16 targets × 2 rounds); open markers are the Koblitz rule at v8 (cold, 64 targets) and the prime planner cells ($w = 0$). $k = 4$ over $\mathbb{F}_{(2^m)^4}$: the characteristic-two ladder against the negation-map rho model (filled) and a measured rho (open); the $n \approx 2^{56}$ point is one target. Curves and units differ between families, so the lines are not ranked against each other. Sources: `figures/data-ledger.json`.

against the theory value $k/(2k-1) = 4/7$. That leaves 5.3× rho at 20 bits and 13.1× at 40 bits. 2,016 runs were verified and replayed; the host was L0, and there is no validation. The source notes that the method is generic, so its limit is baby-step giant-step.

- *A new 20-bit baseline* (validated, `research/prime_base_oracle_tuning_20261029/`). Raising the factor base from 32 to 45 gives 0.818 [0.773, 0.866] of the incumbent at $w = 0$. A second host (L2) replayed all 1,008 records identically and gave 0.836 [0.806, 0.866]. The distance to negation rho falls from 9.47 to 7.75 [6.92, 8.70]. The same rule at 24 and 28 bits was a relabelling, and at 32 bits null.
- *Planner cells.* These are 12.6–25.8× rho.

Algebraic track. The Kummer tower is 28–159× rho (158.9 [121.6, 221.4] at 30 bits). No algebraic prime baseline exists. Its solving degree is the subject of §6.5.

17.3 Binary curves of prime degree

Table B's random-binary rows are 10.6–38.2× matched rho at $\mathbb{F}_{2^{15}}$–$\mathbb{F}_{2^{27}}$, and they are validated. A cofactor-projected base cuts three of them by 1.29–3.36×, leaving 6.1–14.4×. That base is not yet a board row. For Certicom ECC2-131, the $m = 3$ floor with a free oracle is already above rho (Chapter 11).

17.4 Weak extension-field families

Only in these families (Figure 17.3) does index calculus approach rho or go below it.

- *Genus 3 via cubic GHS, hyperelliptic route.* This is the programme's only preregistered decision of IC below the strongest rho: at $q = 2^{16}$, IC over the ideal rho reference is 0.70, 0.71 and 0.70. At six other sizes the decision is unresolved, and the pooled slope is flat (0.034 [−0.047, 0.105] per bit). Pricing one subgroup multiplication at nine $\mathbb{F}_q$ multiplications puts every instance at 2.7–12.4× rho (Chapter 14).
- *Genus 3 via cubic GHS, plane-quartic route (Diem).* Double large primes, structured elimination and Lanczos give a fitted exponent of 0.983 [0.975, 0.993] against negation rho's 1.543. The ratio is 0.005 [0.004, 0.008] of negation rho at $q = 2{,}097{,}091$, and the interpolated crossover is $q^* = 555$ [389, 752] (`research/quartic_n4v2_20261020/`). The 501 IC units and 2,442 rho runs are all verified. Three

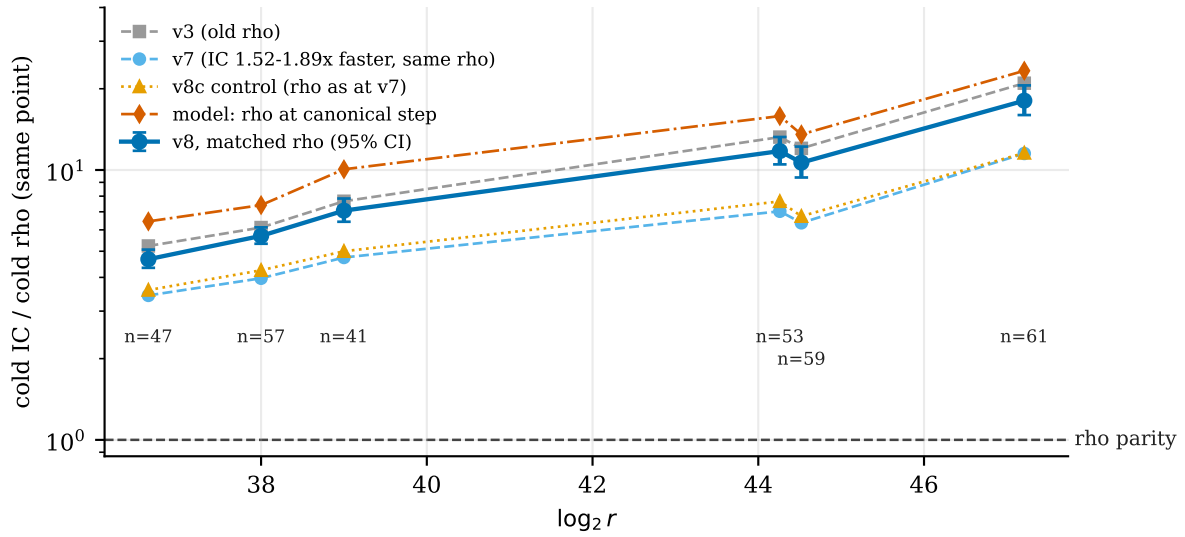


Figure 17.2: The Koblitz rule at three baselines: cold IC over cold rho on the same public point, 64 targets per size. From v3 to v7 the IC improved against the same rho. v8 runs the same IC against a matched, faster rho; its same-binary control, v8c, reproduces v7. The dash-dot line prices rho at the canonical step. Source: `research/ic_tool_program/rule/v8/README.md`.

caveats: the host was L0 under heavy load, there is no validation file, and the frozen baseline's records do not replay. This is a measurement of a known result (Diem, Diem–Thomé).

- *Joux–Vitse cover of* $E(\mathbb{F}_{p^6})$. IC is 1.59–1.98× a pooled rho constant at $p = 1009$, closing as about p^{-2}. This is a lower-bound comparison.
- $k = 4$ *over* $\mathbb{F}_{(2^m)^4}$. IC is 5.2× rho at $m = 15$ after corrections; the crossover is unsupported (Chapter 13).
- *GHS covers over* $\mathbb{F}_{4096}$. Exact genus-3 covers found by search are 549–2,039× rho in full transfer cost (`research/ghs_q16_genus3_subcovers_20261028/REPORT.md`).

17.5 Summary

Table 17.1: Smallest measured distances to rho, by family.

Family	IC/rho	Qualifier
Plane quartic via cubic GHS, $q \approx 2^{21}$	0.005	negation rho; weak family; L0; no validation file
Genus-3 hyperelliptic via GHS, $q = 2^{16}$	0.70	ideal strong rho; one size of seven; pricing-sensitive
Joux–Vitse cover, $p = 1009$	1.59–1.98	pooled rho constant; lower bound
$k = 4$, $\mathbb{F}_{(2^m)^4}$, $m = 15$	5.2	two targets; crossover unsupported
Koblitz, fully priced orbit (best cell)	1.566	never below 1; grows as $r^{0.165}$
Koblitz rule v8, $2^{36.6}$	4.67	cold, matched, 64 targets
Prime, k-sum table, 20 bits	5.3	generic track; not validated
Prime, Table B, 12 bits	5.47	validated
Random binary, Table B	10.6	validated
Koblitz $n = 83$	55.4	cold, five points

On every prime-order curve of cryptographic shape that the programme measured, rho is cheaper. Index calculus goes below rho only in weak extension-field families, which is where theory said it would. There the programme's contribution is matched, priced, verified measurement of known algorithms.

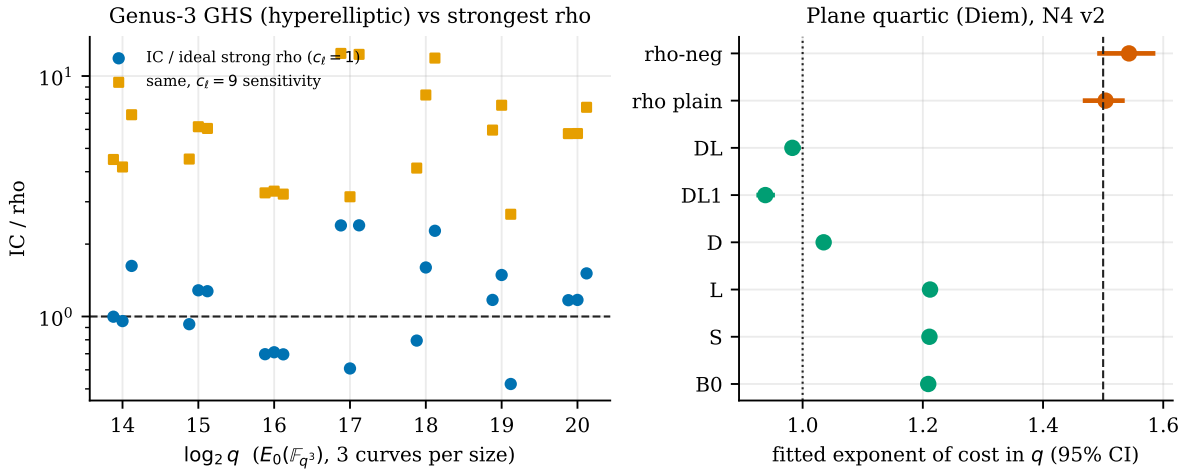


Figure 17.3: Weak families. *Left:* genus-3 GHS index calculus (hyperelliptic, double large primes, round 4) over the ideal strong rho on 21 instances of $E_0(\mathbb{F}_{q^3})$, in round 4's unit. Blue: one subgroup multiplication priced at one $\mathbb{F}_q$ multiplication, the registered unit. Orange: priced at nine, the protocol's sensitivity. Below the dashed line IC is cheaper. Source: `research/ghs3_vs_rho_20261023/TABLES.md`. *Right:* plane-quartic (Diem) index calculus, N4 v2. Fitted exponents of cold cost in q over $k = 8$–22 against plain and negation rho, at $w = 0$. Dotted line: $\tilde{O}(q)$. Dashed line: rho's $q^{3/2}$. Source: `research/quartic_n4v2_20261020/README.md`. Data: `figures/data-new.json`.

18 Corrections and validation

A ratchet is worth only as much as its pawl. Several recorded results were corrected once the reference was matched, native work was priced, or the evidence was replayed. Each correction is recorded in the source with a dated note; frozen evidence is never rewritten. Figure 18.1 shows the before and after values. Every correction moved IC's position against rho *down*, and several moved it from apparently ahead of rho to behind it.

18.1 Claims that did not survive

1. Every online win is retired. The one-target online ladder compared IC *with* a retained table against rho *without* one:

- 13.94× [9.81, 19.13] over strong rho at $n = 61$, on 64 targets;
- 18.53× at $n = 83$ with a dense root index;
- 7.29× and 12.29× on hosted single targets at $n = 71$ and $n = 83$.

Given a distinguished-point table no larger than IC's pair table, Bernstein–Lange preprocessing rho [4] is 53–317× faster online at $n = 41, 53, 61$ on the same 64 targets (benchmarks row `20261014-ic-next10-09-equal-advice`). At $n = 83$, rho with a table matched to IC's set-up time or memory wins every pair: 3.74× time-matched and 40.61× memory-matched against K3300, and 81.5–395.5× against K600 (benchmarks row `20261016-n83-precomputed-rho-control`). Cold, the hosted $n = 83$ point is 133.7× rho. Those online rows measured what a table buys, and generic rho buys more per byte. The within-IC result survives: K1798's probe reduction over K1500 at $n = 83$ (from 300,021.9 to 169,069.8 probes per relation).

2. Pricing native work moved two arms.

- *Inherited F4 at $n = 13$.* Its $S = 16.7$, close to rho's 14.4, left its Boolean word XORs unpriced. Priced, $S = 421.4$ (`research/ecbench_rootfind_faster_20261014/`).
- *F4 pricing change.* The pricing change (#1774, 0.002304 GAE per word XOR) made every earlier F4 S a lower bound. A separate algorithmic change (#1862) raised counted work (`research/f4_pricing_jump_20261018/README.md`).
- *Compact orbit at $n = 13$.* Earlier quoted at 1.08× rho as a lower bound, it reads 0.93, 1.53 and 5.7× the strong rho at $w = 0, 0.1, 1$, so the pair is unresolved inside the band (`research/ic_frontier_campaign_20261015/select/SELECTION.md`).

3. Matching the rho reference.

- *Rule v8.* Matching the rho reference raised the Koblitz gap by 1.30–1.58× at every size (Chapter 17).
- *The standing rho improved.* Ecbench's standing `rho.signed_frobenius_strong` now runs at 0.802× its former instruction count with identical GAE, and the strong Koblitz rho executes 718 instructions per step instead of 1,196. Counted-unit comparisons are unaffected. Wall-time comparisons against the older rho overstate IC by up to that factor.
- *Two rho studies on the prime side.* Choosing the better of $P \pm R$ cuts steps by 1.129×. Signed σ steps on ECC2K-130 cut GPU time to 0.961–0.970. Neither is yet in a campaign arm.

4. A replay regression, and its root cause. The root-finder ecbench session was recorded with a replay certificate of 120 of 168 identical records; the 48 mismatches were all in inherited F4's word-XOR counts. On `main` of 19 October it replayed only 72 of 168, with mismatches in all four index-calculus arms.

- *First diagnosis* (20 October). A cross-ISA replay found that with the x86-only BMI2 branch disabled, x86 reproduces all 168 of 168 records, and so does an aarch64 build under emulation (`research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`).
- *Cause, stated precisely* (26 October addendum). At the recorded commit, BMI2 availability decided whether F4's row specialisation was *eager* (every kept row rewritten at once and charged per row) or *lazy* (rewritten on first read). The two differ in the quantity charged, for example 301,961 against 300,531 word XORs. Since #2043 every host is eager and the two eager paths charge identical counts.

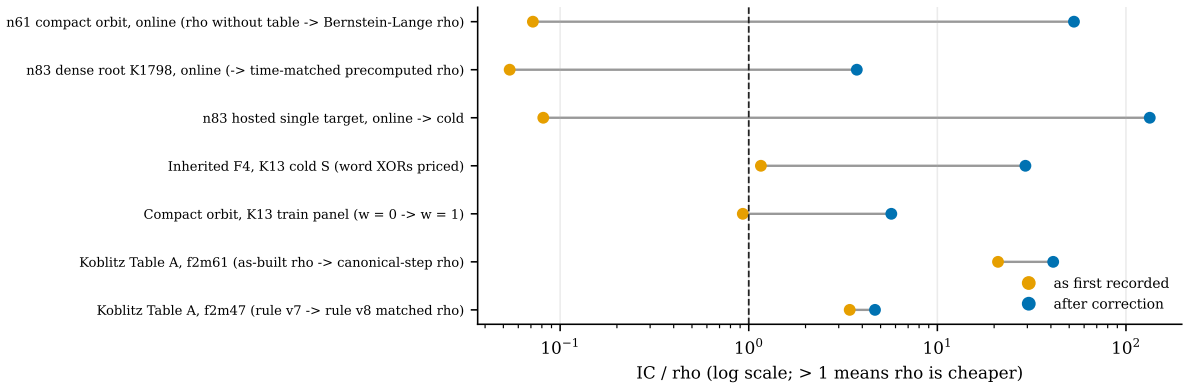


Figure 18.1: IC/rho as first recorded (orange) and after correction (blue), on the boundary the correction used. Values left of the dashed line had claimed IC ahead of rho. Sources: `figures/data-ledger.json`.

Sessions now record the version of the F4 accounting they were made under (`research/validation_sweep_20261019/D-known-issues/ADDENDUM-20261026-item1-accounting.md`).

Archived sessions recorded with the lazy charge cannot replay on `main`. That is a change of definition, not a dependence on the host. Scalars, trial counts and charged group operations are unaffected.

5. Reported cells that compared unlike statistics. The u64-rootbatch rank and per-target cells compared a fastest run with a median, and were corrected from 2.95× and 6.09× to 2.47× and 5.00× (Chapter 10). The $n = 83$ stacked producer's "360 → 94.4" cold-gap history mixed a ratio from another run; paired within the run it is 305.8 → 94.4 (Chapter 10).

6. Results resting on faulty code or wrong references.

- *Plane-quartic IC.* Its S values of 650,896 to $3.79 \cdot 10^7$ rest on a degree-3 reduction in `research/ghs_cubic_quartic_ic_20261012/quartic_ic.rs` that merges distinct classes (reported in an open pull request). Recoveries are still checked against the planted scalar, but relation counts and S are unverified.
- *Volcano census crater.* A claim that IC is 22–25% cheaper on the volcano crater was withdrawn, because the crater's rho reference should have been the cubic-automorphism rho. Against it, IC is 84.9× rho at $w = 0$ (`research/volcano_ecdlp_census_20261017/REPORT.md`, erratum A3).

Reading. None of these corrections removes an IC improvement against its own incumbent. They correct the *distance to rho*, which is the quantity most exposed to an unmatched reference or an unpriced counter. This is why the gate of §18.2 asks for a matched, band-priced rho on every row.

18.2 Validation status

Since 19 October no row may enter the index-calculus leaderboard or `benchmarks/README.md` without three pieces of evidence, named by the evidence directory's `VALIDATION.json` (`docs/VALIDATION_GATE.md`):

- an independent replay certificate with verified scalars;
- an independent validator report that recomputes every headline number;
- a matched rho reference: the strongest available, in the same unit and the same binary, priced across the §5a band.

On 29 October, **19 of the 28 leaderboard rows are validated** (`docs/ic/leaderboard.json`, `validation_gate.status_counts`).

- *Table B, all 13 rows (prime and random binary, one target).* These validated on 24 October (`research/ic_validation_gate_20261024/`):
  - 1,257 runs replay identically in a fresh process;
  - 1,578 of 1,578 negative controls are rejected by an independent verifier;
  - 52 headline numbers recompute to relative 10^{-9};
  - index calculus stays above rho at every band weight.

Its distance is 5.47–38.21× the matched negation-map rho, or 4.05–54.76× across the band.

- *Table A, all 6 rows (Koblitz descent, one target, 64 targets per size).* These validated at baseline v7 (`research/ic_tool_program/rule/v7/`): IC is 3.43–11.5× the as-built rho in the same process. The sweep flags the reference as only partly adequate. Against the matched rho of rule v8, the same rows read 4.67–18.07× (Chapter 17).
- *Table C, all 9 rows (32-target batch).* These stay pre-gate. Their replays and recomputations pass, but no matched rho with a band exists for them.
- *Components outside Tables A–C.* Two are validated: the prime-field base-size tuning and the post-Gröbner linear stage (`research/prime_base_oracle_tuning_20261029/VALIDATION.json`, `research/postgb_linear_stage_20261026/validation/VALIDATION.json`).

Validation says the numbers are what the archive says, and that the reference is fair. **No validated row is below its reference.**

The theory of Part II has a different standard of evidence:

- written proofs, which still need expert review;
- independently checked certificates;
- Lean for the finite and algebraic identities only.

Each theorem there names its tier.

Part V

Outlook

19 Open problems and proposed experiments

1. The solving degree of Weil-descended summation systems. Every subexponential claim for prime-degree binary curves rests on a bound for this degree (§3); none is proved and the experimental evidence is contested. *Experiment:* measure first-fall, last-fall and solving degree of S_{m+1} restricted to ℓ-dimensional subspaces over a grid of (n, ℓ, m) with $m \geq 3$, at least four values of n per (ℓ, m), against the semi-regular prediction, with multi-graded regularity (one block per summand) as a second model. *Settles:* whether the degree grows with n at fixed (ℓ, m); a growth consistent with zero would revive the Petit–Quisquater and Semaev heuristics, a positive slope would close them for this model.

2. Cold index calculus against rho with every operation priced. No measured arm beats matched rho on the cold one-target boundary once native work is priced on both sides (§16). *Experiment:* the frontier campaign of §16 extended to the largest sizes each arm can reach, with exponents fitted per arm and the crossover reported only where the fitted exponents separate across the whole pricing band. *Settles:* whether any implemented arm has an exponent below $1/2$ in r on these families.

3. Relation yield is now the dominant stage. Root finding made the decomposition oracle cheap; what remains is the number of trials per relation, governed by the factor-base density. *Experiment:* for each factor-base family, the measured trials-per-relation law against r and base size, compared with the counting prediction $m!\, r/N^{m-1}$; the residual (e.g. the unexplained factor 5 in the ECC2K-130 extrapolation, §10) is the target.

4. Root finding beyond two summands on prime curves. The two-summand oracle is prior art in mechanism (McGuire–Mueller) and costs $\Theta(N)$ per trial; three summands cost a pair table of N^2 entries. *Question:* is there a sub-N^2 structure for S_4 membership over a prime-field base—for example from the D_m-invariant coordinates of 2-torsion curves—and does it change the $r^{2/3}$ cost of two-summand prime index calculus? A negative answer with a counting argument would be as useful as a positive one.

5. The preprocessing regime. Index calculus with a retained table beats rho without one, and loses by 53–317× to Bernstein–Lange rho with an equal or smaller table (§16.3). *Experiment:* measure the advice–time curve (S, T) of the compact-orbit arm across sizes against preprocessing rho's $S\,T^2 \approx N$; an index-calculus online result counts only if it lies below that curve at equal S.

6. Symmetry at $k = 4$, end to end. The D_4 reduction and its four-unknown weighted form move a *modelled* handover over $\mathbb{F}_{p^4}$ from about 2^{150} to 2^{119}–2^{137} depending on frame (§16.3). Since then the odd-characteristic oracle is 370× cheaper again (index calculus still $1.39 \cdot 10^6\times$ rho), and in characteristic two a Frobenius-reduced form brings the measured gap to 6.1× rho at $n \approx 2^{56}$, closing as $n^{-1/4}$ (§16.4). *Experiment:* continue the characteristic-two ladder to $m = 15$–17 with a measured negation-map rho at every rung, cold and priced across the band; a measured crossover near $n \approx 2^{68}$ would confirm the fit, and its absence would locate the term the model misses. A red-team review (Chapter 13) found the only clean point, $m = 15$, at 5.2× rho after corrections and placed the crossover at $m^* \approx 17.6$ (16.3–19.2); its frozen replication plan (eight targets per arm, $m = 14$–17, about 30 h) is the experiment to run.

7. Pricing and calibration infrastructure. Rankings now require every arm's native work to be counted (AGENTS.md §5a). *Work:* count the three uncounted categories of the shared-rank arm; calibrate every ladder size on one fixed host; record the inherited-F4 code path (BMI2 or portable) in its method identity, now that the cross-host counter drift is traced to it; register the $n = 83$ dense root index in the harness.

8. Genus three after cubic GHS, to the crossover. The one family whose fitted exponent (about $q^{1.4}$) lies below rho's $q^{3/2}$ was 1.15× a measured negation-map rho at $q = 2^{20}$ before rounds three and four cut a further 0.44× (§§16.3,16.4), and is now below the strongest rho at $q = 2^{16}$ under the registered pricing, unresolved at the other sizes to 2^{20}, and above it under a heavier sensitivity pricing, with no

measurable pooled trend (§14). What would settle it is the same comparison at $q = 2^{21}$–2^{24} on instances of one cofactor, with the subgroup multiplication's price measured rather than chosen; its local exponent is down to 1.356, still above 4/3 because the linear algebra grows faster than $q^{4/3}$, and $q = 2^{23}$ needs more than the 4.5 GB cap. *Experiment:* measure negation-map rho on the same instances in the latest unit (inversions and mod-ℓ multiplications priced), and run both past $q = 2^{22}$, in the harness and across the pricing band. A measured crossover there would be the programme's first ADVANCE against rho, on a narrow and undeployed family.

9. Plane quartics: Diem's $\tilde{O}(q)$. Genus-3 GHS covers whose Jacobian is that of a smooth plane quartic admit Diem's index calculus [8, 10] at $\tilde{O}(q)$, against $\tilde{O}(q^{4/3})$ for the hyperelliptic route and $q^{3/2}$ for rho on $E(\mathbb{F}_{q^3})$. The repository builds such quartics directly ($m = 2$ GHS, toy index calculus at $q \leq 11$) and through Smith's $(2, 2, 2)$-isogeny from the hyperelliptic case (18.6% of random curves); a four-workstream programme was scoped on 18 October (`research/plane_quartic_program_20261018/PROGRAM.md`). *Experiment:* the hyperelliptic ladder's instances, carried to plane quartics where Smith's conditions pass, with Diem's relation search and the same linear algebra, against the same matched rho. The 23 October literature round ranks this branch as its one exponent candidate (LEAD-15, `research/pdp_literature_leads_20261022/`). *Since then:* the arithmetic has been rebuilt with canonical class keys, which resolve the degree-3 merges (§4.5). On the direct quartic route, index calculus is below a matched plain rho from $q = 4051$, with fitted exponent 1.083 against 1.485, and N4 v2 extends this against a negation-map rho to $q \approx 2^{22}$, at exponent 0.983 and 0.005× rho (§14). *Still open:* a validation of N4 v2 on an isolated host with a replayable baseline; index calculus on the quartics reached through Smith's isogeny; and a record-scale run (for example $q \approx 2^{40}$, $\ell \approx 2^{120}$) where rho is out of reach and the published Diem–Thomé timings are the comparator.

10. A published index calculus as the reference arm. Every ratio in this treatise is against rho or the programme's own previous arm; none is against the state of the art in index calculus at the same (q, n, m) (§16.5). *Work:* fix, per curve and size, the standard base $\{P : x(P) \in \mathbb{F}_q\}$, the FGHR-symmetrised summation polynomial (2-torsion and, on Koblitz curves, Frobenius invariants), msolve as the solver and WDSat for binary decompositions, and run it in ecbench on the existing ladders (binary $n = 13$–37, Koblitz $n = 41$–61, $k = 4$ extension fields). *Settles:* whether any engineering result of §§8–14 improves on published index calculus, as opposed to on this programme's own baseline. Stage ratios quoted against the in-house F4 say nothing about published Gröbner engines until this is done.

11. The Joux–Vitse cover, to the crossover. The cover of $E(\mathbb{F}_{p^6})$ closes on rho as about p^{-2} and is 1.6–2.0× a pooled rho constant at $p = 1009$, with the line meeting rho near $p \approx 1330$ by extrapolation (§16.5). *Experiment:* emit the cover arm's unpriced F4 and Wiedemann counters, measure rho at the largest p it can run (or with a memory-light rho at $p \geq 503$), and run both at $p \approx 1300$–2000, across the band. A measured crossover would be the programme's first ADVANCE against rho, on a weak and undeployed class; its absence at the predicted p would locate the term the cost model misses.

12. A growing lower bound on the last fall degree. Chapter 6 proves the first fall degree constant and certifies $d_F \geq 5$ on named binary instances, but no presentation has a proved lower bound that grows with n or k. *Proof route:* construct degree-c pseudo-solutions, i.e. linear functionals on $B_{\leq c}$ that kill $W_c(F)$ and not 1, for a family of instances, generalising the closure certificates of Proposition 6.4. The regular closure pattern observed (exactly $4n$ new cubic dimensions per round, then a stall) and the iterated-system lower-bound technique of [36] are the starting points. *Settles:* the open problem of HKY [22] and Kousidis–Wiemers [27]; a bound $\omega(\sqrt{n \ln n})$ would invalidate the subexponential conclusion of Semaev's analysis for these systems.

13. Certificates at scale. *Experiment:* certify $1 \notin W_4$ (and, where it holds, $1 \notin W_5$) on about ten random and ten polynomial subspaces per n for $n = 33$–51, publish the certificates, and extend the guarded chain to $B = 4$–8 by computation. *Settles:* whether the step $4 \to 5$ is a population property, and where $5 \to 6$ occurs; the exact d_F of the guarded chain beyond $B = 3$.

14. The prime-field Kummer tower, re-measured. The session values $d_F = 3, 4, 4, 4, 4, 5, 5, 5, 5, 6$ at $t = 1..10$ were never committed. *Experiment:* regenerate stage A at $t = 1..14$ with closure certificates and score the registered prediction $d_F = 4 + \lfloor (t-2)/4 \rfloor$; extrapolated F4 times are about 1.7 h, 17 h and

7 days at $t = 12$–14 (`research/prime_ic_program_20261029/`). *Settles:* whether the proved constant first fall degree 4 sits beside a last fall degree that keeps growing in the setting PKM introduced.

20 Conclusion

Four months of work on elliptic-curve index calculus produced four kinds of result.

Theory.

- *What is fixed for every curve.* The summation cover's monodromy is $(\mathbb{Z}/2)^{m-2}$ everywhere. Its fibres split completely where relation search operates. The mean yield at a given base size is a double count. So no choice of curve or base geometry can raise the *mean* decomposition yield at matched base size. Coverage, the cost of the membership oracle and the difficulty of the solve remain free, and the engineering of Part III works on those.
- *What is open.* For three presentations of the decomposition system the first fall degree is a constant of the presentation, and the last fall degree is not bounded by it:
  - proved for the guarded chain;
  - certified on named instances of binary Weil descent;
  - measured for prime-field Kummer towers.

  The subexponential claims for binary curves assume exactly the step these results show is not automatic. Whether the last fall degree grows without bound is unproved.

Engineering. The decomposition oracle became two to five orders of magnitude cheaper where root finding applies. The Gröbner engines improved by one to two orders in counted work. Factor bases closed under known-logarithm maps, a k-sum table, and set-up and linear algebra engineering each cut a constant. None of this changed an exponent on a prime field or a prime-degree binary field. The prime-field k-sum oracle moved its exponent from 0.62 to 0.52, but it is generic, and its limit is baby-step giant-step.

Measurement. At a matched reference, priced across a band and replayed, rho is cheaper on every prime-order curve the programme measured:

- 4.7–18× on Koblitz curves of 37–47 bits;
- 5.3–102× on prime curves of 20–40 bits;
- 10.6–38× on random binary curves.

Nineteen leaderboard rows passed a validation gate that demands independent replay, recomputation and a matched rho. None of them is below its reference. Every online win disappeared under an equal-advice audit, and the corrections forced by matching, pricing and replaying lengthened the measured distance more often than any single improvement shortened it.

Where index calculus wins. In weak extension-field families ($E_0/\mathbb{F}_{q^3}$ through genus-3 covers, Joux–Vitse covers of $E(\mathbb{F}_{p^6})$, and perhaps $k = 4$ in characteristic two) the measured cost crosses or approaches rho, as theory predicts. The plane-quartic route measured exponent ≈ 1 against rho's 1.5, at 0.005× negation rho near $q = 2^{21}$. These are known algorithms; the programme's contribution there is fair measurement.

What would change the picture. Three items, in order of what they would settle:

1. *A proved lower bound on the last fall degree that grows with n*, for any presentation of the binary decomposition system. It would answer the question left open by Huang, Kosters and Yeo and by Kousidis and Wiemers, and it is the only result that bears on the subexponential claims.
2. *More certified instances* at $n = 33$–51, with public certificates, to show whether the last fall degree reaches 6.
3. *The frozen $k = 4$ replication* at $m = 14$–17, the cheapest experiment that could put a measured crossover on record. It would be in a weak family.

Nothing here concerns a deployed curve. The open problems of Chapter 19 state each remaining question as the experiment or proof that would settle it.

A Artefacts

Every number in this treatise is copied from a committed artefact. The treatise ran one re-verification (§6.3) and no new measurement. Artefact paths are relative to the `crypto` repository unless prefixed `crypto-autoresearcher:`.

Companion documents.

- `papers/ecdlp-index-calculus-program/` [3] is the chronological record, with study catalogues for both repositories (about 1,400 directories) and the list of open pull requests. Where this treatise reuses its chapters (background, literature, measurement, root finding, Gröbner engines, Koblitz, binary and prime curves, transfers, frontier, open problems), the text is that record's as of 26 October, updated where later evidence changed a number.
- `papers/ecdlp-ic-improvement-ledger/` [2] is the ratchet of incumbents to 23 October.
- `research/paper_opportunities_20261029/` is the assessment that identified the fall-degree results of Chapter 6 as the programme's strongest theory, and that holds the certificate re-verification.

Theory (Part II).

- `crypto-autoresearcher:papers/semaev-conservation-specialization/` (Chapter 5).
- `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/`: `theorem-dossier.md`, `results/`, and `verify/` (`cert_check.py`, `b_sweep.py`).
- `research/guarded_fall_degree_20261028/`: `proofs/`, `extensions/`, `native/`, `formal/`.
- `research/prime_fall_degree_20261027/DERIVED.md`.
- `crypto-autoresearcher:papers/nagao-lemma4-note/` and `crypto-autoresearcher:docs/reports/2026-10-22-semaev-first-fall-replay/`.
- `crypto-autoresearcher:knowledge/leads/ecdlp/` (Chapter 7), `research/twist_obstruction_20261026/` and `research/lift_dense_factor_base_20261027/`.

The summer record (Chapter 15).

- `research/ecdlp_jul_sep_index_20261029/`: `INDEX.json`, `INDEX.tsv`, `SUMMARY.json`, `build_index.py`. Each row cites its `crypto-autoresearcher:ledger/decisions/` file and the `experiments/EXP-*/runs/` directories of its targets.

Late measurements (Chapters 13–18).

- `research/gaudry_k4_char2_redteam_20261026/REPORT.md`.
- `research/quartic_n4v2_20261020/` and `research/ghs3_vs_rho_20261023/`.
- `research/prime_ksum_table_20261028/` and `research/prime_base_oracle_tuning_20261029/`.
- `research/prime_ic_program_20261029/`.
- `research/n83_k1107_bl_reproduction_20261028/`.
- `research/ghs_q16_genus3_subcovers_20261028/`.
- `research/postgb_linear_stage_20261026/`.
- `research/ic_validation_gate_20261024/`, `docs/ic/LEADERBOARD.md` and `docs/ic/leaderboard.json`.

Figures. The figures of the programme record and the ledger are copied with their data (`figures/data-program.json`, `figures/data-ledger.json`); their plotting scripts live in those papers' directories. The new figures (fall degrees, weak families) are rendered by `figures/make_fig_fall_degrees.py` and `figures/make_fig_weak.py` from `figures/data-new.json`. That file holds recorded values only, each with its source.

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